Analysis note for the paper System size and energy dependence of near-side jet-like correlations

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For the STAR collaboration

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Contents

Chapter 1

Particle selection and identification

1.1 Event selection

Data in this paper are from, $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV from RHIC's 2002 run, $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV from RHIC's fourth run (2004) and $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV from RHIC's fifth run (2005.) For all systems and energies except for $d + Au$ collisions, the vertex is limited to within 30 cm of the center of the TPC along the beam axis. The further an event is from the center of the TPC the lower the geometric acceptance; a 30 cm cut allows reasonably uniform geometric acceptance. For $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV the vertex was limited to within 50 cm of the center of the TPC. The production used and the number of good events is listed in Table 1.1 for all data sets.

The centrality bins used in this analysis and their corresponding reference multiplicities, N_{bin}, and N_{part} are given in Appendix D.

Events from $Au + Au$ collisions have a more clearly defined vertex because there are more tracks produced in an $Au+Au$ collision which can be used to reconstruct the vertex position. The 80% most central events were usable in $Au+Au$ collisions. Fewer tracks are produced in $Cu+Cu$ collisions and the luminosity was significantly higher in 2005 than in 2006. The higher luminosity meant that particles from a previous event could still be traversing the TPC when a new event triggered the detector to begin reading out data, a problem called pile-up. For higher multiplicities, it is still

Table 1.1: Information about the productions used in the analysis

collision system and energy	production used	number of events
$d + Au$ $\sqrt{s_{NN}} =$ 200 GeV	P03ih	1M
$Cu + Cu$ $\sqrt{s_{NN}}$ = 200 GeV	P06ib	38M
$Au + Au$ $\sqrt{s_{NN}}$ = 200 GeV	P05ic	24M central, 25M minimum bias

possible to determine the primary vertex, but the lower the multiplicity the harder it is to determine where the vertex from the event which triggered the detector is. For this reason only the 60% most central data from these events was usable. For more peripheral events, the vertex which triggered the event could not be found in more than 2% of events, in large part because vertices from pile-up events usually had a greater multipicity. Full details of the separation of the vertices from the triggered event and from pile-up events can be found in [1].

1.2 Selection of events and charged tracks

All particles are restricted to $|\eta| < 1.0$. All tracks used in these studies are required to have at least 15 hits in the TPC associated with the track. Tracks are restricted to a DCA < 1.0 cm, limiting contributions from decay particles.

Unidentified hadron efficiency

To determine the efficiency for reconstructing tracks, the response to the particle as it traverses the STAR detector is simulated. For unidentified hadrons, the response of the detector is simulated for pions. There are significant contributions from protons in the data, however, the efficiency for the reconstruction of pions and protons is similar

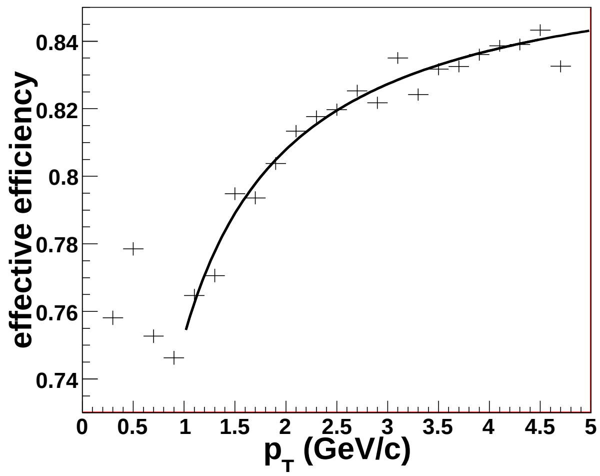

Figure 1.1: The effective efficiency, including acceptance effects, for reconstructing unidentified hadrons in 0-10% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The curve is a fit to the data.

for particles with $p_T > 1$ GeV/c. The reconstructed track must have at least 10 of the simulated hits for the simulated track to be considered reconstructed. The ratio of the number of reconstructed tracks to the number of simulated tracks gives the effective efficiency, including acceptance effects. The effective reconstruction efficiency was determined for tracks meeting the quality requirements applied in the analysis and was determined as a function of p_T and centrality for each data set. This was fit to a curve $a\ exp((\frac{b}{p_T})^c)$. An example of the effective efficiency for the reconstruction of unidentified hadrons in 0-10% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV is shown in Figure 1.1. For $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV, a constant efficiency of 0.89 was used. The parameters for the fits to the efficiency as a function of p_T for each centrality bin for every other data set are in Appendix C.

1.3 V^0 reconstruction

A schematic diagram of the geometry of a Λ decay is shown in Figure 1.2. The K_S^0 decay is similar except for the identity of the daughters. This shows the different parameters describing the reconstructed decay vertex. Restrictions of these parameters can be varied to alter the purity and the efficiency of reconstructed V^0 candidates.

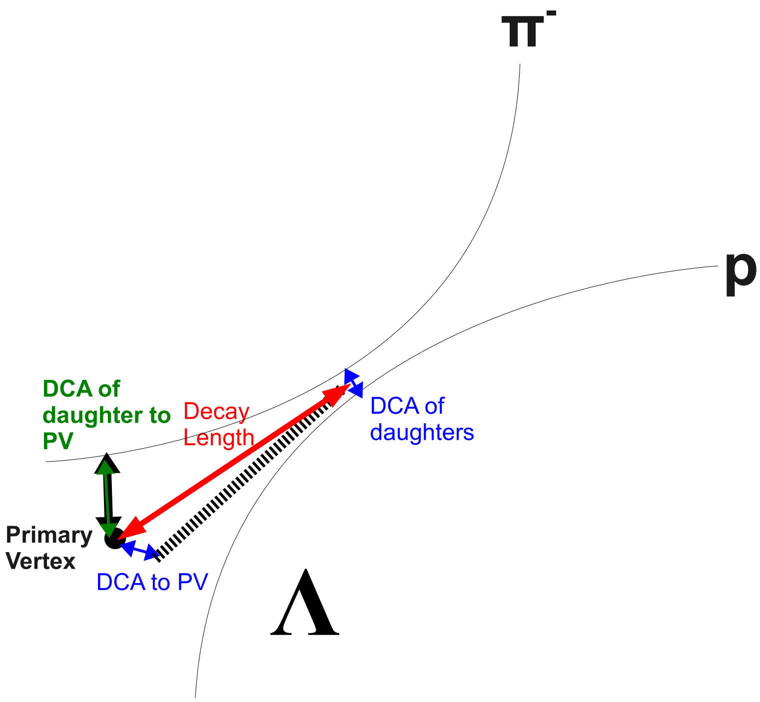

Figure 1.2: Schematic diagram of a Λ decay showing the different parameters of a V^0 candidate's reconstructed decay vertex. Primary vertex is abbreviated PV.

Table 1.2: Default geometric cuts for V^0 candidates.

	$p_T^{V^0} < 3.5$ GeV/c	$p_T^{V^0} > 3.5$ GeV/c
DCA of V^0 to PV $<$	0.8 cm	0.8 cm
DCA of daughters $<$	0.8 cm	0.8 cm
DCA of daughters to PV $>$	0.7 cm	0 cm
Decay length $>$	2 cm	2 cm
N_{hits} on daughter track $>$	15	15
$\mid n_{\sigma,dE/dx} \mid \geq$	3	3

The list of default cuts is given in Table 1.2.

The dE/dx of daughter tracks can also be used to reduce the background. The number of standard deviations away from the center of the band is calculated and a 3σ cut was applied by default. The dE/dx bands from different particles begin to merge above 1 GeV/c in the STAR TPC, but at higher momenta the bands begin to

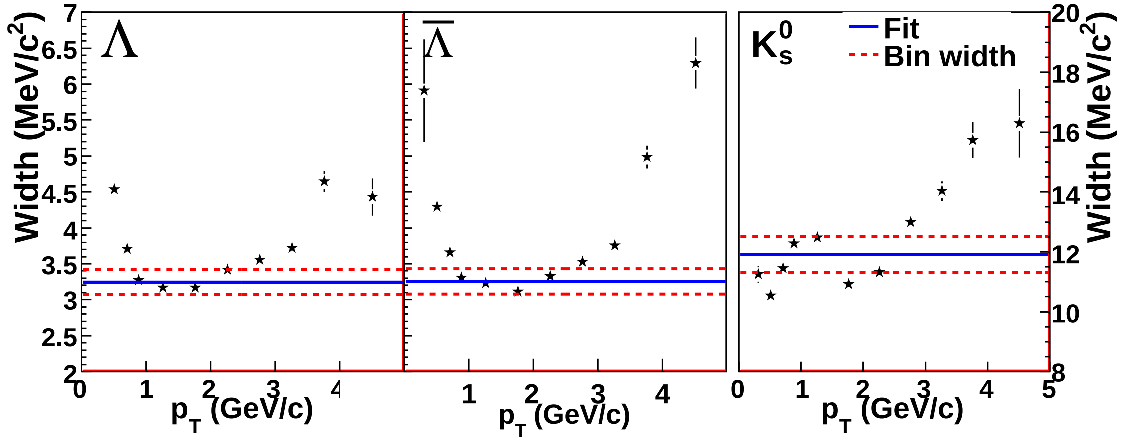

Figure 1.3: V^0 mass widths as a function of p_T in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The solid black line indicates the PDG value and the red dashed lines indicate the PDG value $\pm$ the bin width.

separate so some resolution is possible. It is more effective for the p and $\bar{p}$ daughters of the Λ and $\bar{\Lambda}$ since there are more primary pions than protons produced.

The widths of the mass peaks as a function of p_T are shown in Figure 1.3 and the centers of the mass peaks as a function of p_T are shown in Figure 1.4. These parameters are determined from a fit of a Breit-Wigner distribution

$$f(M_{reco}) = N \frac{(\frac{\Gamma}{2})^2}{(M_{reco} - M_{center})^2 + (\frac{\Gamma}{2})^2} \tag{1.1}$$

plus a linear background to the mass peak, where Γ is the width of the distribution, M_{center} is the center of the distribution, M_{reco} is the reconstructed mass, and N is the total number of particles. The mass resolution is dominated by detector effects from the single track momentum resolution and the shape of the reconstructed mass peak is altered by the geometric cuts. A Breit-Wigner is used because this function fits the mass peak best; the fits were only used for determining the purity of the sample. Since the purpose of these studies was only identification of V^0s for di-hadron correlations, the p_T dependence of the centers and widths of the mass peaks was not investigated in detail.

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Starting with the geometric cuts given in Table 1.2, the restrictions on the DCA

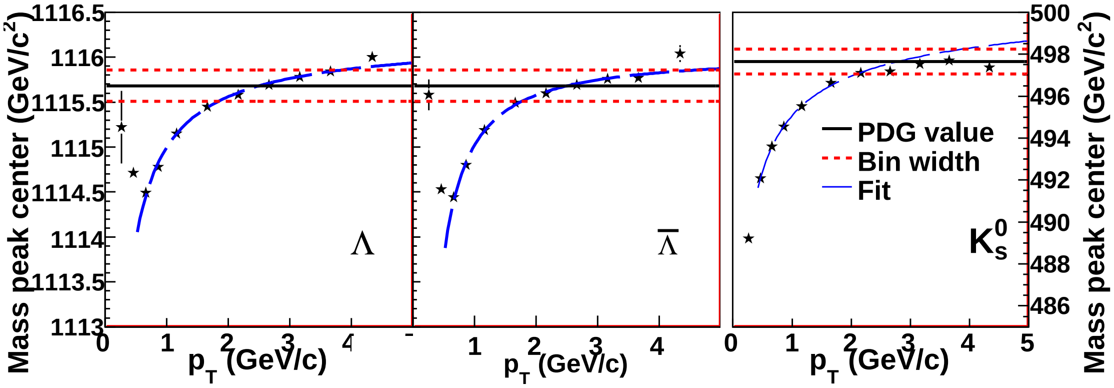

Figure 1.4: V^0 mass peak centers as a function of p_T in $Cu+Cu$ collisions at $\sqrt{s_{NN}} =$ 200 GeV. These were fit to $Ae^{-(b/p_T)^c}$, shown as a black dashed line. These parameters are given in Table 1.3. The solid black line indicates the PDG value and the red dashed lines indicate the PDG value ± the bin width.

Table 1.3: V^0 center fit parameters from fit of data in Figure 1.4 to $Ae^{-(b/p_T)^c}$ in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

particle	a	b	c
Λ	1116.25 ± 0.02	$3.9 \cdot 10^{-4} \pm 0.4 \cdot 10^{-4}$	0.86 ± 0.01
$\bar{\Lambda}$	1116.03 ± 0.02	$2.4 \cdot 10^{-3} \pm 0.3 \cdot 10^{-3}$	1.16 ± 0.03
K^0_S	501.5 ± 0.2	$1.7 \cdot 10^{-4} \pm 0.3 \cdot 10^{-4}$	0.50 ± 0.02

of the V^0 to the primary vertex, the DCA of the daughters to each other, the DCA of the daughters to the primary vertex, the decay length, the number of TPC hits on each daughter, the $n_{\sigma,dE/dx}$ and the distance from the center of the mass peak for each daughter were tuned until a purity of about 95% was reached in each centrality bin and at each momentum. The final cuts are given in Appendix B. These tight restrictions on the purity were used to reduce the impact of impurities on di-hadron correlations. The purity as a function of p_T, shown in Figure 1.5, was determined by fitting the mass peak with a Breit-Wigner distribution. The center was determined from a fit to the center as a function of p_T to $Ae^{-(b/p_T)^c}$. The parameters of these fits are given in Table 1.3.

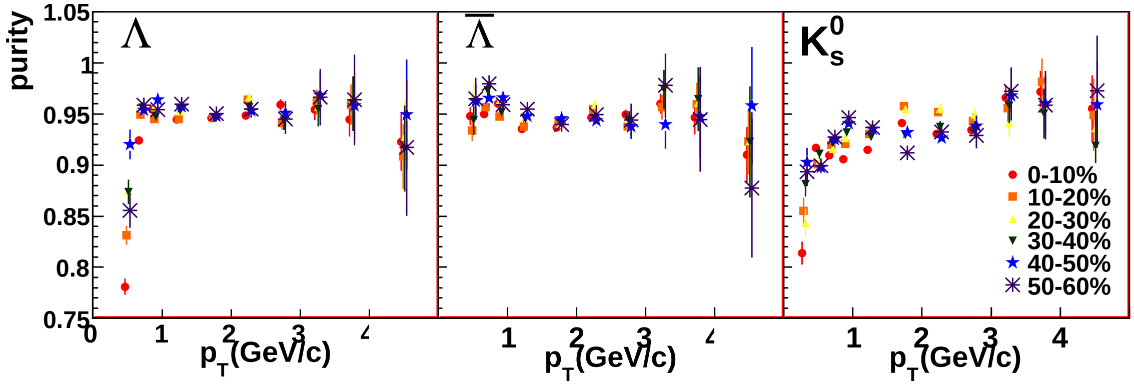

Figure 1.5: V^0 purity as a function of p_T after tuning the geometric cuts in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

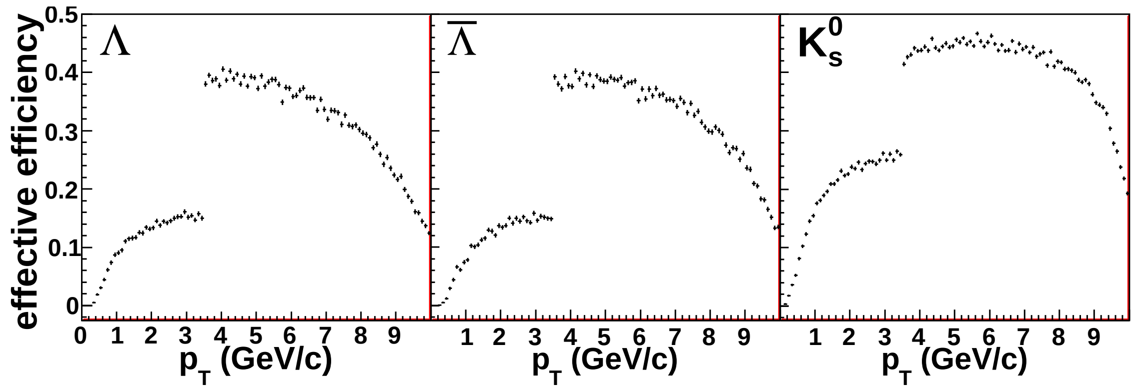

Figure 1.6: Sample effective V^0 reconstruction efficiency in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV for the default cuts listed in Table 1.2.

V^0 efficiency

The effective V^0 reconstruction efficiencies were determined as for unidentified hadrons. The efficiency of reconstructing a V^0 in the STAR detector is limited by the single track efficiency ($\leq$ 0.90) and the branching ratio (69.2% for K^0_S and 64.1% for Λ and $\bar{\Lambda}$.) These effects combined put an upper limit on the efficiency of about 55% before applying any geometric cuts. As for charged hadrons, the cuts were applied to the embedded, reconstructed data, compared to the number of simulated tracks, and the ratio fit to give an efficiency. A sample effective efficiency, including acceptance effects and the branching ratio, for the default cuts listed in Table 1.2 is shown in Figure 1.6. The large jump in the efficiency at $p_T = 3.5$ GeV/c is from the change

Table 1.4: Default geometric cuts for Ξ and Ω candidates in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

	$p_T < 3.5$ GeV/c	$p_T < 3.5$ GeV/c
DCA of V^0 daughters	< 0.8 cm	< 0.8 cm
DCA of V^0 daughters to PV	> 0.4 cm	> 0.0 cm
DCA of Ξ to PV	< 0.8 cm	< 0.8 cm
DCA of Ξ daughters	< 0.8 cm	< 0.8 cm
Ξ Decay length	> 2 cm	> 2 cm
N_{hits} on daughter track	> 15	> 15
$n_{\sigma,dE/dx}$	$-3 < n_{\sigma,dE/dx} < 3$	$-3 < n_{\sigma,dE/dx} < 3$

in the cuts at $p_T = 3.5$ GeV/c. The fit parameters in each p_T and centrality bin are given in Appendix C.

1.4 Ξ reconstruction

The identification of a Ξ or Ω through their decay vertices is a straightforward extension of V^0 identification. Instead of π or p daughters, one of the daughter tracks is a Λ. There are more parameters describing the geometry of the reconstructed decay, those describing the Λ decay vertex and those describing the Ξ or Ω decay vertex. The efficiency is also significantly lower than for the reconstruction of a V^0 because three tracks must be reconstructed rather than two and the branching ratios of both the Λ and the Ξ or Ω are relevant.

As with V^0 reconstruction, default cuts were applied and these cuts were tightened to reach the desired purity. Most of the cuts applied are exactly analogous to those applied for V^0 recontruction and only cuts unique to Ξ and Ω recontruction will be reviewed here. Table 1.4 lists the default cuts used for Ξ and Ω reconstruction.

As for V^0 reconstruction, the background for the Ξ is dominated by random combinations of primary tracks. This means that on average the distance of closest approach to the primary vertex of the Ξ will be lower on average than to the background and the DCA to the primary vertex of the Λ will be higher on average than

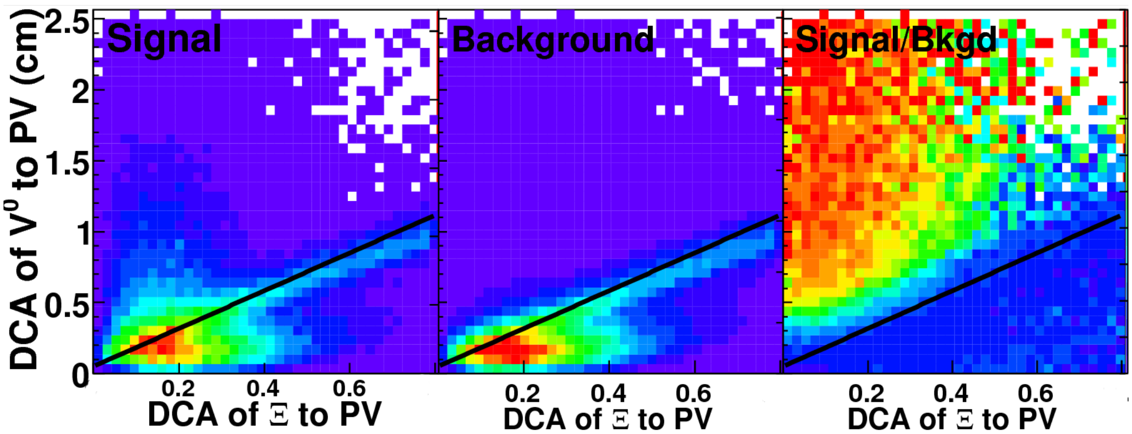

Figure 1.7: The DCA of the Λ daughter to the primary vertex versus the DCA of the Ξ candidate to the primary vertex in the signal region (1315< m_{ξ} <1325), the background region (1270< m_{Ξ} <1312 and 1133< m_{Ξ} <1350 MeV/c) and the ratio of the two for all Ξ candidates with $p_T > 1$ GeV/c in the $Cu+Cu$ data at $\sqrt{s_{NN}} = 200$ GeV. The line shows where the cut is applied to reduce the background in Figure 1.8.

the background. The DCA of the Λ daughter to the primary vertex and the DCA of the Ξ candidate are strongly correlated for real Ξ.

Figure 1.7 shows the DCA of the Λ daughter to the primary vertex versus the DCA of the Ξ candidate in the signal region (1315< m_{Ξ} <1325), the background region (1270< m_{Ξ} <1312 and 1133< m_{Ξ} <1350 MeV/c) and the ratio of the two for all Ξ candidates with $p_T > 1$ GeV/c in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The strong correlation between the DCA of the Ξ candidate to the primary vertex and of the DCA of the V^0 daughter to the primary vertex was demonstrated in [2] to be an efficient way to reduce the background in the Ξ mass peak. The line drawn in Figure 1.7 shows an example of a correlated cut

$$\mathrm{DCA\,of\,V^0\,to\,PV} > A * \sqrt{\mathrm{DCA\,of\,\Xi\,to\,PV}} + B \tag{1.2}$$

which can be used to eliminate background. A tighter cut is applied by either increasing either A or B.

Figure 1.8 shows the mass peak for Ξ candidates in $Cu+Cu$ at $\sqrt{s_{NN}} = 200$ GeV above $p_T > 1.0$ GeV/c before applying the cut in Equation 1.2, after applying the cut

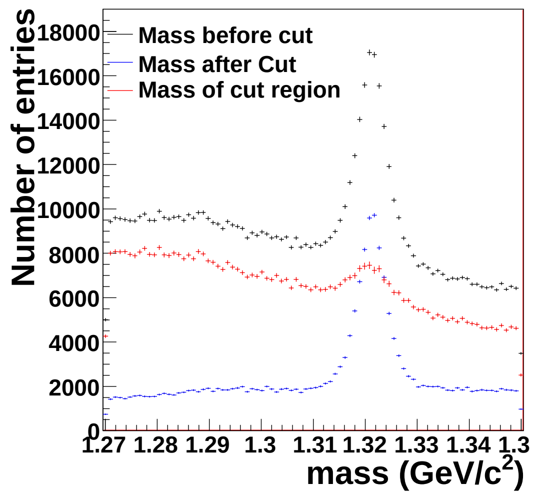

Figure 1.8: Ξ mass peak before (black) and after (blue) applying the correlated cut. The difference is shown in red. The cut applied, DCA of the V^0 to the primary vertex $>$ 1.33 times the DCA of the Ξ candidate to the primary vertex, is shown in Figure 1.7.

Table 1.5: Ξ center fit parameters from fits to the data in Figure 1.9

particle	a	b	c
Ξ^-	1323.1 ± 0.3	$8.0 \cdot 10^{-5} \pm 9.0 \cdot 10^{-5}$	0.643 ± 0.088
Ξ^+	1323.1 ± 0.1	$3.8 \cdot 10^{-5} \pm 0.5 \cdot 10^{-5}$	0.595 ± 0.009

in Equation 1.2, and the difference between the two. This cut clearly eliminates mostly background. This was one of the most efficient cuts for reducing the background.

The width and centers of the mass peaks as a function of p_T as determined from fits of the mass peaks to Breit-Wigner distributions are given in Figure 1.9. The center was then fit as a function of p_T to $Ae^{-(b/p_T)^c}$. The parameters from these fits are given in Table 1.5. The purity was calculated as the number of particles in the mass peak as determined from the fit divided by the total in the mass peak. The geometric cuts were tuned until the purity was around 90%. The final geometric cuts are given in Appendix B and the purity is shown in Figure 1.10. Ξ baryons are only used as trigger particles so no reconstruction efficiency is necessary.

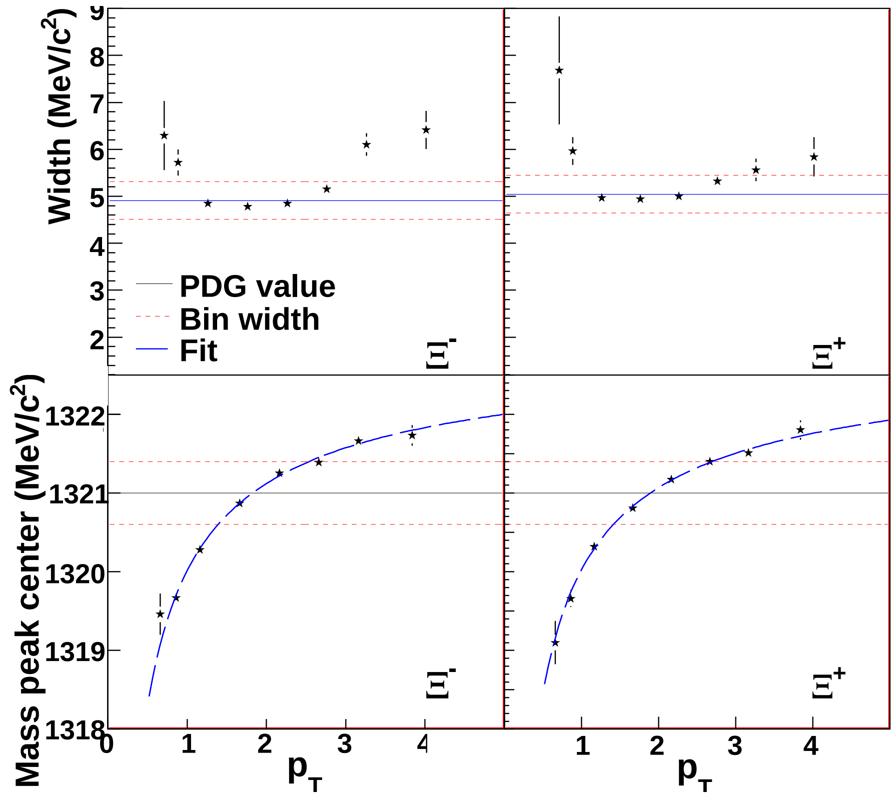

Figure 1.9: Centers of Ξ mass peaks in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV as determined from fits of the mass peaks to a Breit-Wigner distribution

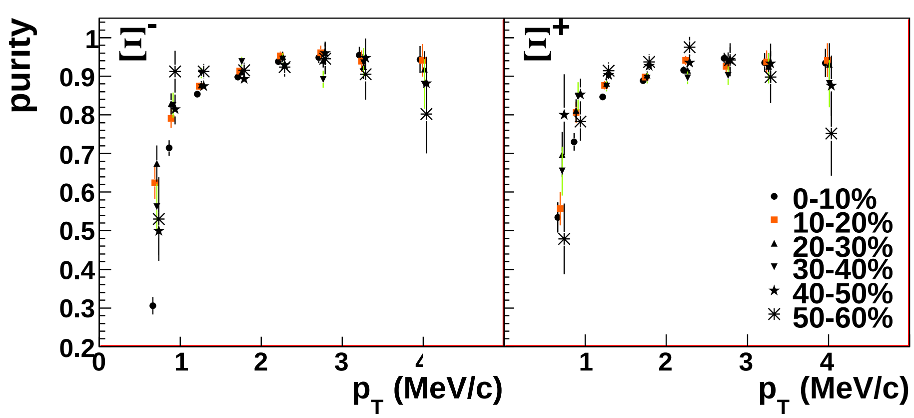

Figure 1.10: Ξ center

Chapter 2

Correlation Method

2.1 Correlation technique

In a given event, a high-p_T track, called the trigger particle, is selected and the distribution of all lower momentum particles, called associated particles, relative to that track in $\Delta\eta$ and $\Delta\phi$ is determined. Tracks are selected as described in Chapter 1, with no additional restrictions on the high momentum track other than the p_T. Multiple tracks in a single event may be counted as trigger particles. The data are symmetric about $\Delta\phi = 0$ and $\Delta\eta = 0$ because the sign of $\Delta\phi$ and $\Delta\eta$ are determined by an arbitrary choice of coordinates. Since many of these measurements are limited by statistics, the data are reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$ to minimize statistical fluctuations. The single track efficiency, described in Chapter 1, is dependent on particle type, collision system and energy, p_T, and centrality and the correction for associated particles is applied on a track-by-track basis. The final distribution is normalized by the number of trigger particles, so no correction for the efficiency of reconstructing the trigger particle is necessary. Two corrections for the acceptance of track pairs in $\Delta\phi$ and $\Delta\eta$, described below, are applied. This is done for the full sample of events to give the average distribution of associated particles relative to a high-p_T trigger particle $\frac{1}{N_{trigger}}\frac{d^2N}{d\Delta\phi d\Delta\eta}$.

Two key features of the STAR TPC affect the geometric acceptance of track pairs. Tracks in the TPC can only be resolved well for -1 $< \eta <$ 1 because the lower

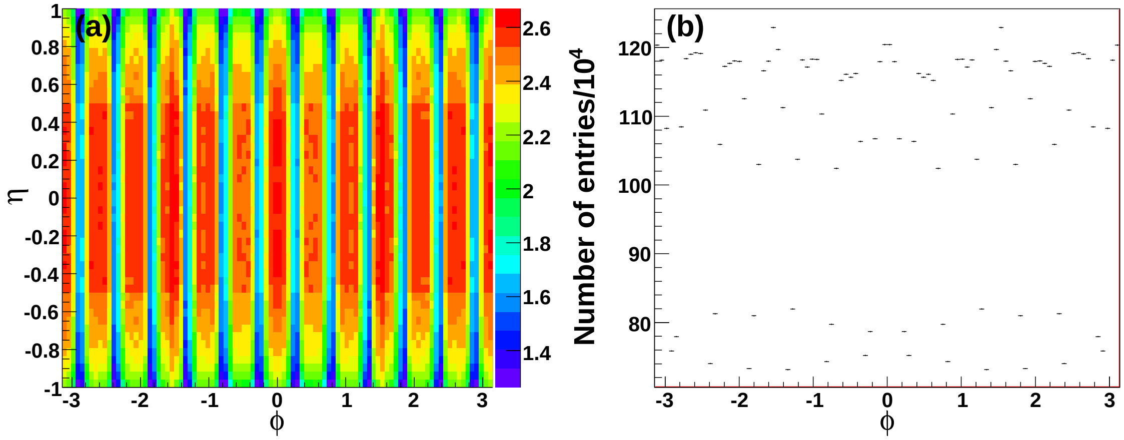

Figure 2.1: Distribution of tracks in (a) ϕ and η and (b) ϕ in $Cu+Cu$ events at $\sqrt{s_{NN}} = 200$ GeV for 1.5 GeV/c < p_T <6.0 GeV/c. Note the suppressed zero.

geometric acceptance leads to tracks with fewer hits in the TPC, so only tracks within -1 < η < 1 are used for this analysis. Track pairs with $\Delta\eta \approx 0$ are nearly always accepted, however, track pairs with $|\Delta\eta\,| \approx 2$ are only accepted if the tracks are near opposite edges of the detector. Therefore acceptance is roughly 100% near $\Delta\eta \approx 0$ but roughly 0% near $|\Delta\eta\,| \approx 2$. This leads to an acceptance in $\Delta\eta$ with a triangular shape.

In addition, there are gaps between each of the 12 TPC sectors. This leads to holes in the acceptance in azimuth. Most tracks will have hits in multiple sectors, however, the loss of hits leads to a decrease in the reconstruction efficiency for tracks at the angles of the sector boundaries. A sample distribution of tracks reconstructed in the TPC is shown in Figure 2.1 showing the lower efficiency at the TPC sector boundaries.

To correct for these effects, the distributions of tracks for both the associated and the trigger particles are recorded. A random trigger (ϕ,η) is chosen from this distribution of trigger particles and a random associated (ϕ,η) is chosen from the distribution of associated particles. In this way a simulated distribution of uncorrelated track pairs that includes detector acceptance effects is attained. This distribution is normalized to 1 at the peak at $\Delta\eta \approx 0$ to give an efficiency for finding a pair of tracks as a function of ($\Delta\phi$,$\Delta\eta$). A sample of the efficiency due to detector effects is shown

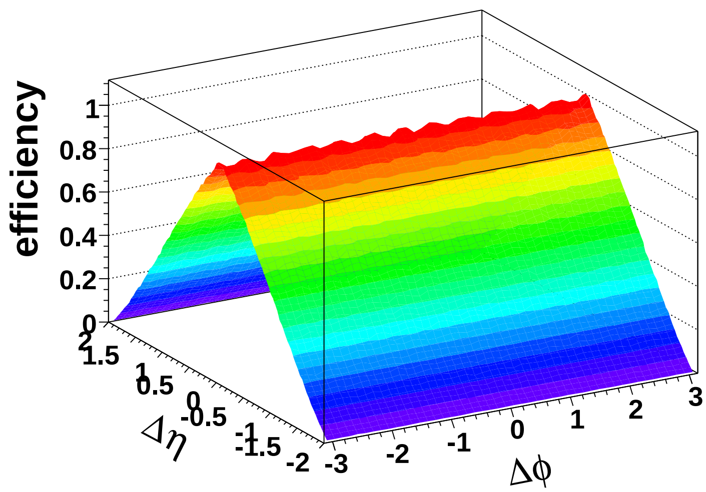

Figure 2.2: Sample efficiency due to detector effects for $Cu + Cu$ events at $\sqrt{s_{NN}} =$ 200 GeV with 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ and $3.0 < p_T^{trigger} < 6.0$ GeV/c

in Figure 2.2 and the effects of the geometric acceptance are evident.

2.1.1 Track merging correction

If two tracks have small separations in pseudorapidity and azimuth, they are more difficult to resolve because hits from each particle may not be resolved by the TPC. Therefore, if the particles have nearly identical momenta, their tracks are not distinguished. This effect is called track merging and causes an artificial dip in the raw correlations centered at ($\Delta\phi$,$\Delta\eta$) = (0,0). Another similar effect is evident in the TPC from the crossing of high-p_T tracks. While the hits are not lost in this case, one track will lose hits near the crossing point and this particle's track gets split into two shorter tracks. Shorter tracks are less likely to meet the track quality cuts in Chapter 1 because they have fewer hits. This effect causes artificial dips near ($\Delta\phi$,$\Delta\eta$) = (0,0) but slightly displaced in $\Delta\phi$. The location of these dips in $\Delta\phi$ is dependent on the relative helicities of the trigger and associated particles. The helicity is given by

$$h = -\frac{qB}{|qB|} \tag{2.1}$$

where q is the charge of the particle and B is the magnetic field [3].

A method for correcting high-p_T triggered di-hadron correlations was developed in [3, 4], building on a method for correcting for track merging in studies of Hanbury-Brown-Twiss correlations [5].

Correction for unidentified hadrons

Figure 2.3 shows a sample of raw di-hadron correlations between unidentified hadrons for all four helicity combinations in $Au + Au$ events at $\sqrt{s_{NN}} = 62$ GeV. When the helicities of the trigger and associated particles are the same, the percentage of overlapping hits is greater. Track pairs are lost whether the pair is part of the combinatorial background or part of the signal so the magnitude of the dip is greater when the background is greater. $Au + Au$ collisions were chosen to demonstrate the effect because the background is largest for these collisions, making the effect easy to see.

The number of hits shared by two tracks is calculated based on the trajectory of the track and if more than 10% of hits are shared between two tracks, the tracks are considered merged and the track pair is discarded. To get an acceptance correction for this effect, the trigger particle is mixed with associated particles from another event. This procedure is called event mixing. The trigger and associated particles from separate events will have no real correlations so any structures in the resulting ($\Delta\phi$,$\Delta\eta$) distribution will result only from detector acceptance effects, including track merging. In the mixed events, the origin of the associated track must be shifted to the new vertex so that the percentage of merged hits can be calculated accurately. In addition, mixed events are required to have vertices within 2 cm of each other along the beam axis in order to ensure similar geometric acceptance in both events. The reference multiplicities of both events are required to be within 10. The calculation of the number of hits which may be shared by the tracks does not describe the performance of the TPC perfectly and some track pairs which were calculated to share more than 10% of their hits are reconstructed. These pairs are discarded even if they were reconstructed so that the percentage of merged track pairs is the same in the data and the mixed events. The mixed events are then used to determine an

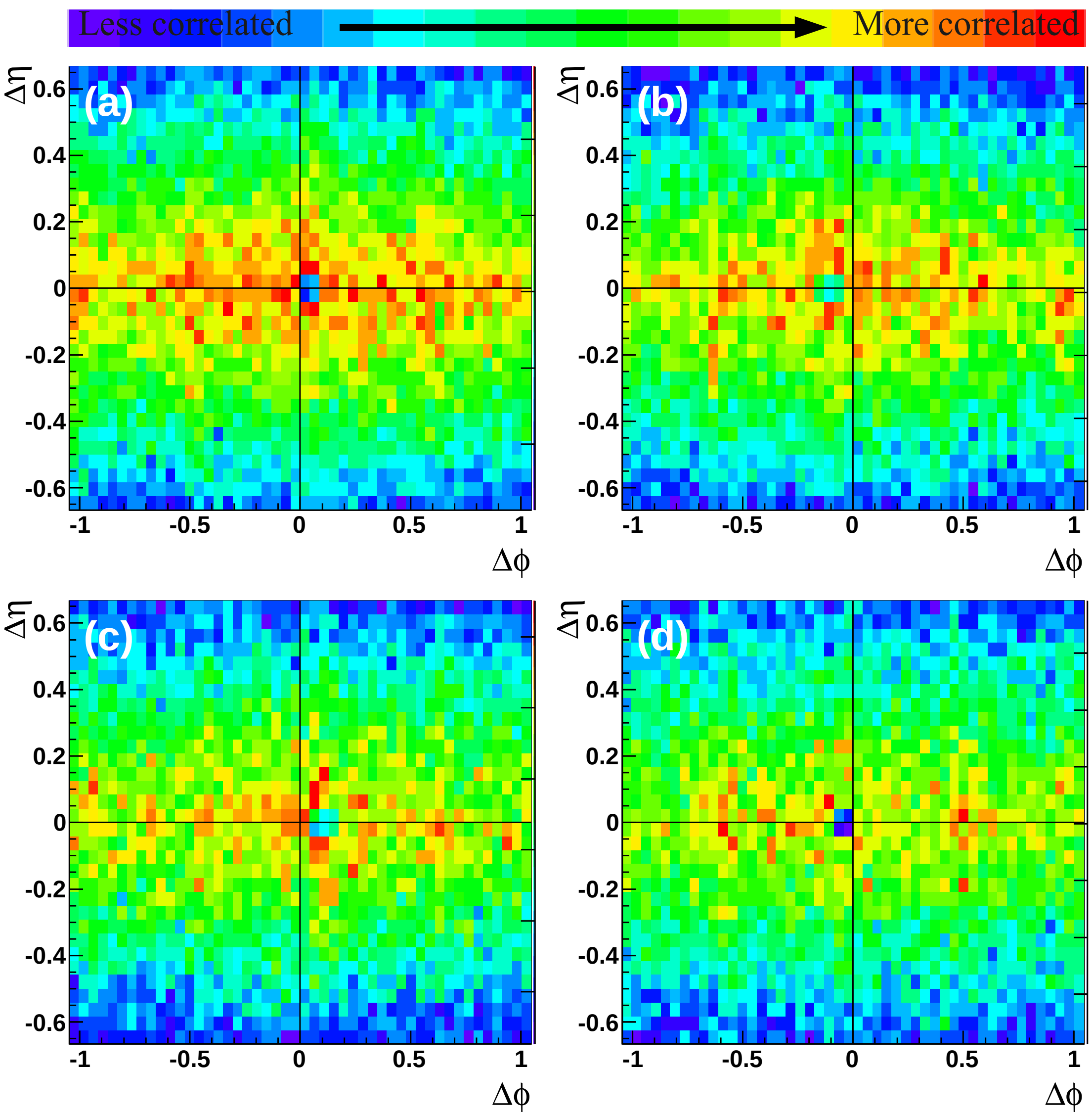

Figure 2.3: Raw correlations for (a) trigger particle helicity 1 and associated particle helicity 1 (b) trigger particle helicity 1 and associated particle helicity -1 (c) trigger particle helicity -1 and associated particle helicity 1 (d) trigger particle helicity -1 and associated particle helicity -1 in h-h correlations in 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV for $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c and $3.0 < p_T^{trigger} < 6.0$ GeV/c.

acceptance correction as a function of $\Delta\phi$ and $\Delta\eta$ for track pairs. Most of the dip is corrected away when the data are divided by the efficiency from mixed events [3, 4].

While the method discused in Section 2.1 corrects for the geometric acceptance of the detector, this method corrects for both the geometric acceptance and for track merging. This method is much more CPU-intensive and therefore was only applied for small ($\Delta\phi$,$\Delta\eta$).

Figure 2.4(a) shows the data and Figure 2.4(b) shows the mixed events for a sample correlation. The dip has a different shape when the helicities of the tracks are the same and when they are opposite. To do the correction the data and mixed events are rotated so that the dip is in the same location. The same helicity data are divided by the acceptance correction from the same helicity events and the opposite helicity data are divided by the acceptance correction from the opposite helicity mixed events. These data are then added after the acceptance correction. A small residual dip remains because mixed events to do not recreate the dip perfectly. The remaining dip is corrected for using the symmetry of the correlations. Since the data are symmetric about $\Delta\phi = 0$, the data on the same side as the dip are discarded and replaced by the data on the side without the dip. Then the data are reflected about $\Delta\eta = 0$ and added to the unreflected data to minimize statistical fluctuations.

This method is only applied to $|\Delta\phi| <1.05$ and $|\Delta\eta| <0.67$ because it is CPU-intensive and track merging only affects small $\Delta\phi$ and small $\Delta\eta$. For large $\Delta\phi$ and $\Delta\eta$, the method described in Section 2.1 is applied. Figure 2.4(c) shows the data after the track merging correction and Figure 2.4(d) shows the data using the method described in Section 2.1 - without the track merging correction - for all helicity combinations for 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

Correction for identified particles

Correlations with an identified V^0 also have a significant loss of tracks at small $\Delta\phi$ and $\Delta\eta$ due to track merging. For correlations with particles identified through decay vertex reconstruction, the associated particle may be merged with any of the daughter particles. The method developed by [3, 4] for unidentified (h-h) correlations was extended to identified particle correlations and is similar to that for h-h correlations. The only correlations studied here are those where either the trigger or the associated particle is an unidentified hadron. A track pair is thrown out if the unidentified

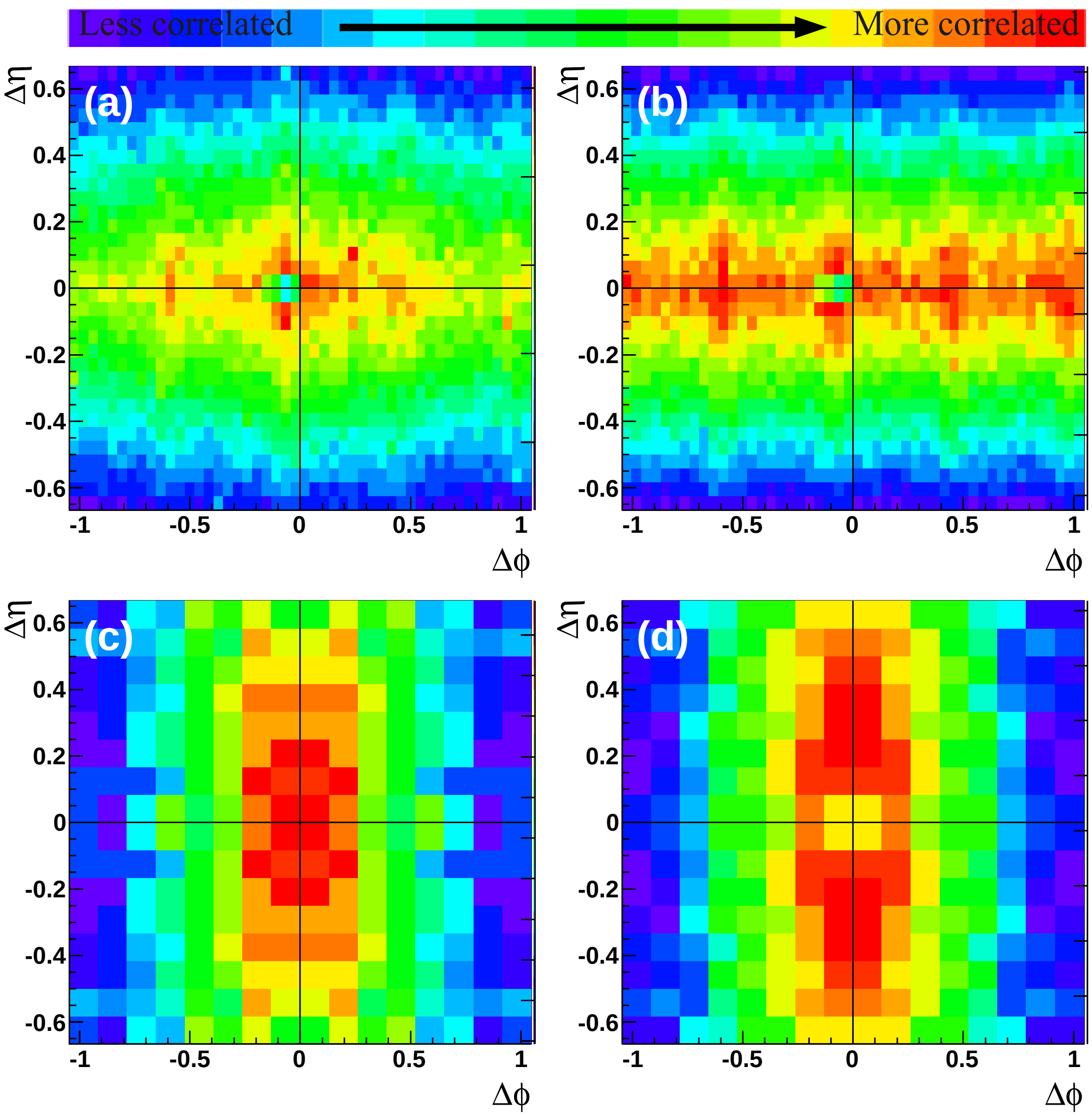

Figure 2.4: The region around the dip for (a) data (b) mixed events (c) data after track merging correction (d) data using the method described in Section 2.1 (no track merging correction) in h-h correlations in 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV for 1.5 < $p_T^{associated} < p_T^{trigger}$ GeV/c and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c.

hadron shares more than 10% of its hits total with any of the decay particles. As for h-h correlations, this is done for both data and mixed events and the efficiency as a function of $\Delta\phi$ and $\Delta\eta$ is used to correct the data for both detector acceptance and

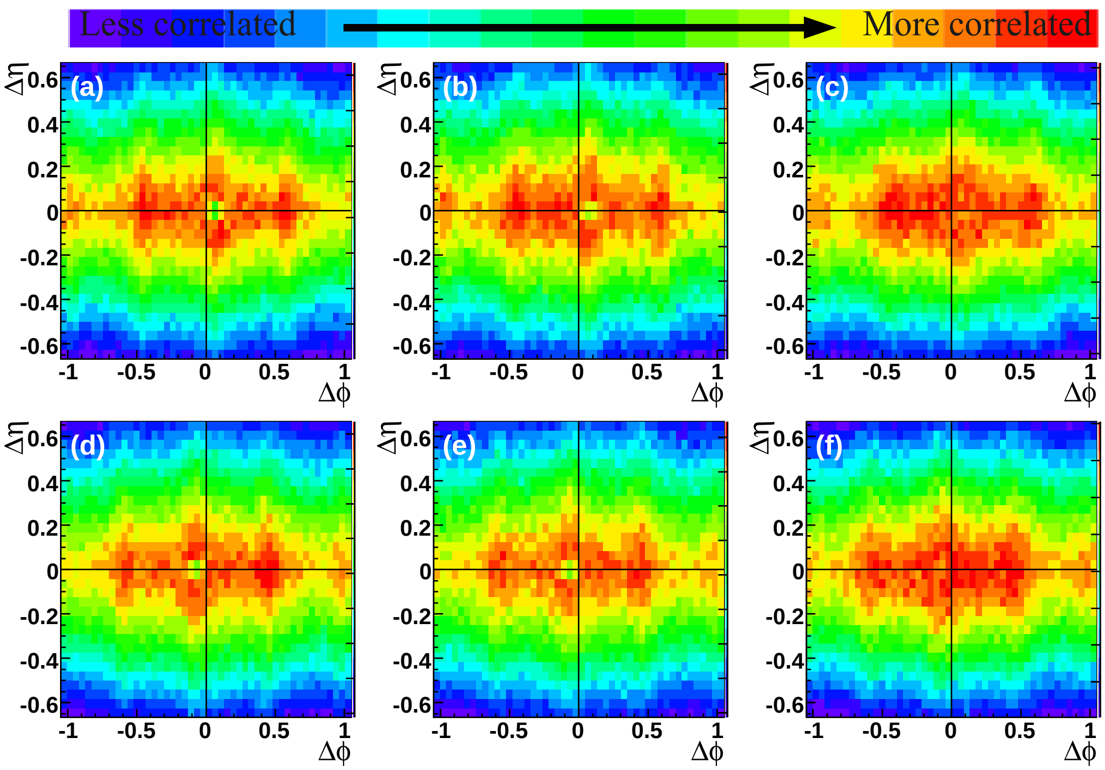

Figure 2.5: The location of the track merging dip in V^0 triggered correlations for a (a) Λ trigger with an associated particle with helicity 1 (b) $\bar{\Lambda}$ trigger with an associated particle with helicity 1 (c) K_S^0 trigger with an associated particle with helicity 1 (d) Λ trigger with an associated particle with helicity -1 (e) $\bar{\Lambda}$ trigger with an associated particle with helicity -1 (f) K_S^0 trigger with an associated particle with helicity -1 in 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

track merging.

The residual dip is corrected by reflection, so the location of the dip must be established for different trigger particles. Figure 2.5 shows the raw correlations for V^0 triggered correlations for associated hadrons with different helicities. Since the V^0 is neutral, its helicity is zero. Data from $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV are used because the multiplicities are higher, making the dip larger, and more statistics were available, making the location of the dip more evident. The event selection and particle identification are as described in [6, 7, 8, 9, 10]. The dip is larger for Λ and $\bar{\Lambda}$ trigger particles than for K_S^0 trigger particles. The location and size of the dip is dependent on the momenta of the decay daughters and the decay kinematics are different for the Λ and $\bar{\Lambda}$ than for the K^0_S.

Figure 2.6 shows the location of the dip for Ξ^- and Ξ^+ trigger particles with different combinations of helicities. As for V^0 trigger particles, data from $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV are used. The magnitude of the dip is greater for a multistrange trigger particle than for a V^0 because of the greater number of decay particles involved.

Figure 2.7 shows the location of the dip for V^0 associated particles. All helicities of associated particles have been added together since the dip is independent of the trigger particle helicity. The dip is clearly visible for Λ and $\bar{\Lambda}$ associated particles, however, the dip is considerably less pronounced for K^0_S associated particles.

2.2 Separation of the jet-like correlation and the ridge

Figure 2.8 shows a sample correlation after detector acceptance and track merging corrections. To study the near-side of di-hadron correlations, the two components are separated and the yield, number of particles in each within limits on $p_T^{associated}$ and $p_T^{trigger}$, is studied. The ridge was previously observed to be roughly independent of $\Delta\eta$ within the acceptance of the STAR TPC while the jet-like correlation is confined to $|\Delta\eta| < 0.75$.

2.2.1 Extraction of the jet-like yield

The raw correlation consists of a combinatorial background modulated by elliptic flow, v_2, and the signal, which consists of the jet-like correlation and the ridge on the near-side. There are two methods used to extract the jet-like yield from the raw correlation. The first method uses different projections in $\Delta\phi$, the fact that the jet-like correlation is not present at large $\Delta\eta$, and the fact that the combinatorial background and the ridge are both roughly independent of $\Delta\eta$ to isolate the jet-like correlation as a function of $\Delta\phi$. The second method involves taking a projection in

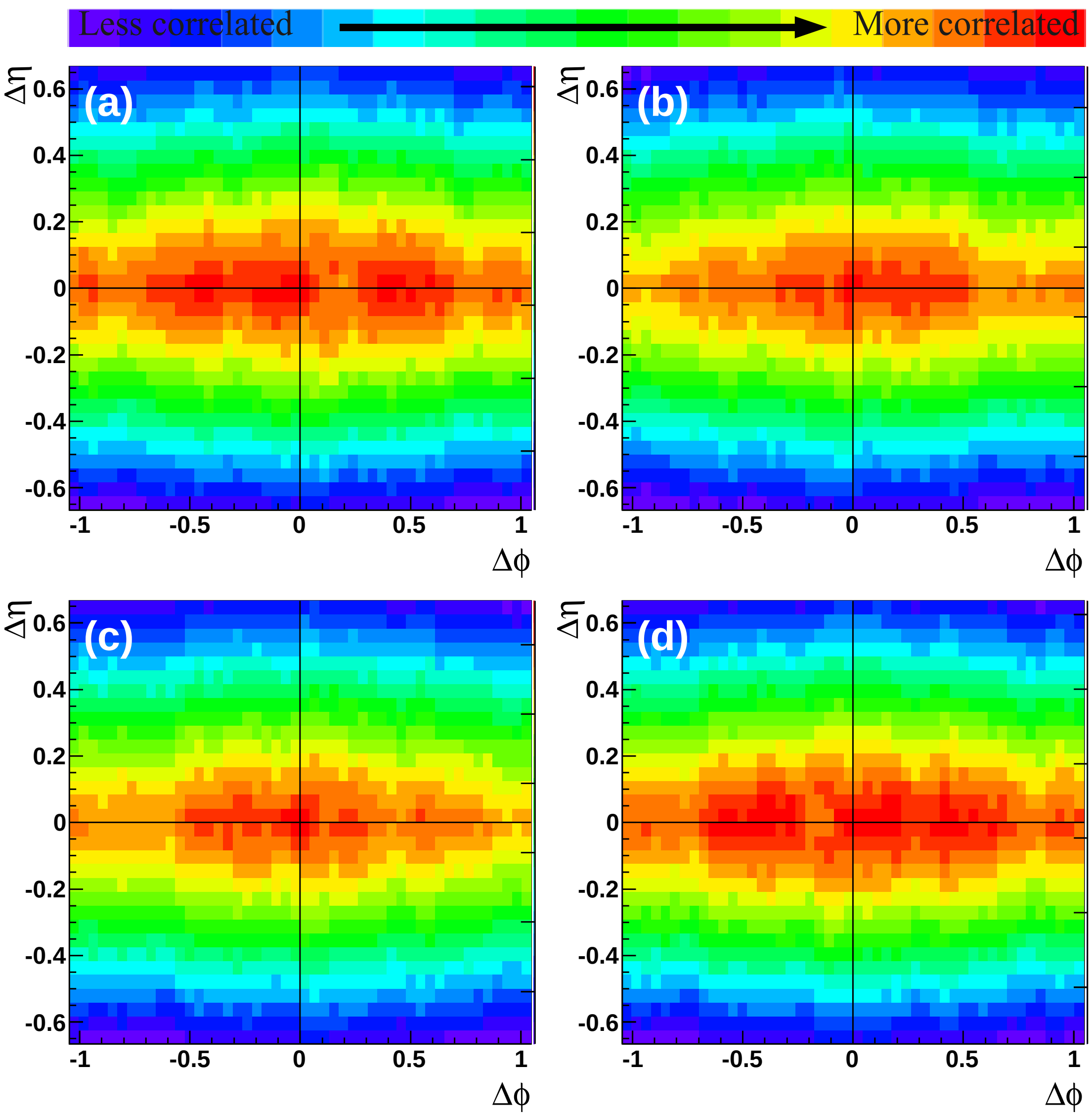

Figure 2.6: Raw Ξ-h correlations for (a) trigger particle helicity 1 and associated particle helicity 1 (b) trigger particle helicity 1 and associated particle helicity -1 (c) trigger particle helicity -1 and associated particle helicity 1 (d) trigger particle helicity -1 and associated particle helicity -1 in 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} =$ 200 GeV for $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c and $3.0 < p_T^{trigger} < 6.0$ GeV/c.

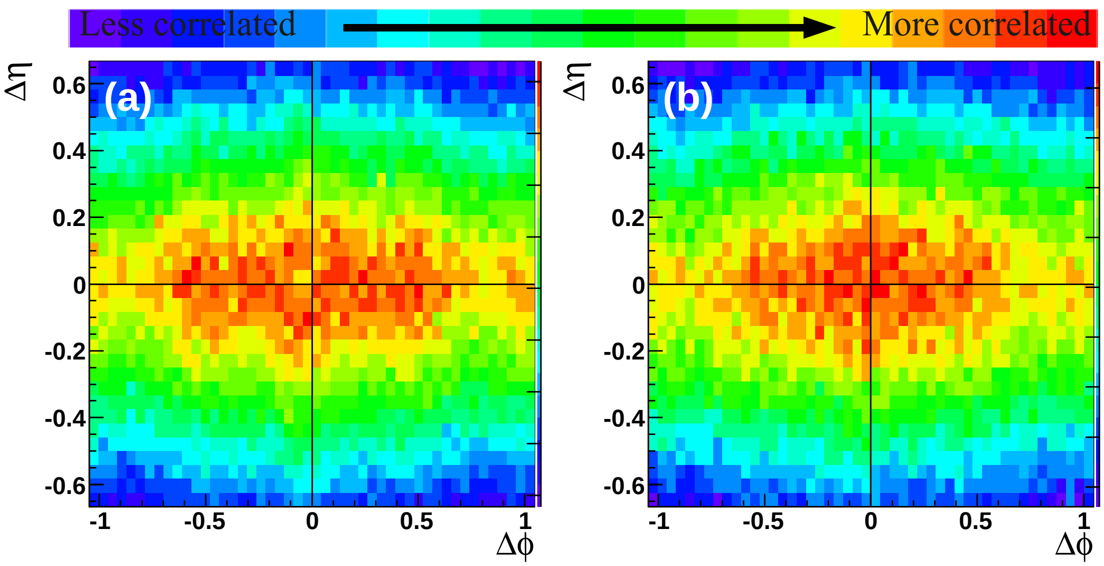

Figure 2.7: The location of the track merging dip in correlations with V^0 associated particles for a (a) Λ and $\bar{\Lambda}$ associated particles (b) (c) K_S^0 associated particles in 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

$\Delta\eta$ for $-1 < \Delta\phi < 1$ and subtracting a constant background to determine the jet-like correlation as a function of $\Delta\eta$. Elliptic flow, v_2, does not lead to a systematic error on the jet-like yield using either of these methods because v_2 is roughly independent of $\Delta\eta$ within the STAR acceptance [11, 12]. Previous studies in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV have found these methods to be consistent at high-p_T [13], with some indications that there may be greater deviations at lower p_T [14]. Both methods are described in greater detail below.

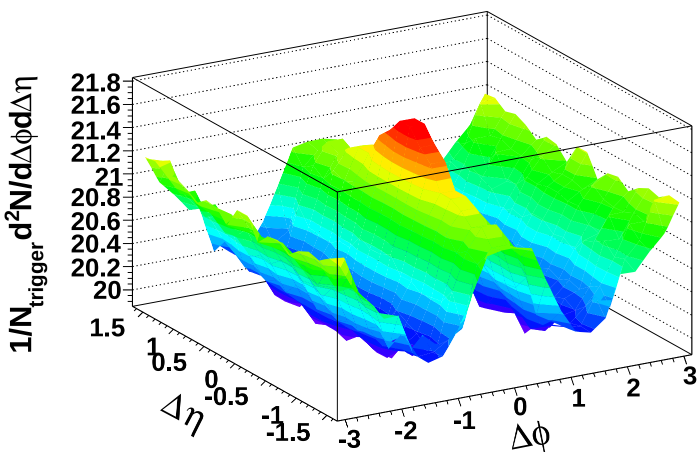

Figure 2.8: Sample correlation after acceptance corrections for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from 0-40% most central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

$\Delta\phi$ method

To determine the jet-like yield, Y_{Jet}, the projection of the distribution of particles $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ is taken in two different ranges in pseudorapidity:

$$\frac{dY_{ridge}}{d\Delta\phi} = \frac{1}{N_{trigger}} \int_{-1.75}^{-0.75} \frac{d^2N}{d\Delta\phi d\Delta\eta} d\Delta\eta + \frac{1}{N_{trigger}} \int_{0.75}^{1.75} \frac{d^2N}{d\Delta\phi d\Delta\eta} d\Delta\eta \quad (2.2)$$

$$\frac{dY_{jet+ridge}}{d\Delta\phi} = \frac{1}{N_{trigger}} \int_{-0.75}^{0.75} \frac{d^2N}{d\Delta\phi d\Delta\eta} d\Delta\eta \quad (2.3)$$

where the former contains only the ridge and the latter contains both the jet-like correlation and the ridge. The jet-like yield, Y_{Jet}, is determined using bin counting in the region $-1 < \Delta\phi < 1$:

$$Y_{jet}^{\Delta\phi} = \int_{-1}^{1} \left(\frac{dY_{jet+ridge}}{d\Delta\phi} - \frac{0.75}{1} \frac{dY_{ridge}}{d\Delta\phi}\right) d\Delta\phi. \quad (2.4)$$

The factor in front of the second term is the ratio of the $\Delta\eta$ width in the region containing the jet-like correlation and the ridge to the width of the region containing only the ridge. In practice because the data have been reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$, the projections in $\Delta\phi$ and $\Delta\eta$ and the bin counting are done over half of the range and then scaled up to ensure that the errors are correct.

$\Delta\eta$ method

In this method the projection of the distribution of particles $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ on the near-side is taken:

$$\frac{1}{N_{trigger}}\frac{dN}{d\Delta\eta} = \frac{1}{N_{trigger}}\int_{-1}^{1}\frac{d^2N}{d\Delta\phi d\Delta\eta}d\Delta\phi. \tag{2.5}$$

This contains both a background due to combinatorics and the ridge, both of which are independent of $\Delta\eta$. The background is determined first by fitting a Gaussian signal plus a constant background, A, to Equation 2.6 and then using the background from that for background subtraction. If statistics permitted, the background could be determined from $|\Delta\eta\ | > 0.75$, however, many of the correlations presented here have limited statistics and using a fit of the whole region to determine the background prevents statistical fluctuations from skewing the background subtraction. The jet-like yield is:

$$Y_{jet}^{\Delta\eta} = \frac{1}{N_{trigger}}\int_{-1}^{1}(\frac{d^2N}{d\Delta\phi d\Delta\eta} - A)d\Delta\phi. \tag{2.6}$$

The yield is determined by bin counting in the range $|\Delta\eta\ | < 1$. To get the proper errors, the projection and the bin counting are also done over half of the range and scaled because the data have been reflected. The statistical error on A from the fit is used to determine the error on the final yield by adding it in quadrature to the statistical error from bin counting. The magnitudes of these two errors are comparable.

Figure 2.9 shows the jet-like correlation from both the $\Delta\phi$ and the $\Delta\eta$ method after background subtraction for all systems and energies using $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$, the limits used for most studies presented here. For all collision systems and energies the jet-like correlation is within $|\Delta\eta\ | <$ 0.75.

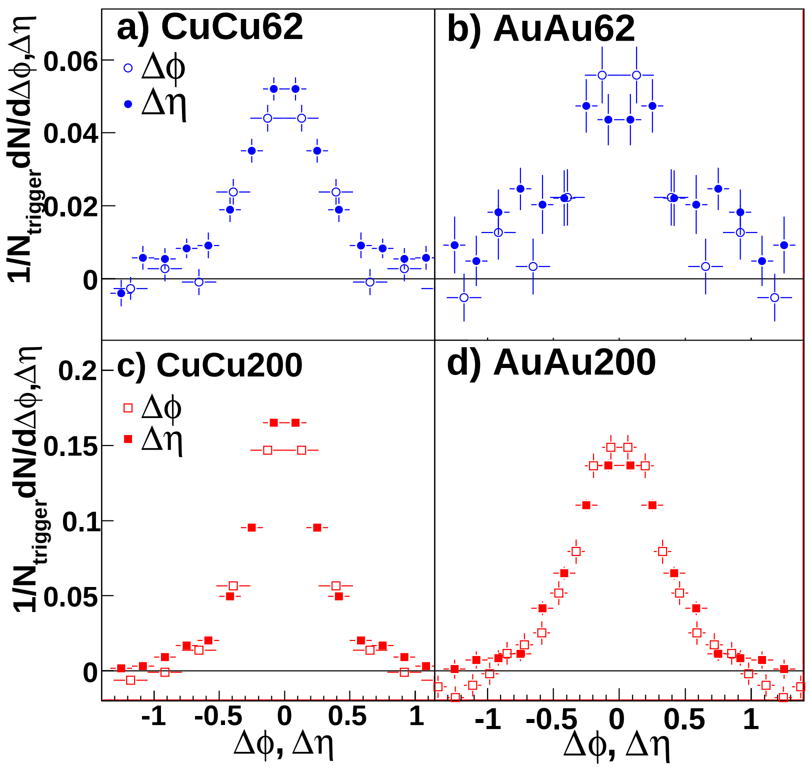

Figure 2.9: Sample jet-like correlations for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from both the $\Delta\phi$ and the $\Delta\eta$ method for (a) 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}}$ $= 62$ GeV (c) 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV and (d) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

2.2.2 Extraction of the ridge yield

There is a large combinatorial background which is dependent on $\Delta\phi$. The raw signal has a background due to particles correlated indirectly with each other in azimuth due to their correlation with the reaction plane. This random background is estimated by

$$\frac{dY_{bkgd}}{d\phi} = B(1 + 2\langle v_2^{trigger}\rangle\langle v_2^{associated}\rangle \cos(2\Delta\phi)) \tag{2.7}$$

where v_2 is the second order harmonic in a Fourier expansion of the momentum anisotropy relative to the reaction plane [15]. The derivation of this parameterization of the background is dependent on the assumption that di-jet production is not correlated with the reaction plane so that the correlation between particles in the jet and random particles in event are not correlated in azimuth through a mutual correlation with the reaction plane. Systematic errors come from the errors on B, $\langle v_2^{trigger} \rangle$ and $\langle v_2^{associated} \rangle$. v_2 is determined from independent measurements. It is assumed that v_2 is the same for events with a trigger particle as for minimum bias events and that v_2 is independent of $\Delta\eta$. For each data set $v_2(p_T)$ was fit in centrality bins to determine $\langle v_2^{trigger} \rangle$ and $\langle v_2^{associated} \rangle$.

Measurements of v_2

The best method for the determination of v_2 depends on the collision system. Since v_2 is a momentum anisotropy in azimuth, the tracks produced in an event can be used to reconstruct the reaction plane. v_2 in Equation 2.7 is the azimuthal momentum anisotropy due to the collective flow of particles, however, there are other sources of azimuthal momentum anisotropy. For particles above 1 GeV/c, jets are the largest known non-flow source of momentum anisotropy. Various methods for measuring v_2 have different non-flow contributions. The systematic error is determined by comparing these different methods. Generally, the higher the multiplicity of an event and the greater the eccentricity of the overlap region, the easier it is to reconstruct the event plane. Therefore that the event plane is more difficult to reconstruct in $Cu + Cu$ collisions.

For $Cu + Cu$ collisions, the nominal for v_2 was taken from a measurement of v_2 using tracks from the FTPCs for reconstruction of the event plane. v_2 measured in $Cu + Cu$ using the event plane from the FTPCs is shown as a function of p_T and centrality in Figure 2.10 [16]. These data were fit to a parameterization which was used for the subtraction of v_2 using Equation 2.7. The fit parameters are given in Table 2.1. The statistical errors on the fits are taken into account and combined with the systematic errors (5% of v_2 for 0-40% central collisions and 10% of v_2 for 40-60% central collisions.) For $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV the v_2 for 40-50% central collisions is used for 50-60% central collisions because the errors in the most peripheral bin were large enough that these data could not be fit and data from the

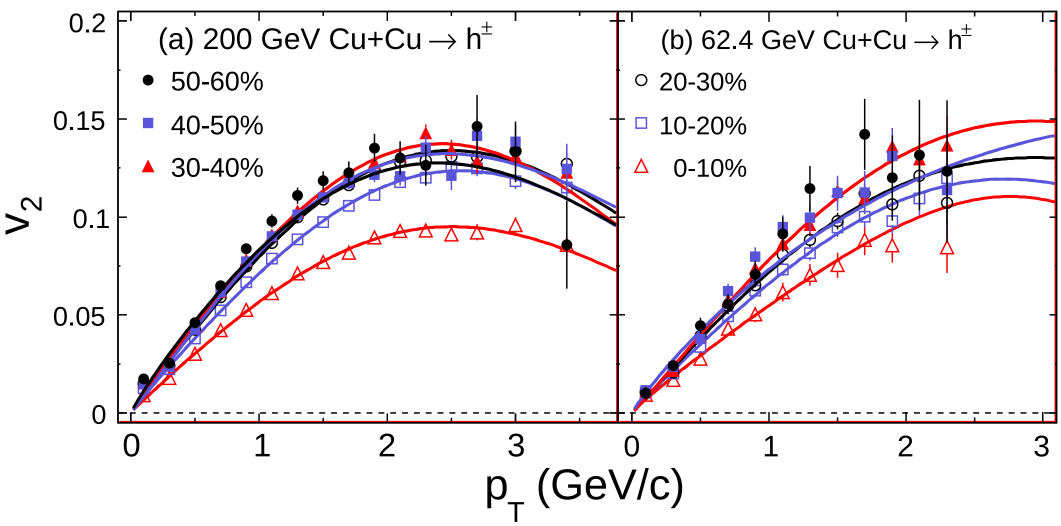

Figure 2.10: v_2 measured using the event plane from the FTPCs in $Cu+Cu$ collisions at (a) $\sqrt{s_{NN}}$ = 200 GeV and (b) $\sqrt{s_{NN}}$ = 62 GeV. Fit parameters are given in Table 2.1

two centralities were consistent. The event plane reconstructed from tracks measured in both of the FTPCs is expected to be less biased by jets since tracks in the FTPCs are separated by more than two units of pseudorapidity. Particles produced in di-jets in $p+p$ collisions do not exhibit long range correlations in pseudorapidity.

v_2 and the systematic errors on v_2 in $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV were taken from [17]. The dominant systematic errors were determined by comparing the reaction plane method to the four-particle cumulant method. The four-particle cumulant method determines v_2 by looking at correlations between four particles, which significantly reduces its sensitivity to non-flow. v_2 and the magnitude of the negative systematic error on v_2 in $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV as a function of p_T and centrality from [17] are shown in Figure 2.11. These were fit to a parameterization of v_2 and the fit parameters are given in Table 2.2. Statistical errors on the fits are also neglected for $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV since the systematic errors on v_2 are several times larger than the statistical errors. The positive systematic error

Table 2.1: Fit parameters from fits of v_2 measured using the event plane reconstructed using the FTPC in $Cu+Cu$ shown in Figure 2.10 to v_2 (p_T)=$ap_T^b e^{-(p_T/c)^d}$ $exp(\frac{b}{p_T}^c)$

centrality	a	b	c	d
200 GeV				
0-10%	0.0594 ± 0.0011	0.9541 ± 0.0192	3.6778 ± 0.0936	2.3449 ± 0.2438
10-20%	0.0735 ± 0.0008	0.9368 ± 0.0127	3.8002 ± 0.0779	2.5779 ± 0.2030
20-30%	0.0836 ± 0.0011	0.9599 ± 0.0141	3.6625 ± 0.0710	2.3440 ± 0.1711
30-40%	0.0863 ± 0.0013	0.9422 ± 0.0163	3.5843 ± 0.0844	2.5421 ± 0.2262
40-50%	0.0910 ± 0.0038	1.0349 ± 0.0364	3.3741 ± 0.1583	1.8515 ± 0.2546
50-60%	0.0888 ± 0.0022	0.8928 ± 0.0247	3.6000 ± 0.0000	2.1461 ± 0.2273
62 GeV				
0-10%	0.0550 ± 0.0029	0.9076 ± 0.0653	4.0000 ± 0.0000	4.0291 ± 2.7322
10-20%	0.0684 ± 0.0033	0.9539 ± 0.0514	4.0000 ± 0.0000	2.3958 ± 0.6978
20-30%	0.0787 ± 0.0051	1.0126 ± 0.0500	4.0000 ± 0.0000	1.7281 ± 0.4684
30-40%	0.0843 ± 0.0067	1.0519 ± 0.0666	4.0000 ± 0.0000	1.8512 ± 0.6561
40-60%	0.0825 ± 0.0055	0.9039 ± 0.0484	5.6000 ± 0.0000	1.2375 ± 0.2896

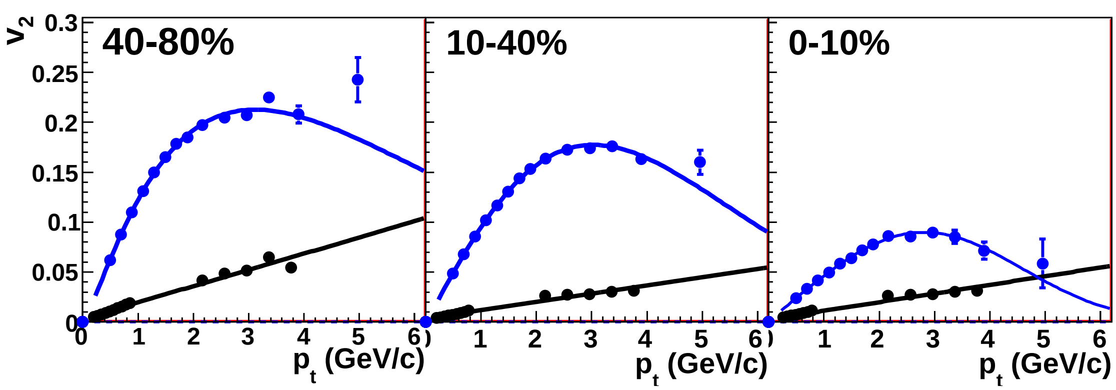

Figure 2.11: v_2 (blue) and the negative systematic error on v_2 (black) in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV. Fit parameters are given in Table 2.2.

in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV is a constant, 0.003, for all centralities and p_T.

v_2 and the systematic errors on v_2 in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV were taken from *http://www.star.bnl.gov/protected/jetcorr/mvl/v2_compare/auau_compare.html*, where the new parametrization which was agreed upon in the jetcorr PWG in 2009

Table 2.2: Fit parameters from fits of v_2 from [17] to v_2 (p_T)=$ap_T^b e^{-(p_T/c)^d}$ $exp(\frac{b}{p_T}^c)$ and of the negative systematic error on v_2 to $mp_T + b$

	v_2				systematic error	
centrality	a	b	c	d	m	b
0-10%	0.205	1.30	2.08	0.90	0.0163	0.0035
10-40%	0.105	1.04	4.03	1.67	0.0082	0.0038
40-80%	0.047	0.96	4.11	2.72	0.0087	0.0026

is described. The v_2 parametrization as a function of p_T and centrality is shown in Figure 2.12. The upper limit is given by the fit to the measurement of $v_2\{FTPC\}$ and the lower limit is the fit to the four-particle cumulant $v_2\{4\}$. For the most peripheral and central bin, the lower limit is determined from an extrapolation. The functional form of the parametrization is $v_2(p_T, cent) = A \cdot p_T^n \cdot e^{B \cdot p_T}$. The values of the parameter B is -0.445 (upper limit) and -0.5 (lower limit). The parameter n depends on centrality as n=1.48-0.5·cent.

v_2 for Λ, Ξ^-, and Ω^- baryons and K_S^0 mesons has been observed to follow quark scaling in $Au+Au$ collisions at both $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV [17] and in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV [18]. The systematic errors on identified particle v_2 in $Cu+Cu$ collisions are more difficult due to low statistics. Quark scaling is assumed and the systematic error is estimated from unidentified hadron v_2. For a hadron comprising n_q quarks:

$$v_2^{n_q} = \frac{n_q}{n_\pi} v_2(\frac{n_\pi}{n_q} p_T) \tag{2.8}$$

where $n_\pi = 2$. It is assumed that unidentified hadron v_2 is dominated by π v_2.

B is determined using the Zero Yield At Minimum (ZYAM) method [19, 12]. This method fixes B by assuming that there is no signal at the minimum of the correlation in azimuth. For measurements with low statistics the statistical error on the minimum data point is a significant contribution to the error on the ridge yield. This error is added in quadrature to the statistical error.

To determine the yield of the ridge, Y_{Ridge}, the di-hadron correlation $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ is projected over the entire $\Delta\eta$ region. To minimize the effects of statistical fluctuations

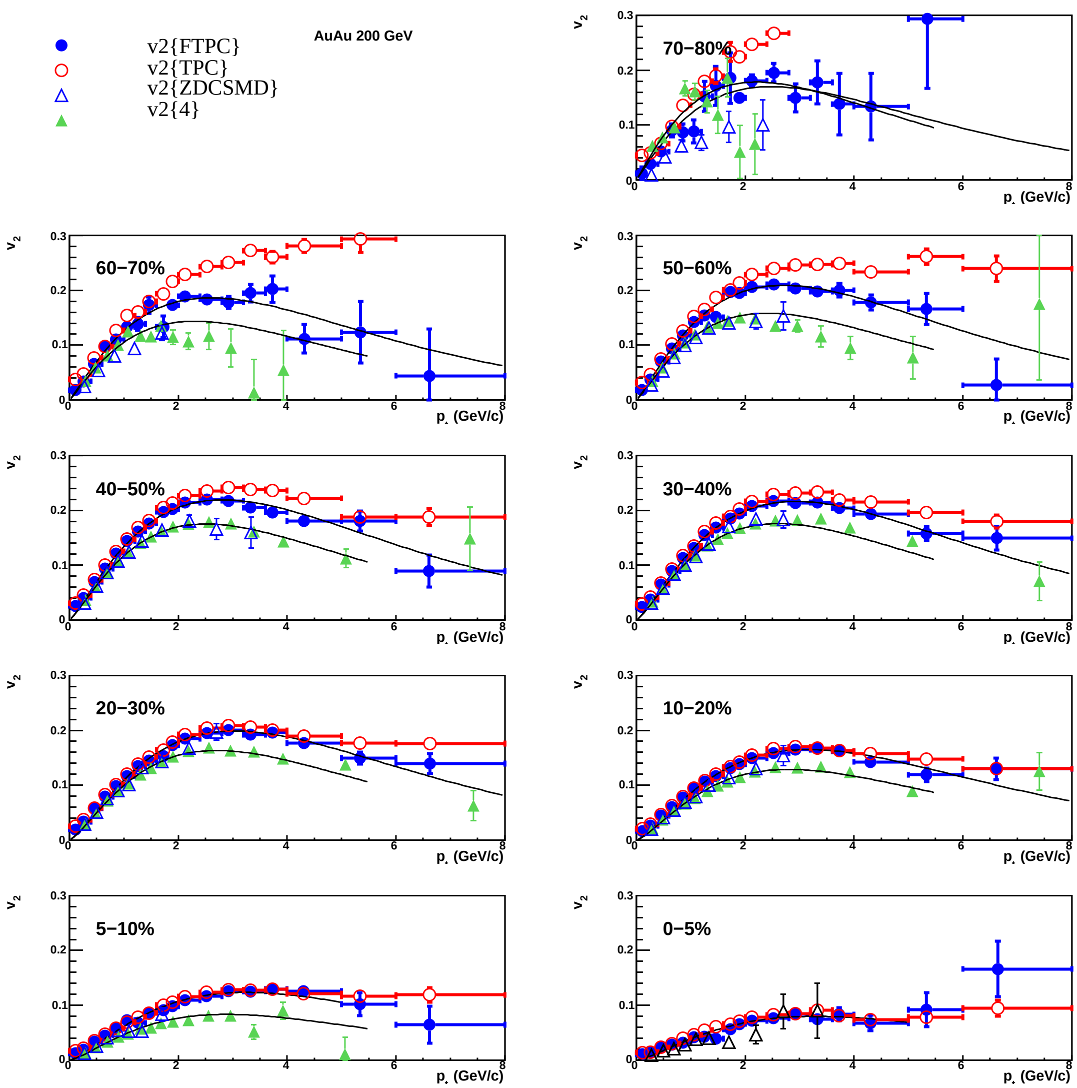

Figure 2.12: Symbols represent v_2 in $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV determined by various methods as described in the legend. The black lines are the upper and lower systematic boundaries on v_2 using the fit function $v_2(p_T, cent) = A{\cdot}p_T^n{\cdot}e^{B\cdot p_T}$. The values of the parameters are B=-0.445 (upper curve), B=-0.5 (lower curve), and n=1.48-0.5·cent.

in the determination of the background, $\frac{dY_{bkgd}}{d\phi}$, Y_{Jet} is subtracted after the projection

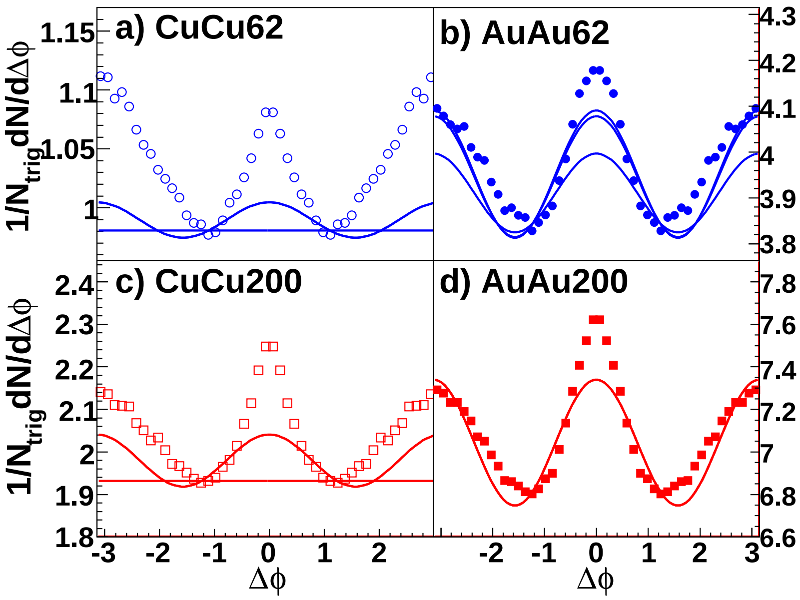

Figure 2.13: Sample correlations in $\Delta\phi$ integrated over $|\Delta\eta\ | < 1.75$ for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from (a) 0-60% central $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV (c) 0-60% central $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV and (d) 0-80% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV showing the background level and shape for high and low bounds on v_2 from each data set.

in $\Delta\eta$:

$$Y_{Ridge} = 1/N_{trigger} \int_{-1.75}^{1.75} \int_{-1}^{1} \left(\frac{d^2N}{d\Delta\phi d\Delta\eta} - \frac{dY_{bkgd}}{d\phi}\right) d\Delta\phi d\Delta\eta - Y_{Jet}^{\Delta\eta}. \tag{2.9}$$

The integration over $\Delta\phi$ is done by bin counting. $Y_{Jet}^{\Delta\eta}$ is used for the subtraction of Y_{Jet} because it is less sensitive to the assumption that the jet-like correlation is contained within $|\Delta\eta\ | < 0.75$. Sample correlations in azimuth after the projection over $\Delta\eta$ are shown in Figure 2.13 for all data sets. These correlations show the background level and shape for the high and low bounds on v_2.

Chapter 3

Results

3.1 The jet-like correlation

If the jet-like correlation is produced by fragmentation, no dependence on collision system or system size would be expected. The particle ratios of the jet-like correlation should be consistent with what was observed in $p+p$ collisions since $p+p$ collisions are expected to be dominated by jet production at high-p_T. The jet-like yield should increase with $p_T^{trigger}$ because higher p_T trigger particles come on average from higher energy jets and the spectra of particles in the jet-like correlation should be harder than that observed in the inclusive. Studies of the jet-like yield as a function of collision system and energy were done for unidentified hadrons for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV to determine whether the data are consistent with the jet-like correlation being produced by fragmentation.

3.1.1 Flavor and system dependence

Differences between baryons and mesons have been observed in R_{AA} and in v_2 and models such as Recombination hypothesizes novel methods for production of baryons and mesons in $A+A$ collisions to explain the observed differences. If particles in jets are produced by vacuum fragmentation, baryon/meson ratios would be expected to be comparable to $p+p$. Jets with leading baryons may have different properties than jets with leading mesons. Particles produced by mechanisms such as Recombination would have baryon/meson ratios comparable to the bulk. No statistically significant differences were observed between Λ and $\bar{\Lambda}$ triggers or Ξ^- and Ξ^+ so these trigger particles were added together to increase the statistical significance of the results for comparisons to K_S^0 trigger particles. The $\Delta\phi$ was used because these measurements were statistically limited and the jet-like correlation is broader for identified particles than for unidentified particles.

Identified trigger particles

Figure 3.1 shows the $p_T^{trigger}$ dependence of Y_{Jet} for identified trigger particles in $d+Au$, $Cu+Cu$, and $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for unidentified hadron, Λ, K_S^0, and Ξ trigger particles. The $Au+Au$ data have not been corrected for the track merging effects discussed in Chapter 2, so the slightly lower yield for identified particles in $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV may be due to track merging. There is no system dependence within errors. The most central K_S^0 trigger in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV is about three sigma above the other data; this data point may be an outlier or may be an indication of a slight particle type dependence. There is no other observed dependence on the trigger particle type.

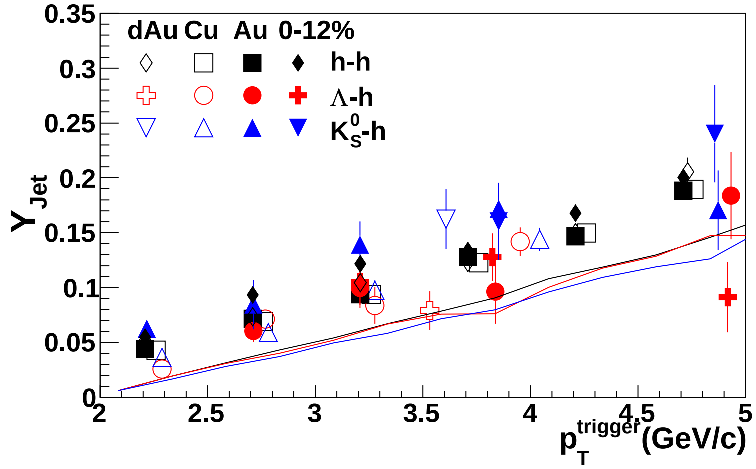

Figure 3.1: $p_T^{trigger}$ dependence of Y_{Jet} for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ in 0-60% central $Cu + Cu$ and 0-12% central $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV for unidentified hadrons, Λ, K^0_S, and Ξ trigger particles. The identified trigger particle data have been horizontally offset for visibility. The $Au + Au$ data have not been corrected for the track merging effects discussed in Chapter 2.

The $p_T^{associated}$ dependence of Y_{Jet} for unidentified hadrons, Λ, K^0_S, and Ξ trigger particles is shown in Figure 3.2. Due to limited statistics, it was only possible to extract Y_{Jet} for two bins in $p_T^{associated}$ for Λ and K^0_S trigger particles and one bin bor Ξ trigger particles. The jet-like yields from identified trigger particles are within errors of the jet-like yields for unidentified hadron trigger particles. The data in $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV also showed no dependence on trigger particle type [8].

The N_{part} dependence of Y_{Jet} is shown in Figure 3.3 for Λ, K^0_S, and Ξ trigger particles. The $Au + Au$ data have not been corrected for track merging effects, and this may explain the difference between identified trigger particles and unidentified trigger particles.

No statistically significant differences between different trigger particles are observed for the jet-like correlations, including non-strange, singly strange, and doubly strange hadrons. Studies in $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV indicated that there is no trigger particle dependence for even the trebly strange Ω [20, 21].

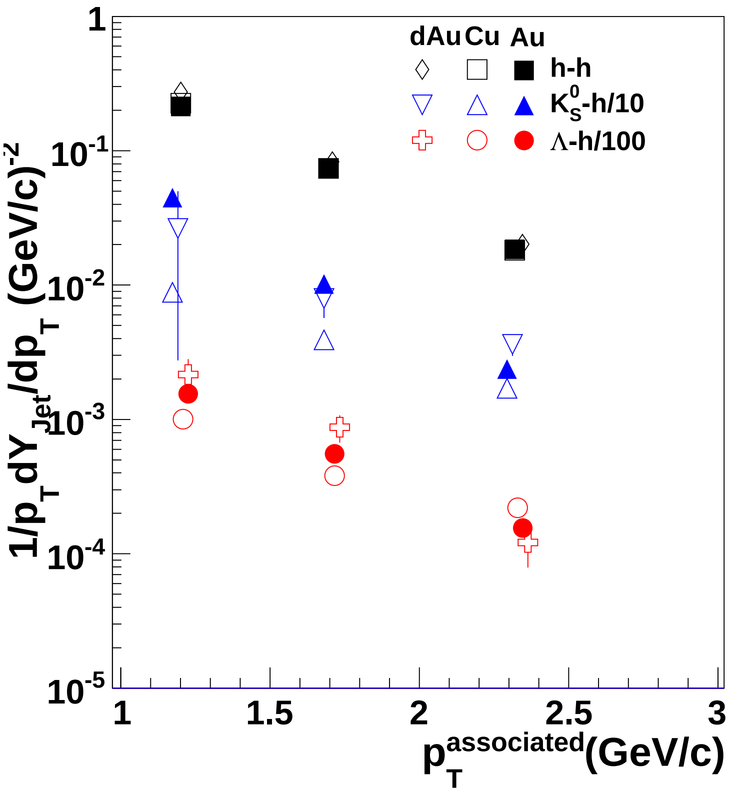

Figure 3.2: $p_T^{associated}$ dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c in 0-60% central $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV for unidentified hadrons, Λ, K_S^0, and Ξ trigger particles. The line is from a fit of the h-h data to A$e^{-p_T/k}$. The identified trigger particle data have been horizontally offset for visibility.

Table 3.1: Inverse slope parameter k (MeV/c) of $p_T^{associated}$ for fits of data from the jet-like correlation in Figure 3.5 to $Ae^{-p_T/k}$. Statistical errors only.

	h-h	h-K_S^0	h-$\Lambda+\bar{\Lambda}$
k (MeV/c)	445 ± 20	505 ± 87	510 ± 55

Identified associated particles

Figure 3.4 shows the $p_T^{trigger}$ dependence of Y_{Jet} for identified associated particles. Similar trends are observed for unidentified associated particles and strange associated particles.

Figure 3.5 shows the dependence of Y_{Jet} on $p_T^{associated}$ for identified associated particles. The ratio $\frac{\Lambda+\bar{\Lambda}}{2K_S^0}$ is roughly 0.5 for the full $p_T^{associated}$ range shown.

3.2 The ridge

The ridge is an unexpected phenomenon first observed in $Au+Au$ collisions. In order to understand its origins and to provide a robust test of models for the production of the ridge, the ridge is studied as a function of as many variables as possible. The

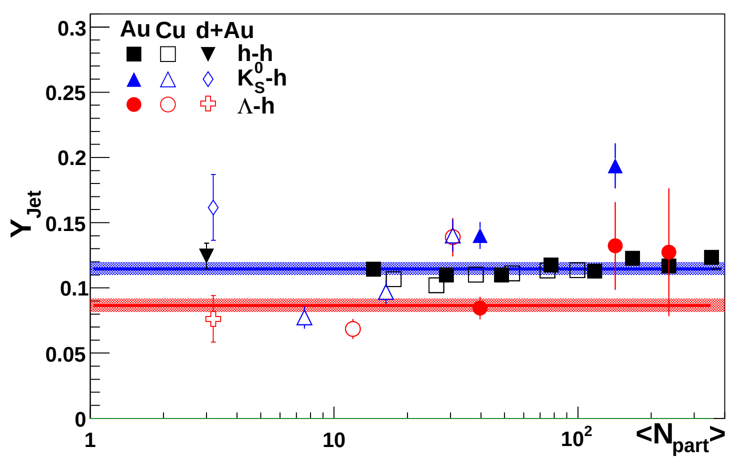

Figure 3.3: N_{part} dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for identified trigger particles in $d+Au$, $Cu+Cu$, and $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for Λ, K_S^0, and Ξ trigger particles.

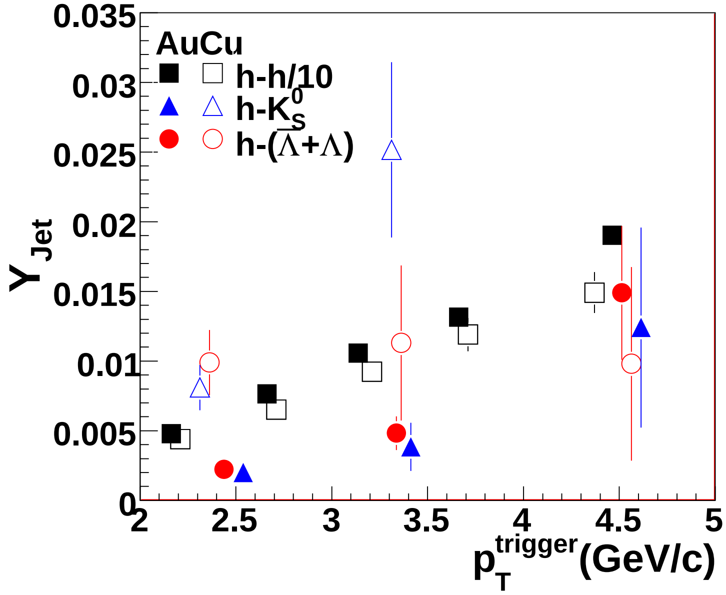

Figure 3.4: $p_T^{trigger}$ dependence of Y_{Jet} for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ of Y_{Jet} in 0-60% central $Cu+Cu$ and 0-12% central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV for identified associated particles. The unidentified hadron associated particles are scaled to fit on the same plot. Data from $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV are from [8].

collision energy, collision system, system size, and trigger particle type dependence of the ridge are presented here.

3.2.1 Flavor dependence

Figure 3.6 shows the dependence of Y_{Ridge} on N_{part} for identified Λ, K_S^0, and Ξ trigger particles for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. The data for identified trigger particles in $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV are consistent, within large errors, of Y_{Ridge} in $Au+Au$ at the same N_{part}.

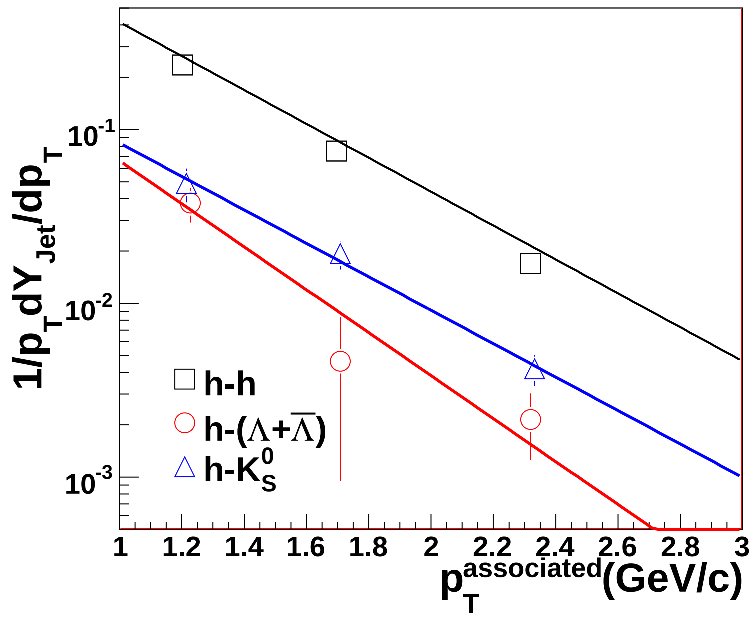

Figure 3.5: $p_T^{associated}$ dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c for $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV. Parameters from fits of the data are listed in Table 3.1.

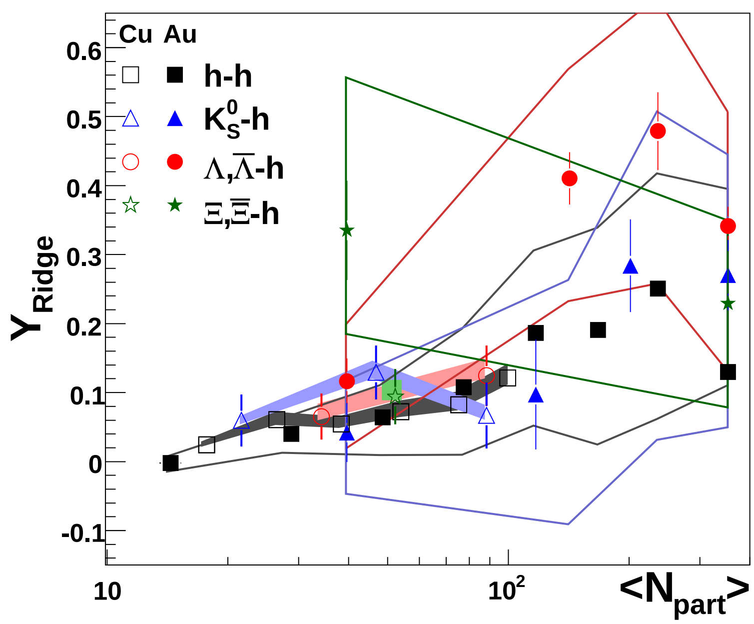

Figure 3.6: N_{part} dependence of Y_{Ridge} for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for identified trigger particles in $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Systematic errors are shown as bands in the same color as the data points.

Appendix A

Code documentation

Code for this analysis is stored in $CVSROOT/offline/paper/psn0530/ . The basic code structure is outlined in Figure A.1, Figure A.2, Figure A.3, and Figure A.4. There are six different libraries used. StHighPtTree, StHighPtLoop, StHighPtEvaluator, and StHighPtTools libraries contain classes and functions which can be used by multiple different analyses and StAnalysis and StAnalysisTools libraries contain classes and functions which are used only for the di-hadron correlation analysis.

Numerical codes were used for particle identification. The codes used in this analysis are defined in Table A.1. In addition, sometimes classes of particles were studied separately to improve CPU efficiency. These classes of particles were given numerical codes which are defined in Table A.2.

A.1 StHighPtTree, StHighPtLoop, StHighPtEvaluator, and StHighPtTools libraries

The StHighPtTree, StHighPtLoop, StHighPtEvaluator, and StHighPtTools libraries are diagramed in Figure A.1 and Figure A.2. Each function or class is briefly defined. StHighPtTree comprises container classes used in TTrees for the analysis. StHighPtLoop contains StMakers which can be used to loop over TTrees or MuDsts. The

Table A.1: Codes used for particle identification. Additional numbers were reserved for other particles these particles were later removed from the analysis.

Number	particle
0	h
1	Λ
2	$\bar{\Lambda}$
3	K_S^0
4	Ξ^-
5	Ω^-
7	π^+
8	π^-
9	p
10	$\bar{p}$
11	K^+
12	K^-
13	e^+
14	e^-
18	Ξ^+
18	Ω^+

Table A.2: Codes used for classes of trigger and associated particles.

Number	particles
0	h
1	Λ, $\bar{\Lambda}$, and K_S^0
2	Ξ^-, Ξ^+, Ω^-, and Ω^+
3	dE/dx identified particles ($\pi^\pm$, p, $\bar{p}$, $K^\pm$, $e^\pm$)

primary class in this library is AnalysisMaker, which is a utility class for other analyses. It manages all track and event cuts and passes only events and tracks which pass quality cuts the next maker. TTreeMaker uses AnalysisMaker to create a TTree from a MuDst; this TTree can also be read by AnalysisMaker. StHighPtEvaluator contains classes which evaluate the quality of events and tracks. StHighPtTools contains tools which are used by multiple other functions.

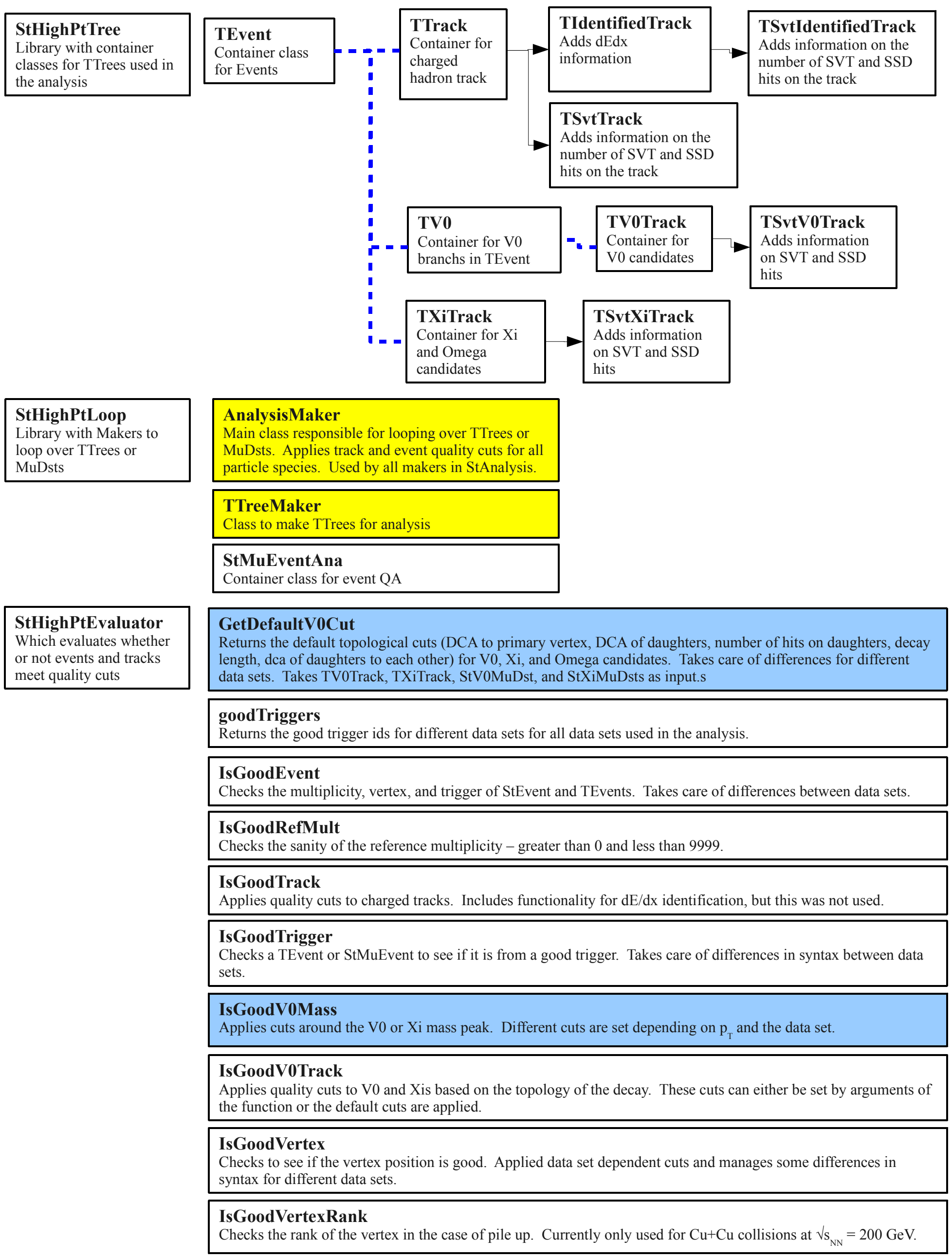

Figure A.1: First part of the code diagram for the StHighPt libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

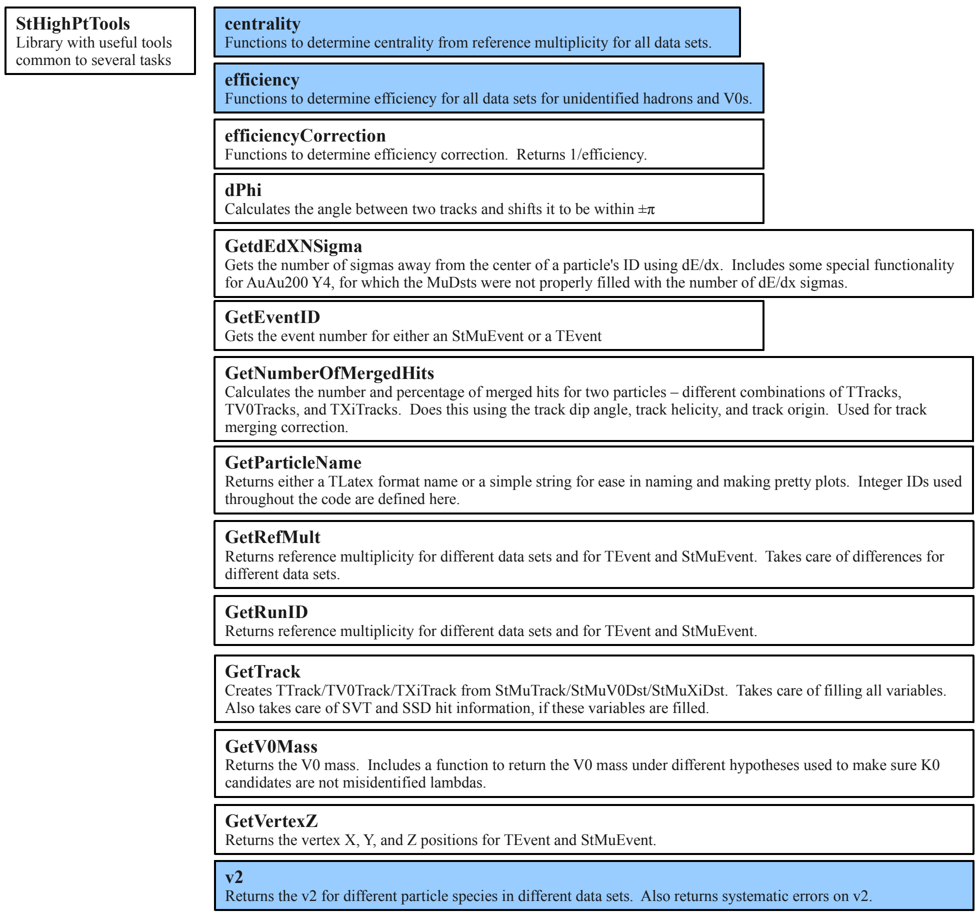

Figure A.2: Second part of the code diagram for the StHighPt libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

A.2 StAnalysis and StAnalysisTools libraries

The StAnalysis and StAnalysisTools libraries are diagramed in Figure A.3 and Figure A.4. StAnalysis contains makers. CorrelationMaker is the maker used for the di-hadron correlation analysis. PhiMaker, RawSpectraMaker, and QAMaker are makers for quality assurance at high-p_T and are provided for convenience. V0CutTuner, XiCutTuner, OmegaCutTuner, and XiNtupleMaker are used for tuning the geometric cuts of V^0s, Ξ^-, and Ω^-. StAnalysisTools contains functions which are used by the makers in StAnalysis. The primary class in this library is CorrelationSummary, which is used to contain all of the relevant information about a di-hadron correlation.

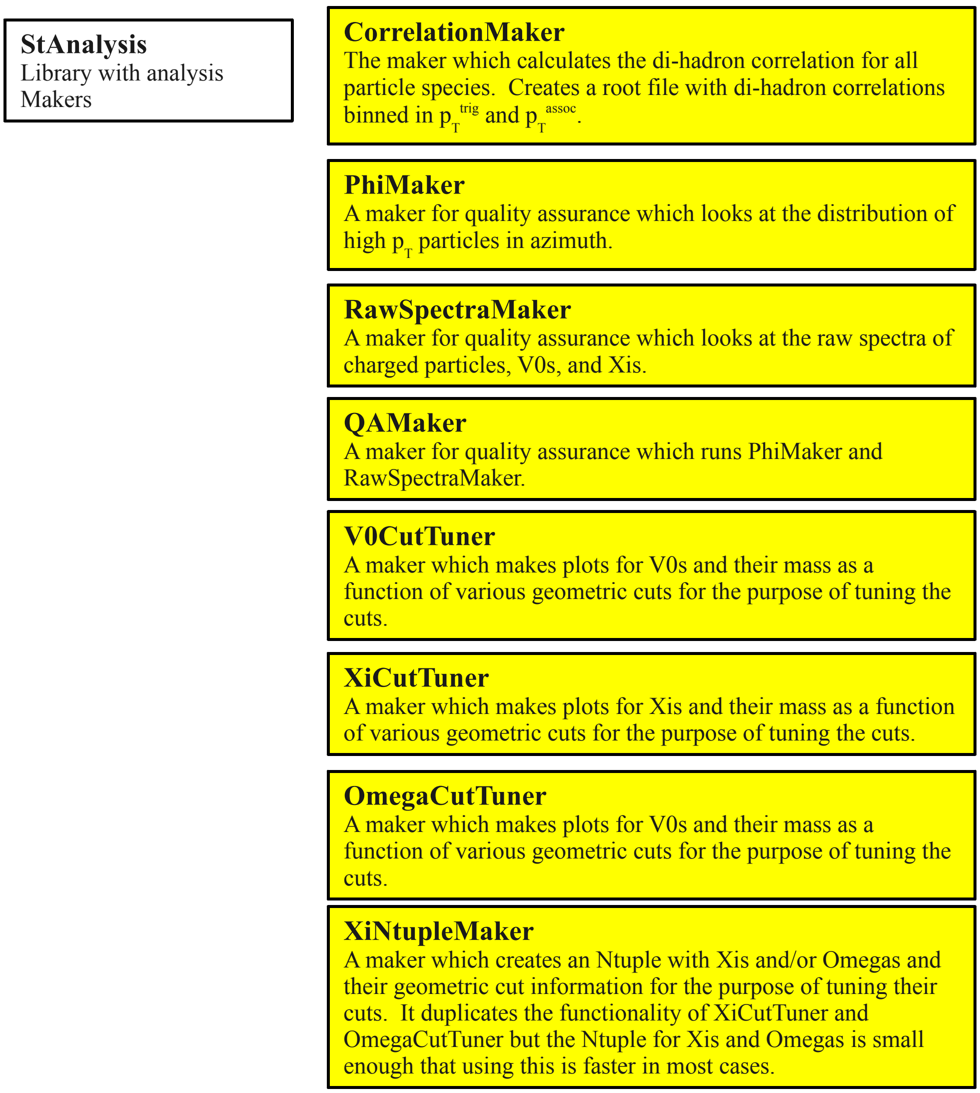

Figure A.3: First part of the code diagram for the StAnalysis libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

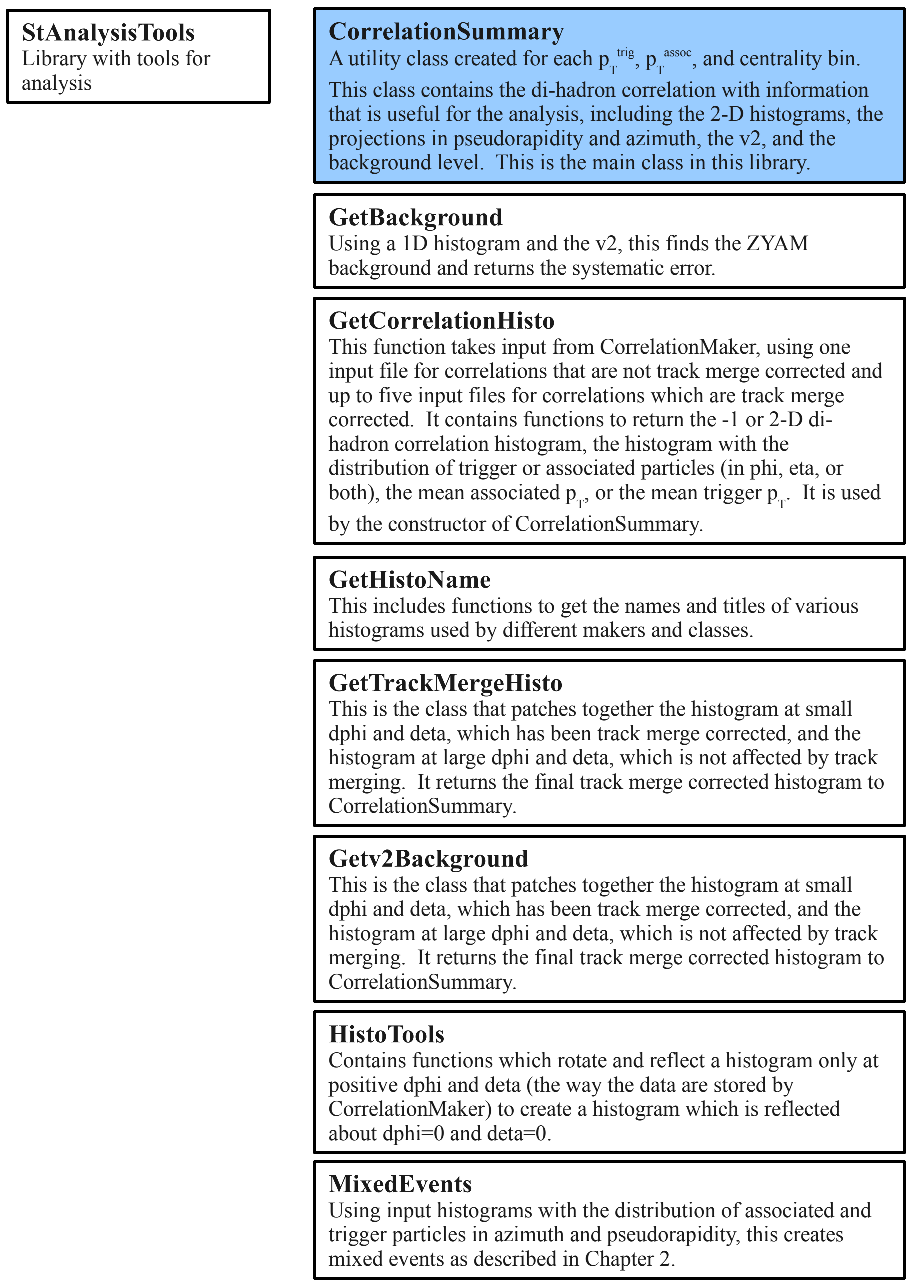

Figure A.4: Second part of the code diagram for the StAnalysis libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

A.3 Sample macros

Sample macros for each maker and for a selection of functions are given in the macros directory. A brief description of how to use each macro is given. As described in the next section, these macros are not all needed to do the analysis, however, the additional macros are provided for future use.

1. **macros/test/testtools.C** Takes no arguments. Tests efficiency, efficiencyCorrection, centrality, and v2 and prints the result to the screen.

2. **macros/test/TestIsGoodV0Mass.C** Takes no arguments. Tests some of the functionality of IsGoodV0Mass and prints the result to the screen.

3. **macros/test/TestGetDefaultV0Cut.C,**
   macros/test/TestGetDefaultXiCut.C Take no arguments. Print the default cuts out to the screen.

4. **macros/test/TestCorrelationSummary.C**
   Example:
   root4star -b -q macros/test/TestCorrelationSummary.C '("AuAu200","tmp/",1.5,6.0,3.0,6.0)'
   This macro takes output files from CorrelationMaker and makes a CorrelationSummary. The arguments are the data set, the output directory for pictures, the minimum associated p_T, the maximum associated p_T, the minimum trigger p_T, and the maximum trigger p_T.

5. **macros/test/TestCorrelationSummaryTrackMergeCorrected.C**
   Example:
   root4star -b -q macros/test/TestCorrelationSummaryTrackmergeCorrected.C '("AuAu200","tmp/",1.5,6.0,3.0,6.0)'
   This macro takes output files from CorrelationMaker and makes a CorrelationSummary with the track merging correction applied. The arguments are the data set, the output directory for pictures, the minimum associated p_T, the maximum associated p_T, the minimum trigger p_T, and the maximum trigger

p_T. Note that the 1D histograms get filled multiple times when running in the track merge correction mode and therefore have an artificial jump in them. The 1D histogram is not corrected for track merging.

6. **macros/MakeCuCu200P06ibTrees.C**

   Example:

   root4star -b -q macros/MakeCuCu200P06ibTrees.C'("inputMuDst.root","output.root")'

   Creates a TTree from a MuDst using TTreeMaker and AnalysisMaker. This is written to run over 200 GeV Cu+Cu collisions.

7. **macros/QA/RunQAMaker.C**

   Example:

   root4star -b -q macros/QA/RunQAMaker.C '("inputMuDst.root","output.phi.root","output.pt.root")'

   This macro runs QAMaker, which in turn runs PhiMaker and RawSpectraMaker. In the background, it uses AnalysisMaker to filter tracks and events. The output is two root files for quality assurance of high-p_T particles.

8. **macros/geoCuts/RunOmegaCutTuner.C**

   Example:

   root4star -b -q macros/geoCuts/RunOmegaCutTuner.C '("inputTTree.root","output.root")'

   Runs OmegaCutTuner with an input TTree and creates an output file with histograms of omega mass peaks.

9. **macros/geoCuts/RunV0CutTuner.C**

   Example:

   root4star -b -q macros/geoCuts/RunV0CutTuner.C'("inputTTree.root","output.root")'

   Runs V0CutTuner with an input TTree and creates an output file with histograms of V^0 mass peaks.

10. **macros/geoCuts/RunXiCutTuner.C**

    Example:

    root4star -b -q macros/geoCuts/RunXiCutTuner.C'("inputTTree.root","output.root")'

Runs XiCutTuner with an input TTree and creates an output file with histograms of Ξ^- mass peaks.

11. **macros/geoCuts/RunXiNtupleMaker.C**
    Example:
    root4star -b -q macros/geoCuts/RunXiNtupleMaker.C'("inputTTree.root","output.root")'
    Runs XiNtupleMaker with an input TTree and creates an output file with an Ntuple with information about the geometric cuts for Ξ^- and/or Ω^-.

12. **macros/correlations/RunCorrelationMaker.C**
    Example:
    root4star -b -q macros/correlations/RunCorrelationMaker.C '("inputTTree.root","output.root",0,0)'
    This runs CorrelationMaker using a TTree as input and creating a file with di-hadron correlation histograms. The last two arguments are particle classes defined in Table A.2.

13. **macros/correlations/RunCorrelationMakerForTrackMerging.C**
    Example:
    root4star -b -q macros/correlations/RunCorrelationMakerForTrackMerging.C '("inputTTree.root","output.root",0,0,1,1)'
    This runs CorrelationMaker using a TTree as input and creates a file with di-hadron correlation histograms but only for small $\Delta\eta$ and $\Delta\phi$ for one combination of trigger and associated helicities at a time. It also eliminates track pairs where the two tracks share more than 10% of their hits, as described in Chapter 2. The third and fourth arguments are particle classes defined in Table A.2. The fifth and sixth arguments are the helicity of the trigger and associated particles, respectively. Note that it is possible to request an impossible helicity combination and then the histograms will be empty. In this example, unidentified trigger and associated particles are requested, each with a positive helicity.

14. **macros/correlations/RunCorrelationMakerForEventMixingCuCu200.C**

Example:
root4star -b -q macros/correlations/RunCorrelationMakerForEventMixingCuCu200.C '("inputTTree.root","output.root",0,0,1,1)'
This runs CorrelationMaker using a TTree as input and creates a file with di-hadron correlation histograms from mixed events for only for small $\Delta\eta$ and $\Delta\phi$ for only one combination of trigger and associated helicities. It also eliminates track pairs where the two tracks share more than 10% of their hits, as described in Chapter 2. The arguments are the same as for RunCorrelationMakerForTrackMerging.C. Note that the 1D histograms get filled multiple times when running in the mixed event mode and therefore have an artificial jump in them. The 1D histogram is not corrected for track merging.

15. **macros/correlations/RunCorrelationMakerForEventMixingCuCu200InPieces.C**
Example:
root4star -b -q macros/correlations/ RunCorrelationMakerForEventMixingCuCu200InPieces.C '(0, "inputTTree.root","output.root",0,0,1,1)'
This does the same thing as RunCorrelationMakerForEventMixingCuCu200.C, however, it will run over only a subset of the input TTree. This is necessary for event mixing to reduce the duration of jobs. This macro is written to run over bunches of 10,000 events. Smaller subsets can be chosen by editing the macro. The arguments are in the same order as RunCorrelationMakerForTrackMerging.C but there is an additional first argument which is an integer defining which part of the TTree to run over. The argument 0 runs over the first 10,000 events.

A.4 Analysis procedure

1. Create TTrees from MuDsts using MakeCuCu200P06ibTrees.C

2. Create output files with di-hadron correlations using RunCorrelationMaker.C. At this point correlations which are not corrected for track merging are available and can be studied with TestCorrelationSummary.C.

3. Create output files with di-hadron correlations at small $\Delta\phi$ and small $\Delta\eta$ using RunCorrelationMakerForTrackMerging.C for each trigger and associated helicity combination. Create corresponding mixed event files using RunCorrelationMakerForEventMixingCuCu200.C or RunCorrelationMakerForEventMixingCuCu200InPieces.C for each trigger and associated helicity combination. For h-h, Ξ^--h, and Ω^--h correlations this corresponds to an additional eight files, since each trigger and each associated particle can have a helicity of either +1 or -1. For V^0-h and h-V^0 correlations there are only four files created at this stage since the V^0 can only have a helicity of 0.

4. Di-hadron correlations can be corrected for track merging are created by using the output file created in the second step and the four or eight input files from the third step to create a track merging corrected di-hadron correlation using TestCorrelationSummaryTrackMergeCorrected.C. CorrelationSummaries can be written to a root file and used later or re-fit.

Appendix B

Geometric cuts

B.1 V^0 Geometric Cuts

Table B.1: Geometric cuts for Λ in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Cuts for $\bar{\Lambda}$ the same with cuts for the p daughter of the Λ applied to the $\bar{p}$ daughter of the $\bar{\Lambda}$ and cuts for the π^- daughter of the Λ applied to the π^+ daughter of the $\bar{\Lambda}$. These are in addition to cuts listed in Table 1.2. Primary vertex is abbreviated to PV.

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.4 GeV/c < p_T < 0.6 GeV/c						
Decay length (cm) >	7.5	7.5	7.5	7.5	7.5	7.5
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	2.3	2.3	2.3	1.5	1.5	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
$\mid n_\Gamma$ from mass peak center $\mid$ ¡	1.0	1.0	1.0	1.0	1.0	1.0
0.6 GeV/c < p_T < 0.8 GeV/c						
Decay length (cm) >	5.0	5.0	5.0	5.0	0.0	0.0
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8

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Table B.1 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	2.2	2.2	2.2	1.5	1.5	1.5
$n_{\sigma,dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
| n_Γ from mass peak center | ¡	2.0	2.0	2.0	2.0	2.0	2.0
0.8 GeV/c < p_T < 1.0 GeV/c						
Decay length (cm) >	6.5	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.4	1.5	1.5	1.1	1.1	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
| n_Γ from mass peak center | ¡	2.0	2.0	2.0	2.0	2.0	2.0
1.0 GeV/c < p_T < 1.5 GeV/c						
Decay length (cm) >	6.5	5.5	5.5	3.5	3.5	7.5
DCA to PV (cm) <	0.7	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.5	1.4	1.4	1.4	1.4	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	2.0	2.0	2.0	2.0	2.0	2.0
1.5 GeV/c < p_T < 2.0 GeV/c						
Decay length (cm) >	9.0	7.0	7.0	0.0	0.0	6.0
DCA to PV (cm) <	0.7	0.7	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.4	1.5	1.5	1.4	1.4	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	2.0	2.0	2.0	2.0	2.0	2.0
2.0 GeV/c < p_T < 2.5 GeV/c						
Decay length (cm) >	7.0	5.5	5.5	7.5	0.0	0.0

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Table B.1 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA to PV (cm) <	0.7	0.4	0.4	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.7	1.4	1.4	1.4	1.4	1.6
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
$\vert n_\Gamma$ from mass peak center $\vert$ ¡	2.0	2.0	2.0	2.0	2.0	2.0
2.5 GeV/c < p_T < 3.0 GeV/c						
Decay length (cm) >	15.0	11.0	12.0	11.0	12.0	0.0
DCA to PV (cm) <	0.5	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.4	1.2	1.8	1.0	1.8	1.5
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
$\vert n_\Gamma$ from mass peak center $\vert$ ¡	2.0	2.0	2.0	2.0	2.0	2.0
3.0 GeV/c < p_T < 3.5 GeV/c						
Decay length (cm) >	17.0	8.0	8.0	8.0	8.0	8.0
DCA to PV (cm) <	0.5	0.5	0.5	0.5	0.5	0.5
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA of π^- to PV (cm) >	1.7	1.6	1.6	1.2	1.2	0.6
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
$\vert n_\Gamma$ from mass peak center $\vert$ ¡	2.0	2.0	2.0	2.0	2.0	2.0
3.5 GeV/c < p_T < 4.0 GeV/c						
Decay length (cm) >	13.0	13.0	13.0	13.0	13.0	13.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.3	0.3	0.3	0.3	0.3	0.3
DCA of π^- to PV (cm) >	1.2	1.2	1.2	1.2	1.2	1.2
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
$\vert n_\Gamma$ from mass peak center $\vert$ ¡	1.5	1.5	1.5	1.5	1.5	1.5

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Table B.1 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
4.0 GeV/c < p_T < 5.0 GeV/c						
Decay length (cm) >	15.0	15.0	15.0	15.0	15.0	15.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	0.3	0.3	0.3	0.3	0.3	0.3
DCA of π^- to PV (cm) >	1.2	1.2	1.2	1.2	1.2	1.2
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	1.5	1.5	1.5	1.5	1.5	1.5
5.0 GeV/c < p_T < 10.0 GeV/c						
Decay length (cm) >	7.5	7.5	7.5	7.5	0.0	7.5
DCA to PV (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of p to PV (cm) >	1.5	1.5	1.5	1.5	1.5	3.0
DCA of π^- to PV (cm) >	1.5	1.5	1.5	1.5	0.0	1.5
$n_{\sigma,dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
| n_Γ from mass peak center | ¡	1.5	1.5	1.5	1.5	1.5	1.5

Table B.2: Geometric cuts for K_S^0 in $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. These are in addition to cuts listed in Table 1.2. Primary vertex is abbreviated to PV.

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.4 GeV/c < p_T < 0.6 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.6	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) >	2.1	1.1	1.1	1.1	1.1	1.1
$n_{\sigma,dE/dx}$ of daughters <	2.0	2.0	2.0	2.0	2.0	2.0
| n_Γ from mass peak center | ¡	1.5	1.5	1.5	1.5	1.5	1.5

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Table B.2 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.6 GeV/c $< p_T <$ 0.8 GeV/c						
Decay length (cm) $>$	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) $<$	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) $<$	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) $>$	1.0	1.0	1.0	1.0	1.0	1.0
$n_{\sigma,dE/dx}$ of daughters $<$	2.0	2.0	2.0	2.0	2.0	2.0
$\vert$ n_Γ from mass peak center $\vert$ ¡	1.5	1.5	1.5	1.5	2.0	2.0
0.8 GeV/c $< p_T <$ 1.0 GeV/c						
Decay length (cm) $>$	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) $<$	0.8	0.8	0.8	0.8	0.8	0.8
DCA of daughters (cm) $<$	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) $>$	0.9	0.9	0.9	0.9	0.9	0.9
$n_{\sigma,dE/dx}$ of daughters $<$	2.0	2.0	2.0	2.0	2.0	2.0
$\vert$ n_Γ from mass peak center $\vert$ ¡	1.5	1.5	1.5	2.0	2.0	2.0
1.0 GeV/c $< p_T <$ 1.5 GeV/c						
Decay length (cm) $>$	4.5	4.5	4.5	4.5	4.5	4.5
DCA to PV (cm) $<$	0.6	0.6	0.6	0.8	0.8	0.8
DCA of daughters (cm) $<$	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) $>$	0.9	0.9	0.9	0.9	0.9	0.9
$n_{\sigma,dE/dx}$ of daughters $<$	2.5	2.5	2.5	2.5	2.5	2.5
$\vert$ n_Γ from mass peak center $\vert$ ¡	1.5	1.5	1.5	1.5	1.5	2.0
1.5 GeV/c $< p_T <$ 2.0 GeV/c						
Decay length (cm) $>$	4.5	4.5	4.5	4.5	4.5	4.5
DCA to PV (cm) $<$	0.4	0.4	0.4	0.8	0.8	0.8
DCA of daughters (cm) $<$	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) $>$	1.5	1.5	1.5	1.2	1.1	0.0
$n_{\sigma,dE/dx}$ of daughters $<$	2.5	2.5	2.5	2.5	2.5	2.5
$\vert$ n_Γ from mass peak center $\vert$ ¡	1.5	2.0	2.0	1.5	1.5	1.5
2.0 GeV/c $< p_T <$ 2.5 GeV/c						
Decay length (cm) $>$	0.0	0.0	0.0	0.0	0.0	0.0

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Table B.2 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA to PV (cm) <	0.4	0.4	0.4	0.8	0.8	0.8
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) >	1.6	1.4	1.4	1.4	1.1	1.1
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	1.5	2.0	2.0	1.5	1.5	1.5
2.5 GeV/c < p_T < 3.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) >	1.2	1.2	1.0	0.0	0.0	0.0
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	1.5	2.0	2.0	1.5	2.0	2.0
3.0 GeV/c < p_T < 3.5 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.5	0.5	0.5
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) >	0.8	0.8	0.8	0.8	0.8	0.8
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	1.5	1.5	1.5	1.5	1.5	2.0
3.5 GeV/c < p_T < 4.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8
DCA of π^-, π^+ to PV (cm) >	0.8	0.8	0.6	0.6	0.6	0.6
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	1.5	1.5	1.5	1.5	1.5	1.5
4.0 GeV/c < p_T < 5.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.0
DCA to PV (cm) <	0.4	0.4	0.4	0.4	0.4	0.4
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	0.8

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Table B.2 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA of π^-, π^+ to PV (cm) >	0.8	0.8	0.6	0.6	0.6	0.6
$n_{\sigma,dE/dx}$ of daughters <	2.5	2.5	2.5	2.5	2.5	2.5
| n_Γ from mass peak center | ¡	2.0	2.0	2.0	2.0	2.0	2.0
5.0 GeV/c < p_T < 10.0 GeV/c						
Decay length (cm) >	0.0	0.0	0.0	0.0	0.0	0.2
DCA to PV (cm) <	0.4	0.4	0.6	0.6	0.8	1.5
DCA of daughters (cm) <	0.8	0.8	0.8	0.8	0.8	2.0
DCA of π^-, π^+ to PV (cm) >	1.4	1.4	1.4	1.4	1.4	1.5
$n_{\sigma,dE/dx}$ of daughters <	1.5	1.5	2.0	2.0	2.0	7.5
| n_Γ from mass peak center | ¡	2.0	2.0	2.0	2.0	2.0	0.0

B.2 Ξ^- Geometric Cuts

Table B.3: Geometric cuts for Ξ^- in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. These are in addition to cuts listed in Table 1.4. Cuts for Ξ^+ the same with cuts for the p daughter of the Λ applied to the $\bar{p}$ daughter of the $\bar{\Lambda}$, cuts for the π^- daughter of the Λ applied to the π^+ daughter of the $\bar{\Lambda}$, and cuts for the Λ applied to the $\bar{\Lambda}$. A and B refer to the parameters in Equation 1.2. Primary vertex is abbreviated to PV.

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
0.6 GeV/c < p_T <0.8 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0

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Table B.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
$\mid n_\Gamma$ from mass peak center $\mid$	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.8333	1.3333	1.8333
B	-0.03	-0.03	-0.03	-0.05	-0.37	-0.06
0.8 GeV/c $< p_T <$1 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
$\mid n_\Gamma$ from mass peak center $\mid$	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.3333	N/A	1.8333
B	-0.2	-0.2	-0.2	-1000	N/A	-0.06
1 GeV/c $< p_T <$1.5 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0

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Table B.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
$\mid n_\Gamma$ from mass peak center $\mid$	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.3333	1.3333	1.3333	N/A	N/A	1.8333
B	-0.35	-0.35	-0.35	N/A	N/A	-0.06
1.5 GeV/c $< p_T <$2 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
$\mid n_\Gamma$ from mass peak center $\mid$	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.3333	1.3333	1.3333	N/A	N/A	1.8333
B	-0.4	-0.4	-0.4	N/A	N/A	-0.06
2 GeV/c $< p_T <$2.5 GeV/c						
Ξ^- Decay length (cm)	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5	> 3.5
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55	< 0.55
DCA of Λ to primary vertex (cm)	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18	> 0.18
DCA of Ξ^- to daughters (cm)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
$\mid n_\Gamma$ from mass peak center $\mid$	> 1.5	> 1.5	> 2	> 2	> 2	> 2

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Table B.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
A	1.8333	1.8333	1.8333	1.3333	1.8333	1.8333
B	0.05	0.05	0.05	0	-0.06	0
2.5 GeV/c $< p_T <$3 GeV/c						
Ξ^- Decay length (cm)	> 3	> 3	> 3	> 3	> 3	> 3
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45
DCA of Λ to primary vertex (cm)	> 0.1	> 0.1	> 0.1	> 0.1	> 0.1	> 0.1
DCA of Ξ^- to daughters (cm)	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45	< 0.45
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
$\mid n_\Gamma$ from mass peak center $\mid$	> 1.5	> 1.5	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.3333	1.8333	1.8333
B	0	0	0.05	0	-0.06	0
3 GeV/c $< p_T <$3.5 GeV/c						
Ξ^- Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to primary vertex (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
$\mid n_\Gamma$ from mass peak center $\mid$	> 2	> 2	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.8333	1.3333	1.8333	1.8333
B	-0.03	-0.03	-0.05	-0.37	-0.06	0

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Table B.3 – continued from previous page

	0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
3.5 GeV/c < p_T <4.5 GeV/c						
Ξ^- Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to primary vertex (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
| n_Γ from mass peak center |	> 2	> 2	> 2	> 2	> 2	> 2
A	1.8333	1.8333	1.3333	N/A	1.8333	N/A
B	-0.2	-0.2	-1000	N/A	-0.06	N/A
p_T >4.5 GeV/c						
Ξ^- Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
Λ Decay length (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to primary vertex (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to primary vertex (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of Ξ^- to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of Λ to daughters (cm)	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8	< 0.8
DCA of p to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
DCA of bachelor π^- to PV (cm)	> 0	> 0	> 0	> 0	> 0	> 0
$n_{\sigma,dE/dx}$ of daughters	> 3	> 3	> 3	> 3	> 3	> 3
| n_Γ from mass peak center |	> 2	> 2	> 2	> 2	> 2	> 2
A	1.3333	1.3333	N/A	N/A	1.8333	N/A
B	-0.35	-0.35	N/A	N/A	-0.06	N/A

Appendix C

Efficiencies

Table C.1: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au+Au$ at $\sqrt{s_{NN}}=62$ GeV to $a\ exp(\frac{b}{p_T}^{c})$

centrality	a	b	c
0-5%	.767	.1498	1.316
5-10%	.7856	.1419	1.33
10-20%	.7975	.1408	1.464
20-30%	.8113	.1374	1.606
30-40%	.822	.1376	1.743
40-50%	.8321	.1327	1.781
50-60%	.8386	.1357	1.912
60-70%	.8387	.1511	2.381
70-80%	.8415	.1455	2.153

Table C.2: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a\ exp(\frac{b}{p_T}^c)$

centrality	a	b	c
0-10%	0.838435	0.0704	0.9409
10-20%	0.843666	0.0681	1.0000
20-30%	0.843211	0.0610	1.0000
30-40%	0.848090	0.0638	1.0000
40-50%	0.856875	0.0606	0.9496
50-60%	0.851112	0.0584	1.0000

Table C.3: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a\ exp(\frac{b}{p_T}^c)$

centrality	a	b	c
0-10%	0.8655	0.1333	0.9890
10-20%	0.8736	0.1207	0.9330
20-30%	0.8643	0.1503	1.1071
30-40%	0.8693	0.1269	0.9922
40-50%	0.8608	0.1511	1.1616
50-60%	0.8606	0.1971	1.3211

Table C.4: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV to $a\ exp(\frac{b}{p_T}^c)$

centrality	a	b	c
0-5%	.767	.1498	1.316
5-10%	.7856	.1419	1.33
10-20%	.7975	.1408	1.464
20-30%	.8113	.1374	1.606
30-40%	.822	.1376	1.743
40-50%	.8321	.1327	1.781
50-60%	.8386	.1357	1.912
60-70%	.8387	.1511	2.381
70-80%	.8415	.1455	2.153

C.1 V^0 Efficiencies

C.1.1 V^0 Efficiencies in Cu+Cu collisions at $\sqrt{s_{NN}}$ = 200 GeV

Table C.5: Fit parameters from fits of the efficiencies for Λ baryons in $Cu + Cu$ at $\sqrt{s_{NN}}$ = 200 GeV to $a\ exp(\frac{b}{p_T}^{c})$

		0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
	a	0.172	0.164	0.168	0.158	0.161	0.166
1< p_T <1.5	b	0.946	0.838	0.819	0.774	0.758	0.861
	c	1.587	1.800	1.825	1.902	2.020	1.895
	a	0.174	0.162	0.169	0.159	0.161	0.164
1.5< p_T <2	b	1.043	0.887	0.855	0.772	0.755	0.821
	c	1.701	1.884	1.835	1.842	2.033	1.947
	a	0.172	0.145	0.152	0.159	0.161	0.163
2< p_T <2.5	b	0.969	0.915	0.913	0.845	0.755	0.789
	c	1.586	2.106	2.066	1.999	2.033	1.790
	a	0.177	0.152	0.157	0.149	0.154	0.159
2.5< p_T <3	b	1.468	0.953	0.924	0.889	0.911	0.926
	c	1.890	2.120	2.149	2.280	2.406	2.023
	a	0.177	0.152	0.157	0.149	0.154	0.159
3< p_T <3.5	b	1.468	0.953	0.924	0.889	0.911	0.926
	c	1.890	2.120	2.149	2.280	2.406	2.023
	a	0.137	0.110	0.198	0.246	0.252	0.140
3.5< p_T <4	b	0.038	0.051	0.023	0.011	0.009	0.048
	c	-0.005	-0.006	-0.004	-0.003	-0.003	-0.006
	a	0.137	0.110	0.198	0.246	0.252	0.140
4< p_T <5	b	0.038	0.051	0.023	0.011	0.009	0.048
	c	-0.005	-0.006	-0.004	-0.003	-0.003	-0.006
	a	0.177	0.152	0.157	0.149	0.154	0.159
p_T >5	b	1.468	0.953	0.924	0.889	0.911	0.926
	c	1.890	2.120	2.149	2.280	2.406	2.023

Table C.6: Fit parameters from fits of the efficiencies for $\bar{\Lambda}$ baryons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a\ exp(\frac{b}{p_T}^c)$

		0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
	a	0.158	0.163	0.169	0.169	0.186	0.196
1< p_T <1.5	b	0.924	0.888	0.884	0.784	0.897	1.015
	c	1.767	1.683	1.752	1.659	1.479	1.444
	a	0.163	0.158	0.174	0.178	0.187	0.205
1.5< p_T <2	b	1.092	1.054	1.076	1.026	1.104	1.200
	c	1.948	2.156	1.950	1.763	1.758	1.527
	a	0.161	0.139	0.152	0.167	0.189	0.198
2< p_T <2.5	b	1.042	0.946	0.968	0.860	0.904	0.981
	c	1.893	2.284	2.084	1.916	1.402	1.294
	a	0.163	0.146	0.155	0.155	0.165	0.183
2.5< p_T <3	b	1.414	0.967	0.953	0.914	1.006	1.058
	c	1.983	2.281	2.155	2.206	2.166	1.632
	a	0.163	0.146	0.155	0.155	0.165	0.183
3< p_T <3.5	b	1.414	0.967	0.953	0.914	1.006	1.058
	c	1.983	2.281	2.155	2.206	2.166	1.632
	a	0.113	0.149	0.227	0.148	0.072	0.224
3.5< p_T <4	b	0.047	0.036	0.010	0.042	0.070	0.013
	c	-0.006	-0.005	-0.002	-0.005	-0.008	-0.003
	a	0.113	0.149	0.227	0.148	0.072	0.224
4< p_T <5	b	0.047	0.036	0.010	0.042	0.070	0.013
	c	-0.006	-0.005	-0.002	-0.005	-0.008	-0.003
	a	0.163	0.146	0.155	0.155	0.165	0.183
p_T >5	b	1.414	0.967	0.953	0.914	1.006	1.058
	c	1.983	2.281	2.155	2.206	2.166	1.632

Table C.7: Fit parameters from fits of the efficiencies for K_S^0 baryons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a\ exp(\frac{b}{p_T}^c)$

		0-10%	10-20%	20-30%	30-40 %	40-50%	50-60%
	a	0.176	0.174	0.280	0.270	0.257	0.294
1< p_T <1.5	b	0.972	0.917	1.065	1.014	0.964	1.129
	c	1.606	1.796	1.589	1.778	1.924	1.563
	a	0.176	0.174	0.280	0.270	0.257	0.299
1.5< p_T <2	b	0.972	0.917	1.065	1.014	0.964	1.000
	c	1.606	1.796	1.589	1.778	1.924	1.298
	a	0.194	0.182	0.284	0.275	0.262	0.299
2< p_T <2.5	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298
	a	0.194	0.182	0.284	0.275	0.262	0.299
2.5< p_T <3	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298
	a	0.194	0.182	0.284	0.275	0.262	0.299
3< p_T <3.5	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298
	a	0.127	0.114	0.164	0.141	0.177	0.234
3.5< p_T <4	b	0.014	0.022	0.041	0.051	0.038	0.014
	c	-0.002	-0.003	-0.004	-0.005	-0.004	-0.002
	a	0.127	0.114	0.164	0.141	0.177	0.234
4< p_T <5	b	0.014	0.022	0.041	0.051	0.038	0.014
	c	-0.002	-0.003	-0.004	-0.005	-0.004	-0.002
	a	0.194	0.182	0.284	0.275	0.262	0.299
p_T >5	b	0.989	0.895	0.936	0.907	0.866	1.000
	c	1.222	1.502	1.329	1.511	1.652	1.298

Table C.8: Fit parameters from fits of the efficiencies for Λ in $Au + Au$ at $\sqrt{s_{NN}}$ = 200 GeV to $a\ exp(\frac{b}{p_T}^{c})+d+e$

centrality	a	b	c	d	e
0-5%	9.73455e-02	1.44024e+00	1.17638e+00	-9.13192e-04	-8.74599e-04
5-10%	1.05882e-01	1.41282e+00	1.23304e+00	-4.41302e-04	-1.03846e-03
10-20%	9.98811e-02	1.21169e+00	1.41522e+00	-1.42956e-04	-9.95265e-04
20-30%	1.23769e-01	1.37160e+00	1.17591e+00	-5.06965e-04	-1.17704e-03
30-40%	1.09135e-01	1.20635e+00	1.34792e+00	1.98552e-03	-1.37347e-03
40-50%	1.07304e-01	1.16138e+00	1.54782e+00	2.08468e-03	-1.35536e-03
50-60%	8.70185e-02	1.02422e+00	2.14817e+00	3.93120e-03	-1.29337e-03
60-70%	7.30531e-02	1.01570e+00	2.43480e+00	8.27065e-03	-1.70979e-03
70-80%	9.63416e-02	1.24434e+00	1.25699e+00	5.21417e-03	-1.63667e-03

Table C.9: Fit parameters from fits of the efficiencies for $\bar{\Lambda}$ in $Au + Au$ at $\sqrt{s_{NN}}$ = 200 GeV to $a\ exp(\frac{b}{p_T}^{c})+d+e$

centrality	a	b	c	d	e
0-5%	9.11053e-02	1.38980e+00	1.24573e+00	3.35935e-04	-9.73930e-04
5-10%	1.00155e-01	1.29565e+00	1.28446e+00	1.75720e-04	-1.06484e-03
10-20%	9.38267e-02	1.22126e+00	1.48764e+00	1.83101e-03	-1.18936e-03
20-30%	9.90359e-02	1.14356e+00	1.48145e+00	9.35607e-04	-1.07923e-03
30-40%	1.01358e-01	1.11225e+00	1.52444e+00	2.63142e-03	-1.36184e-03
40-50%	1.07008e-01	1.11577e+00	1.50770e+00	2.32430e-03	-1.39252e-03
50-60%	9.11440e-02	1.06653e+00	1.91641e+00	4.72654e-03	-1.44842e-03
60-70%	9.09852e-02	1.09978e+00	1.60967e+00	5.31837e-03	-1.54718e-03
70-80%	9.74573e-02	9.86275e-01	1.77140e+00	-2.77767e-03	-6.33058e-04

C.1.2 V^0 Efficiencies in $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

Table C.10: Fit parameters from fits of the efficiencies for K_S^0 in $Au + Au$ at $\sqrt{s_{NN}}$ = 200 GeV to $a\ exp(\frac{b}{p_T}^c)$+d+e

centrality	a	b	c	d	e
0-5%	1.90006e-01	1.46579e+00	1.58672e+00	1.18775e-03	-1.16531e-03
5-10%	1.99357e-01	1.42488e+00	1.62480e+00	2.00574e-03	-1.21733e-03
10-20%	2.11215e-01	1.36336e+00	1.65828e+00	1.71577e-03	-1.28160e-03
20-30%	2.25984e-01	1.35987e+00	1.60081e+00	1.44734e-03	-1.37115e-03
30-40%	2.20745e-01	1.26539e+00	1.75534e+00	2.46239e-03	-1.38532e-03
40-50%	2.01929e-01	1.19933e+00	2.04934e+00	5.35690e-03	-1.44794e-03
50-60%	2.34387e-01	1.25719e+00	1.68057e+00	2.42561e-03	-1.56188e-03
60-70%	1.94564e-01	1.23620e+00	2.26538e+00	5.01216e-03	-1.36288e-03
70-80%	1.80066e-01	1.14058e+00	2.16036e+00	2.52093e-03	-8.10902e-04

Appendix D

Centrality

	R	N_{part}	N_{bin}
0-10%	$101 \leq$ R	96	162
10-20%	$71 \leq$ R < 101	72	108
20-30%	$49 \leq$ R < 71	52	68
30-40%	$33 \leq$ R < 49	36	42
40-50%	$22 \leq$ R < 33	25	26
50-60%	$14 \leq$ R < 22	16	15

Table D.1: Centrality definitions for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

	R	N_{part}	N_{bin}
0-10%	$139 \leq$ R	99	189
10-20%	$98 \leq$ R < 139	75	124
20-30%	$67 \leq$ R < 98	54	78
30-40%	$46 \leq$ R < 67	38	48
40-50%	$30 \leq$ R < 46	26	29
50-60%	$19 \leq$ R < 30	17	17

Table D.2: Centrality definitions for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

	R	N_{part}	N_{bin}
0-5%	$373 \geq$ R	347	904
5-10%	$313 \leq$ R < 373	293	714
10-20%	$222 \leq$ R < 313	229	512
20-30%	$154 \leq$ R < 222	162	321
30-40%	$102 \leq$ R < 154	112	193
40-50%	$65 \leq$ R < 102	74	109
50-60%	$38 \leq$ R < 65	46	57
60-70%	$20 \leq$ R < 38	26	27
70-80%	$9 \leq$ R < 20	13	11

Table D.3: Centrality definitions for $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

	R	N_{part}	N_{bin}
0-5%	$510 \geq$ R	352.4	1051.3
5-10%	$431 \leq$ R < 510	299.3	827.9
10-20%	$312 \leq$ R < 431	234.6	591.3
20-30%	$217 \leq$ R < 312	166.7	368.6
30-40%	$146 \leq$ R < 217	115.5	220.2
40-50%	$94 \leq$ R < 146	76.6	123.4
50-60%	$56 \leq$ R < 94	47.8	63.9
60-70%	$30 \leq$ R < 56	27.4	29.5
70-80%	$14 \leq$ R < 30	14.1	12.3

Table D.4: Centrality definitions for $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

Appendix E

$p_T^{trigger}$ dependence for identified trigger particles

The default method is bin counting. Fits in $\Delta\eta$ are used in the following bins where the track merging effect is still significant:

- $d + Au$ collisions: Λ for $2.0 < p_T^{trigger} < 3.0$ GeV/c

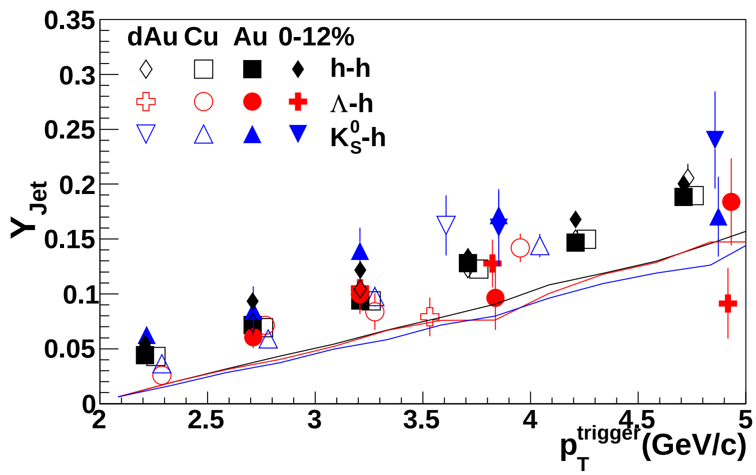

Figure E.1: $p_T^{trigger}$ dependence of Y_{Jet} for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ in 0-60% central $Cu + Cu$ and 0-12% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for unidentified hadrons, Λ, K_S^0, and Ξ trigger particles. The identified trigger particle data have been horizontally offset for visibility.

- $Cu + Cu$ collisions: Λ for $2.0 < p_T^{trigger} < 3.0$ GeV/c
- 0-12% central $Au + Au$ collisions: Λ for $3.0 < p_T^{trigger} < 4.5$ GeV/c, K_S^0 for 3.0 $< p_T^{trigger} < 3.5$ GeV/c
- **Jana the raw plot for lambdas 4.5-5.5 GeV looks a little funny. I think we would not be unjustified in throwing this out.**
- 40-80% central $Au + Au$ collisions: Λ for $2.0 < p_T^{trigger} < 4.5$ GeV/c, K_S^0 for 2.0 $< p_T^{trigger} < 4.5$ GeV/c

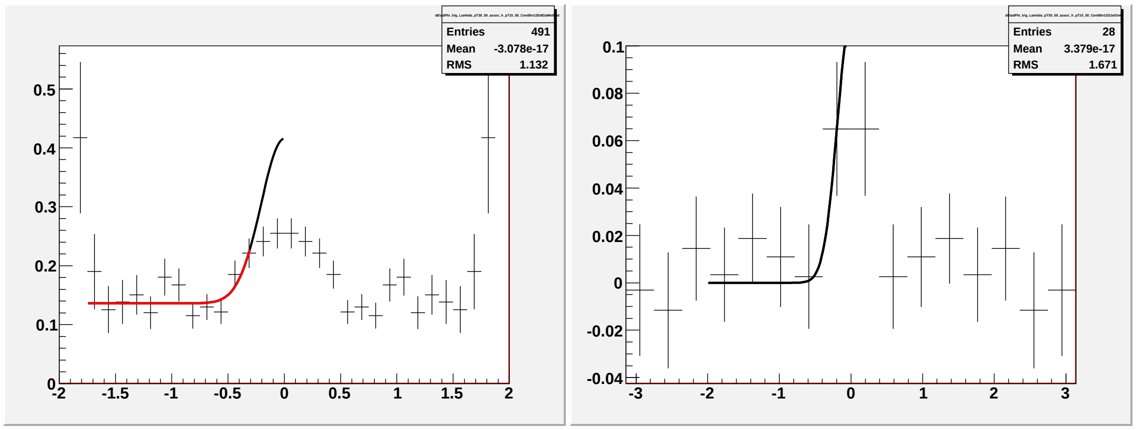

Figure E.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

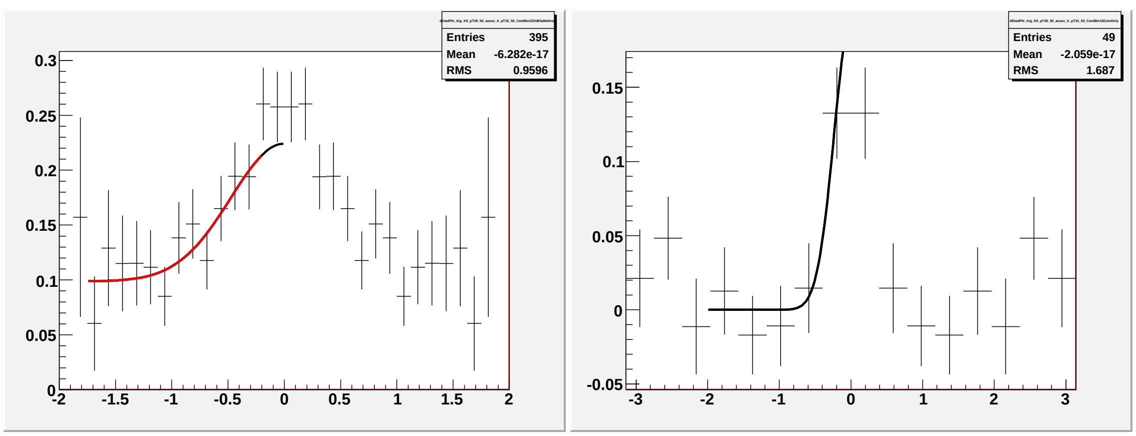

Figure E.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

E.A Fits for $p_T^{trigger}$ dependence in $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

E.B Fits for $p_T^{trigger}$ dependence in $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV for identified trigger particles

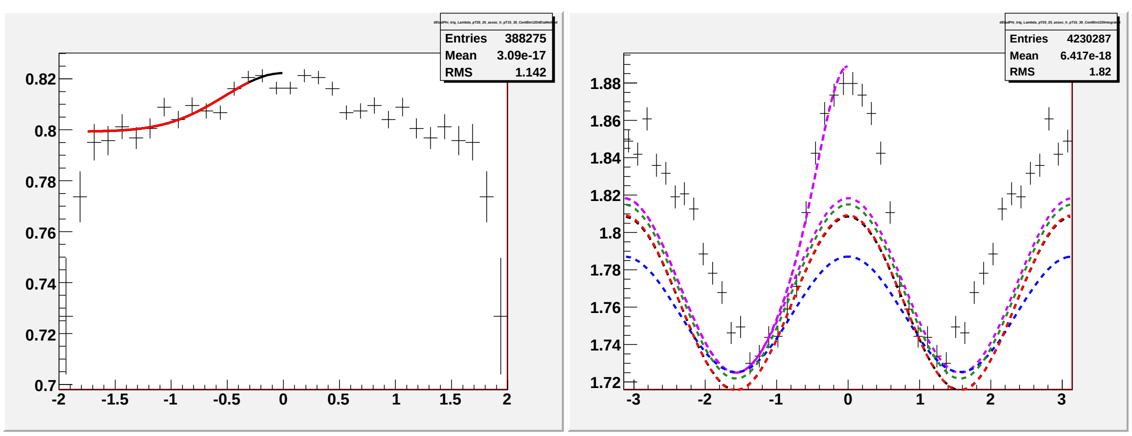

Figure E.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

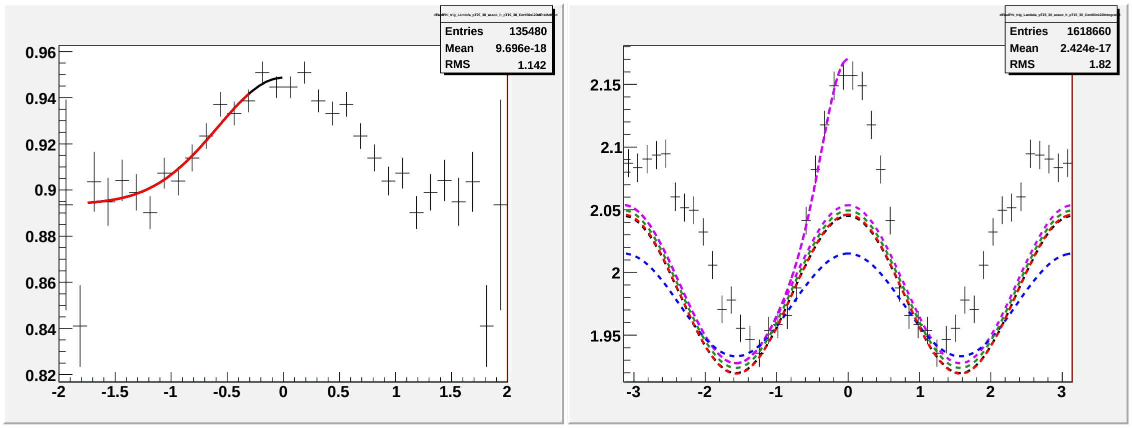

Figure E.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

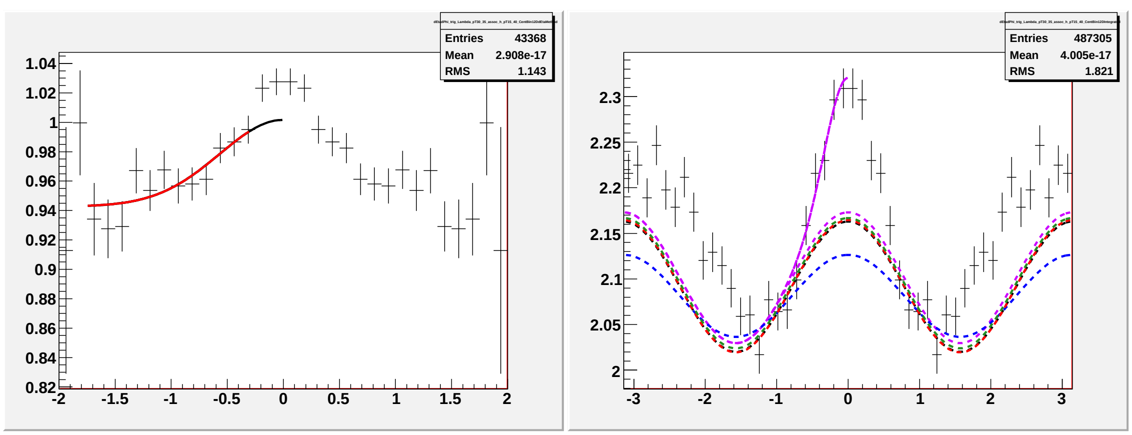

Figure E.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

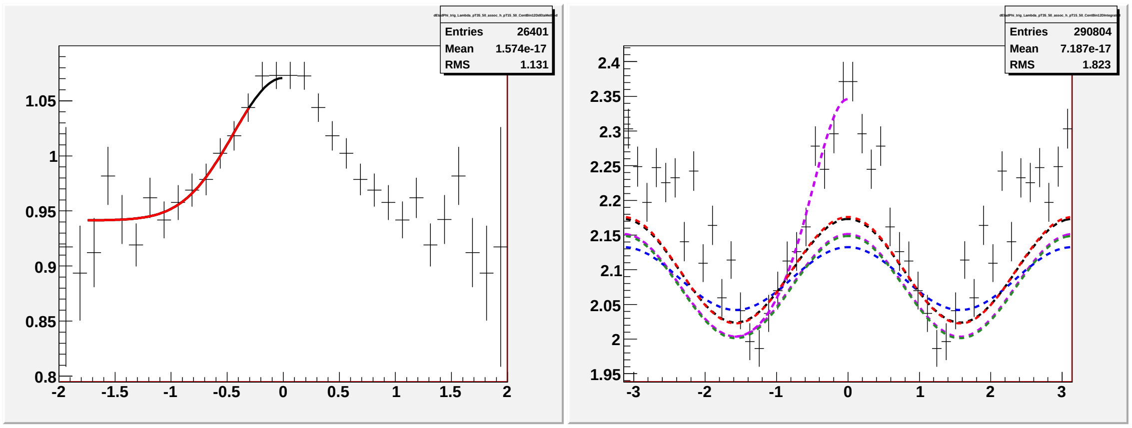

Figure E.7: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

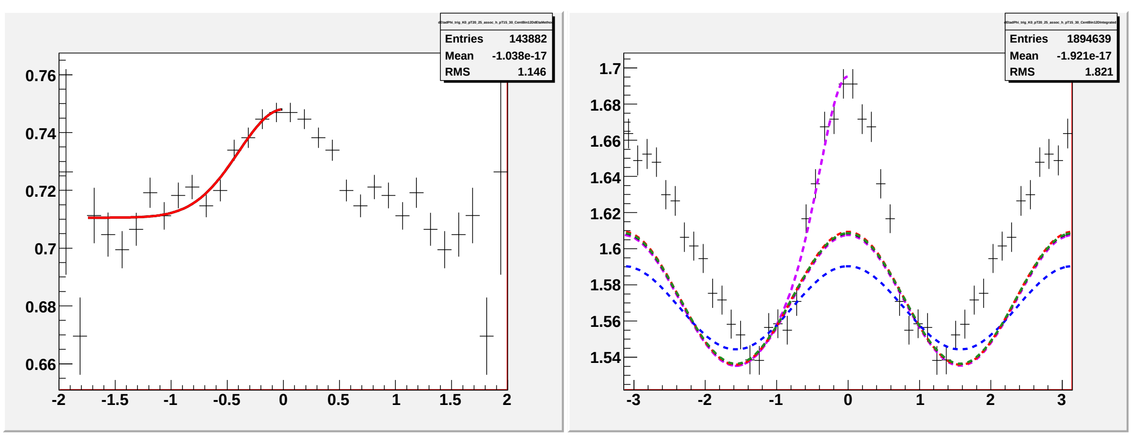

Figure E.8: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

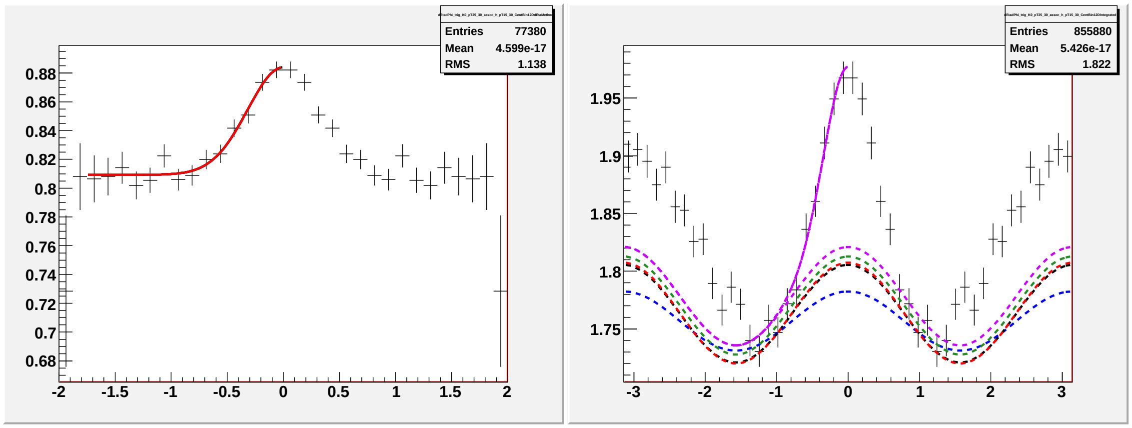

Figure E.9: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

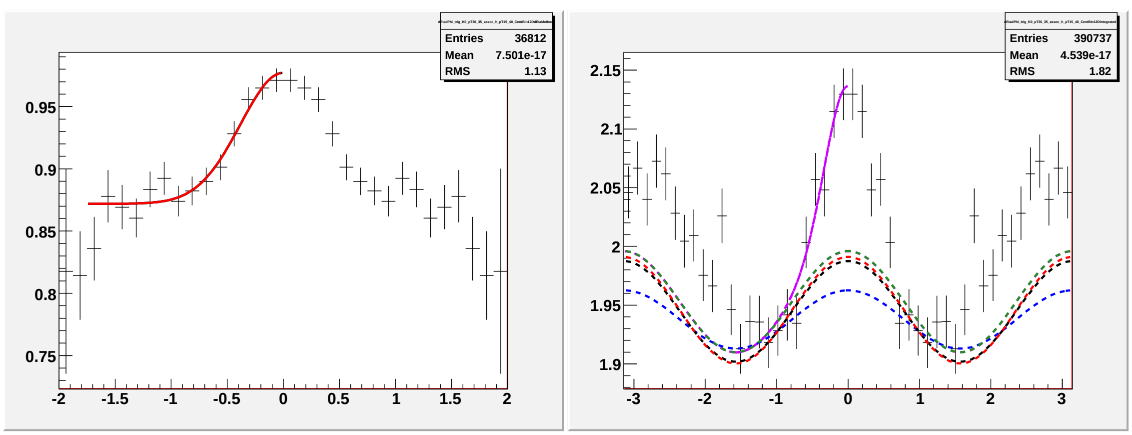

Figure E.10: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

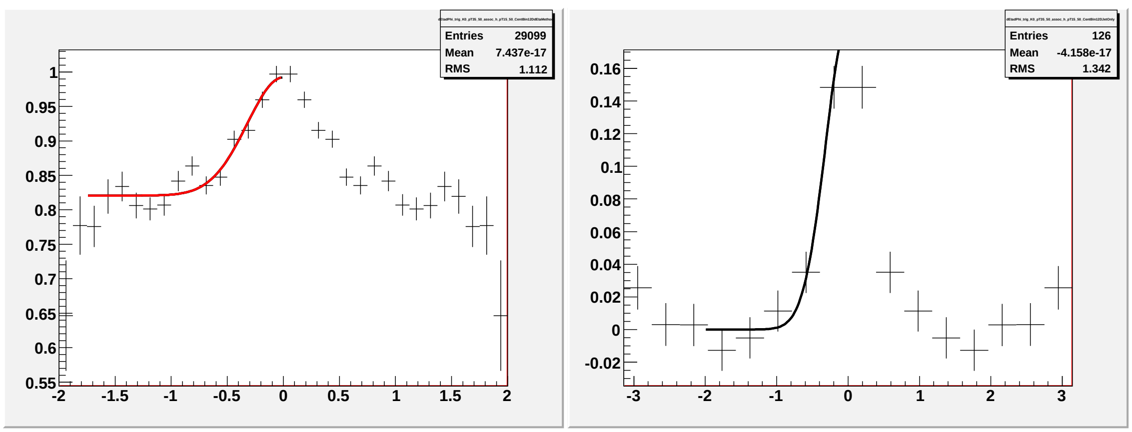

Figure E.11: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

E.C Fits for $p_T^{trigger}$ dependence in 0-12% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for identified trigger particles

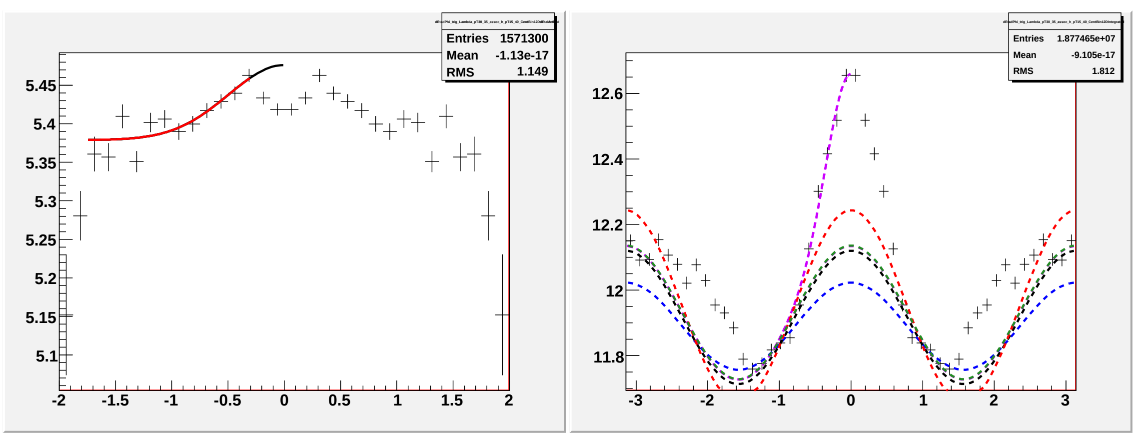

Figure E.12: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

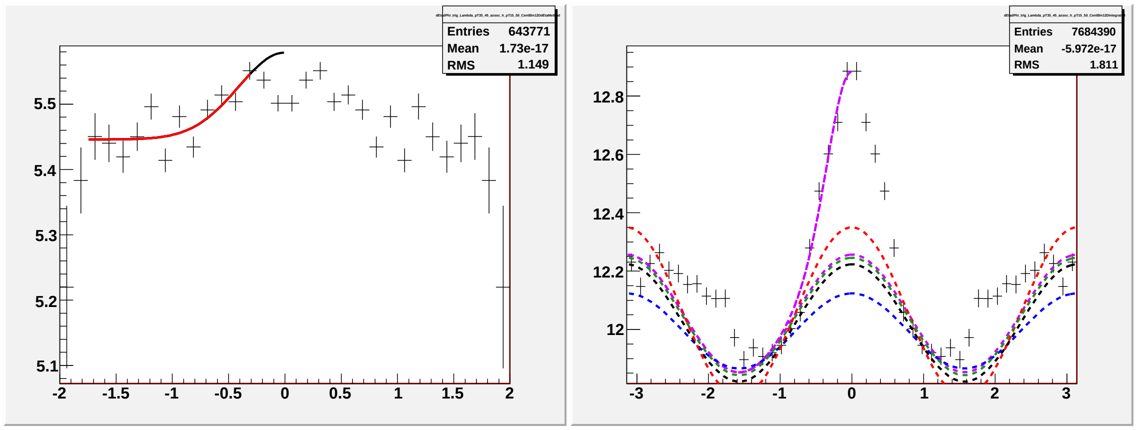

Figure E.13: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

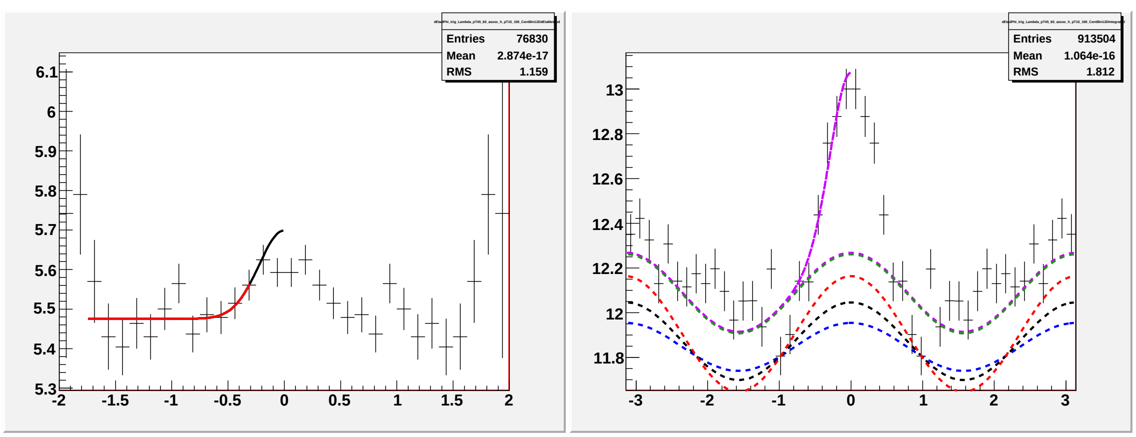

Figure E.14: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.5 < p_T^{trigger} < 5.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

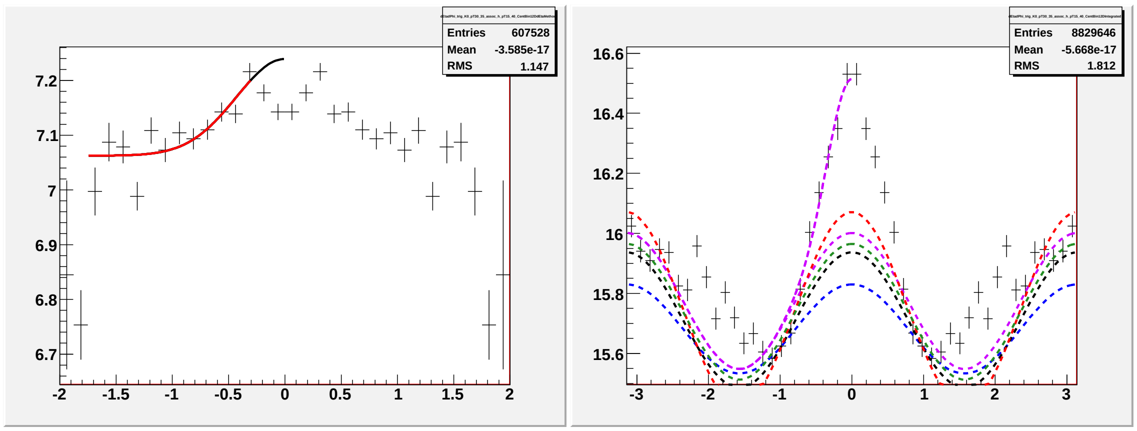

Figure E.15: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

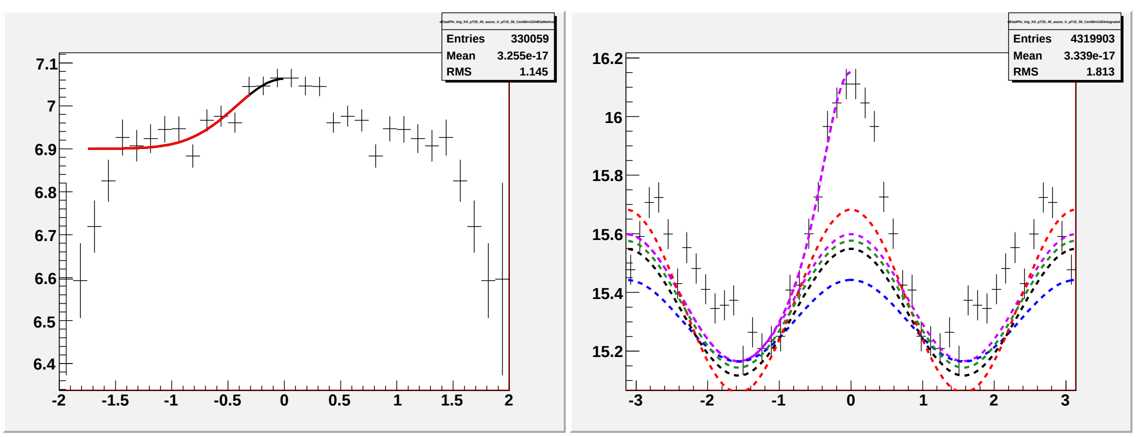

Figure E.16: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.5$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

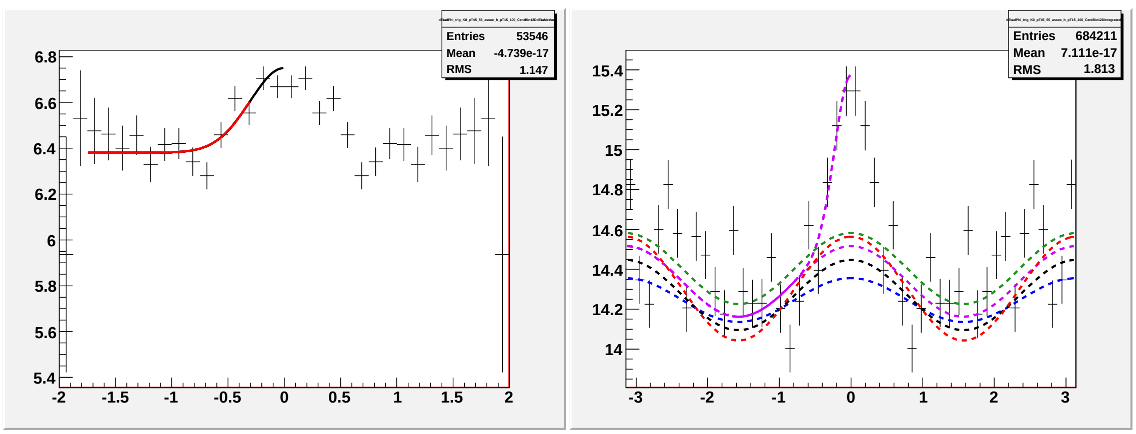

Figure E.17: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.5 < p_T^{trigger} < 5.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

E.D Fits for $p_T^{trigger}$ dependence in 40-80% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for identified trigger particles

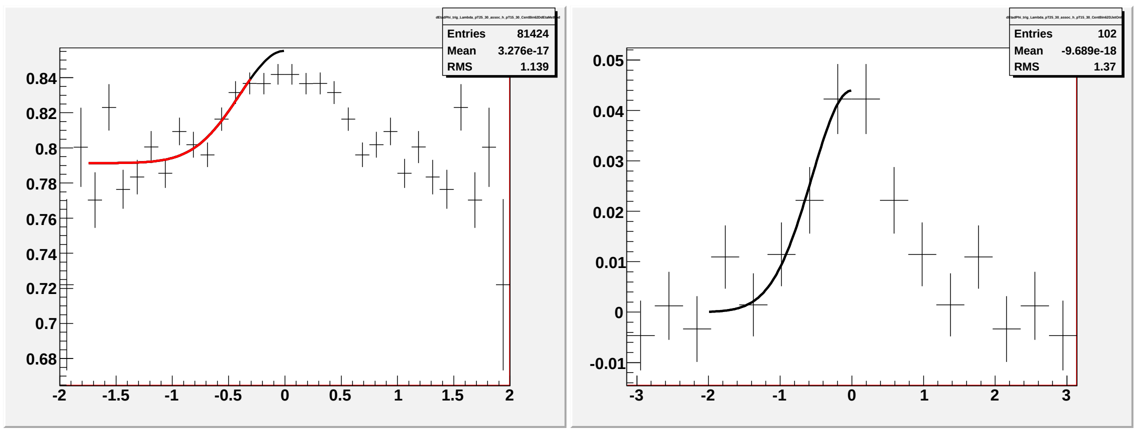

Figure E.18: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

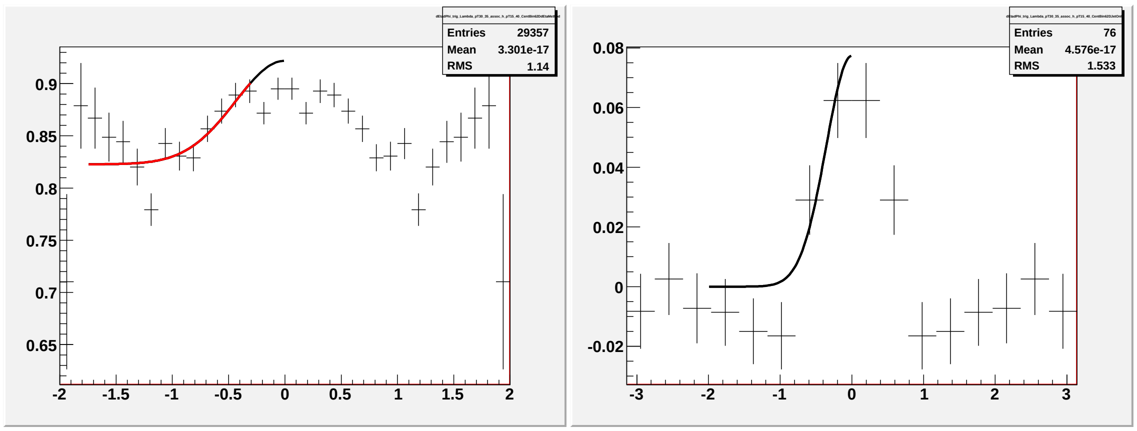

Figure E.19: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

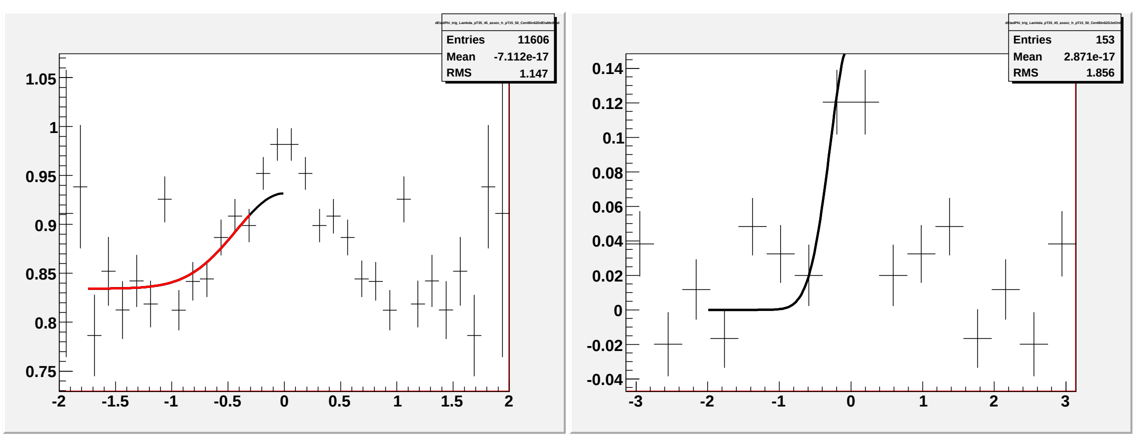

Figure E.20: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

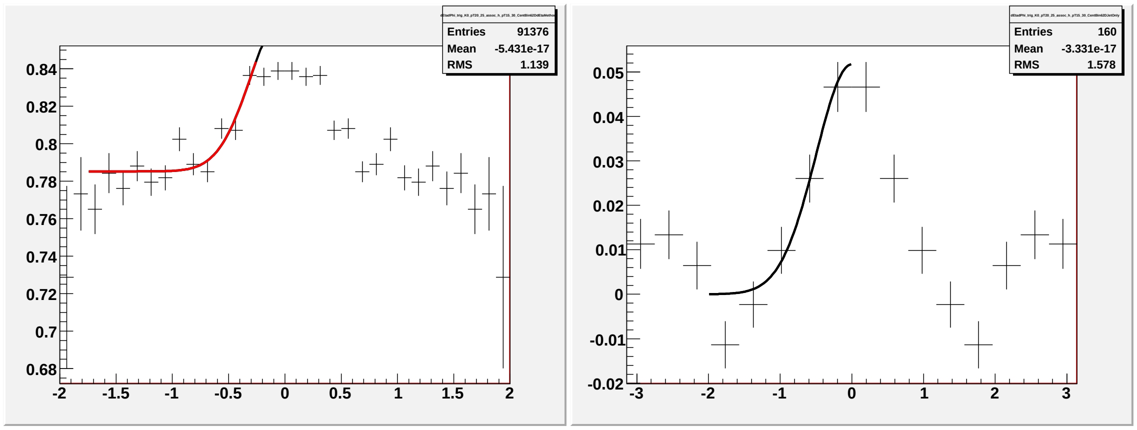

Figure E.21: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

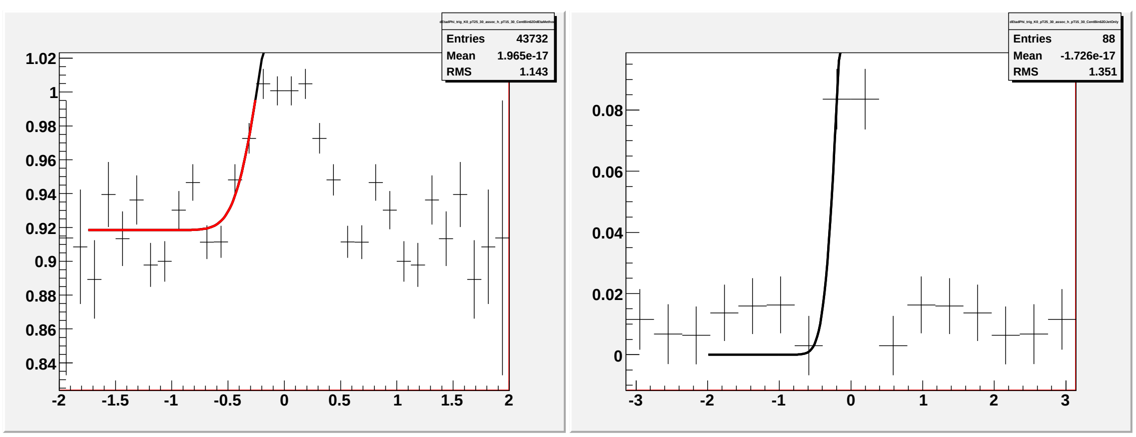

Figure E.22: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

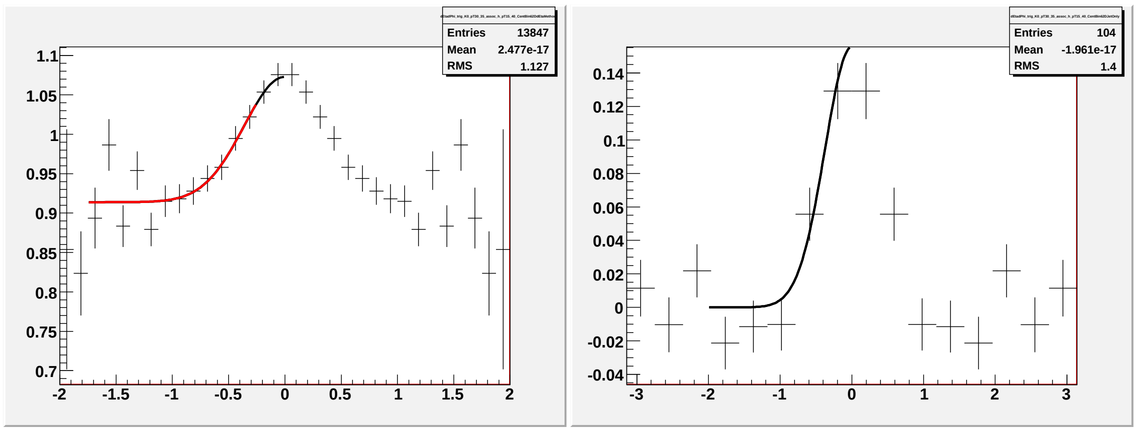

Figure E.23: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

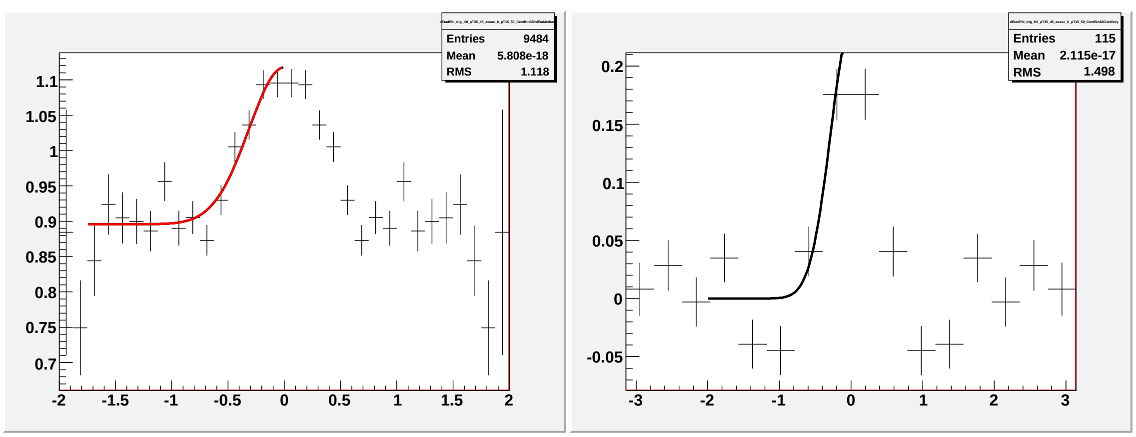

Figure E.24: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

Appendix F

N_{part} dependence for identified trigger particles

Every data point used bin counting except the most central $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV with a Λ trigger.

Each particle was fit to a constant. The central values of the fits are shown as lines with a band for 1σ. Most central $Au+Au$ need to add fit and mark out problem data point for lambda. Need to update plot and table.

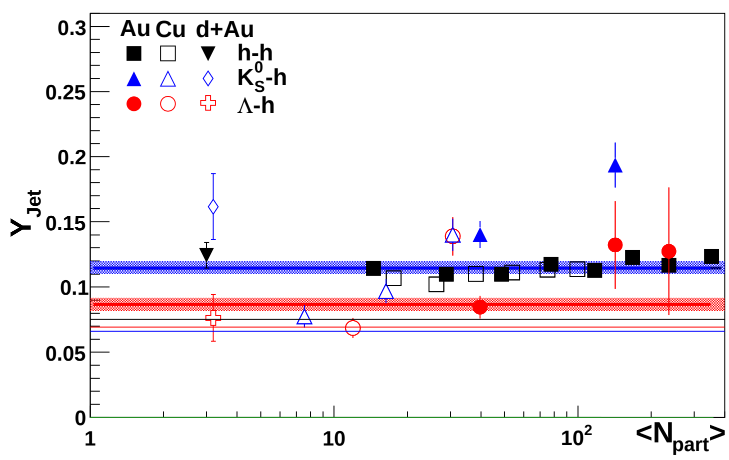

Figure F.1: N_{part} dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for identified trigger particles in $d + Au$, $Cu + Cu$, and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for Λ, K_S^0, and Ξ trigger particles.

Table F.1: Fit Error bars

particle	slope
h	$0.1150 \pm .0006$
Λ	0.0967 ± 0.0043
K_S^0	0.1262 ± 0.0045
Ξ^-	0.0902 ± 0.0083

The default method is bin counting. Fits in $\Delta\eta$ are used in the following bins where the track merging effect is still significant:

- $Au + Au$ 0-12% Λ
- **Jana I think we need to discuss the most central data point**

F.A Fits for N_{part} dependence in $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

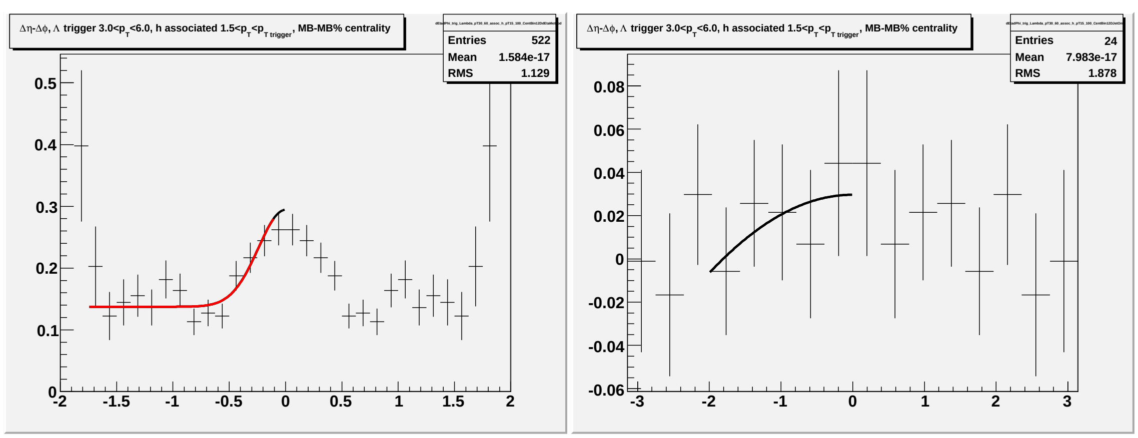

Figure F.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for $\Lambda + \bar{\Lambda}$ triggers for minimum bias $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

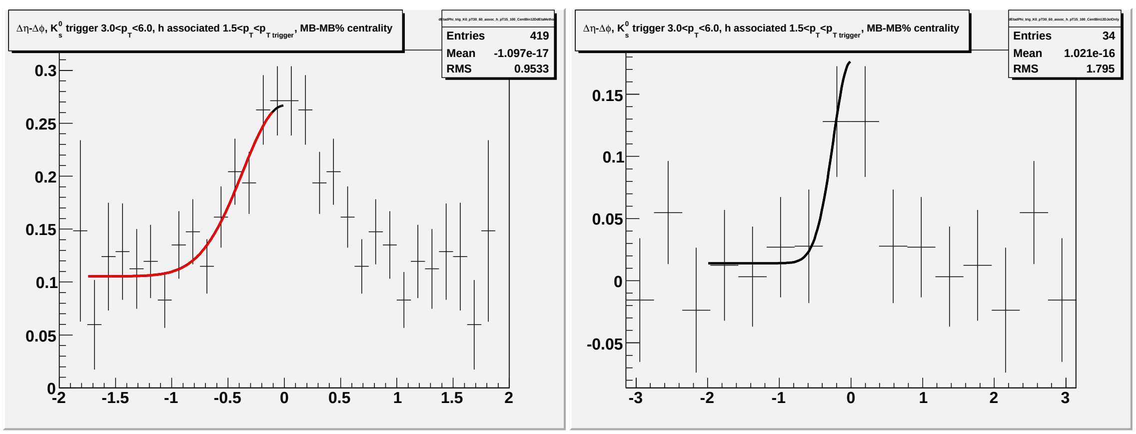

Figure F.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 triggers for minimum bias $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

F.B Fits for N_{part} dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV for identified trigger particles

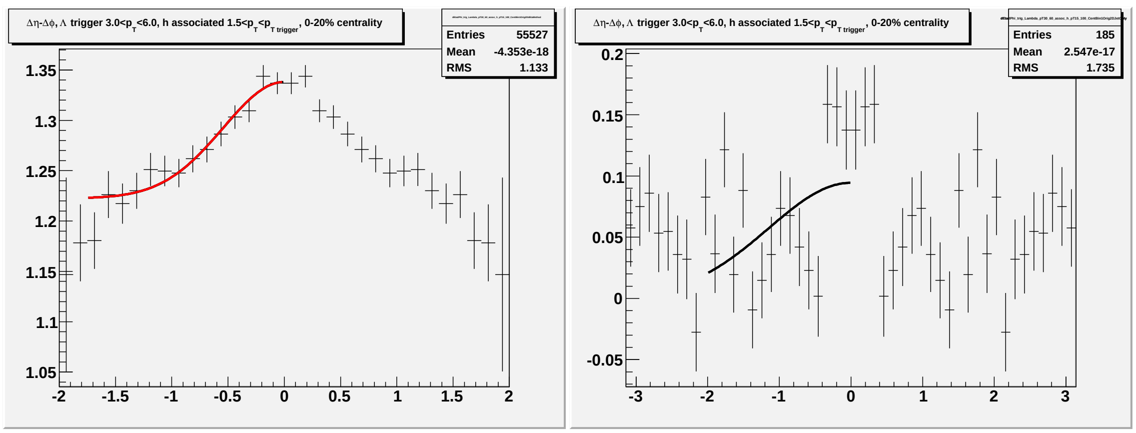

Figure F.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

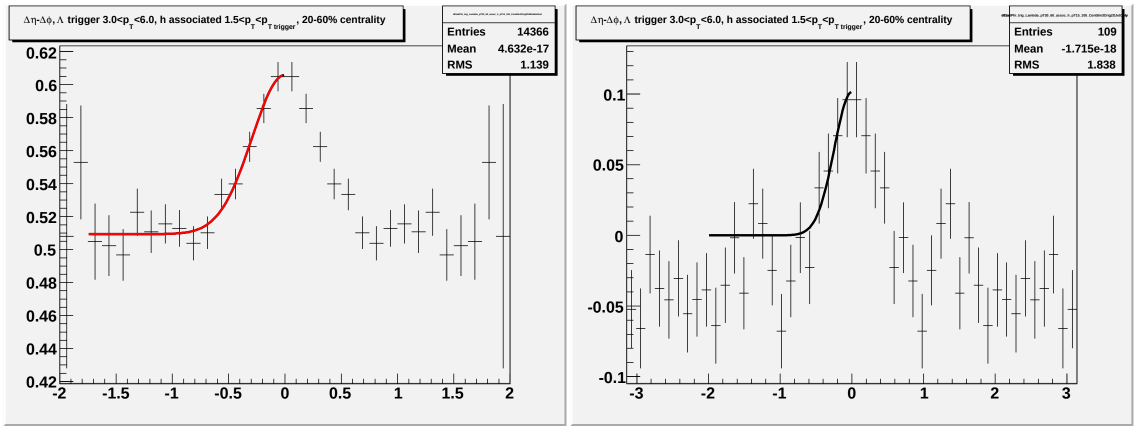

Figure F.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

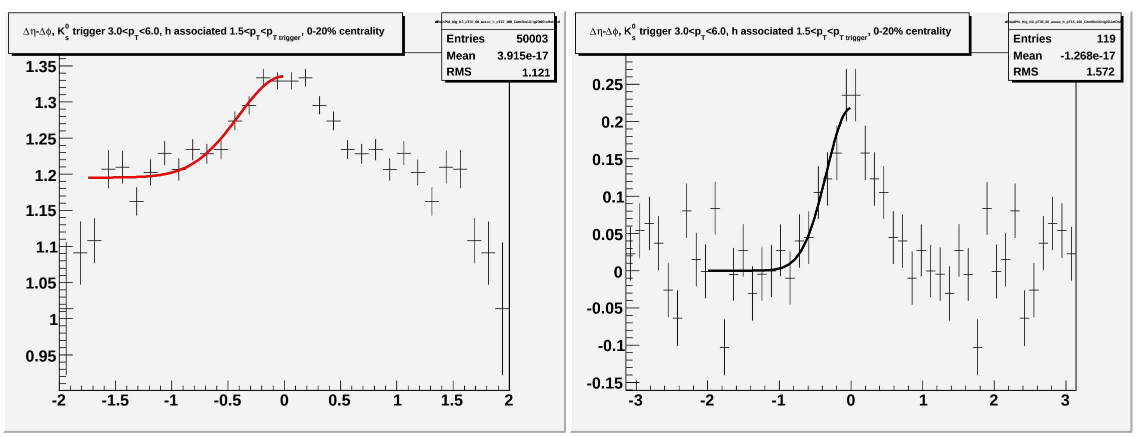

Figure F.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

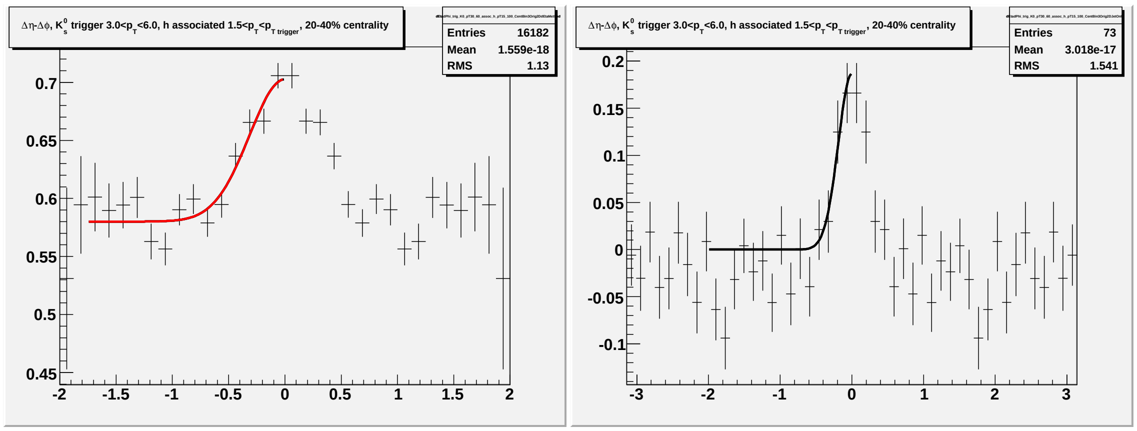

Figure F.7: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

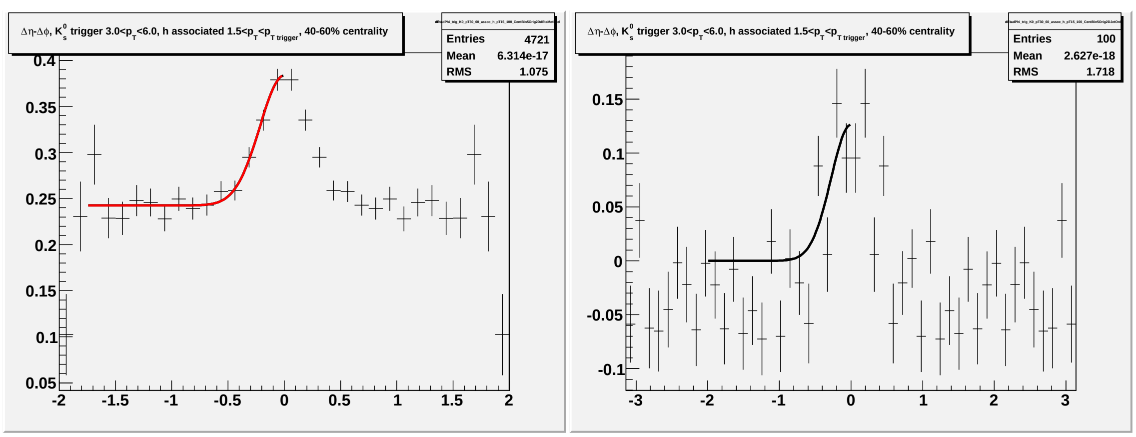

Figure F.8: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

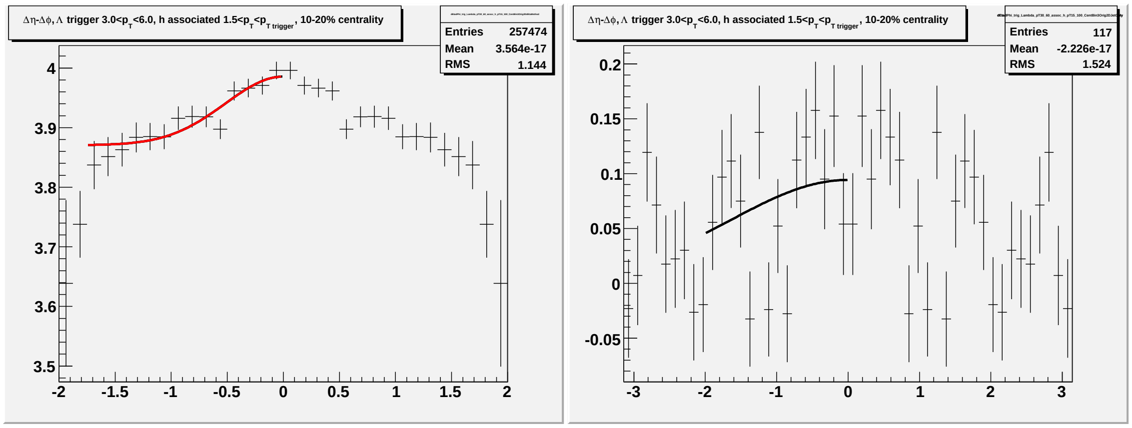

Figure F.9: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

F.C Fits for N_{part} dependence in $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV for identified trigger particles

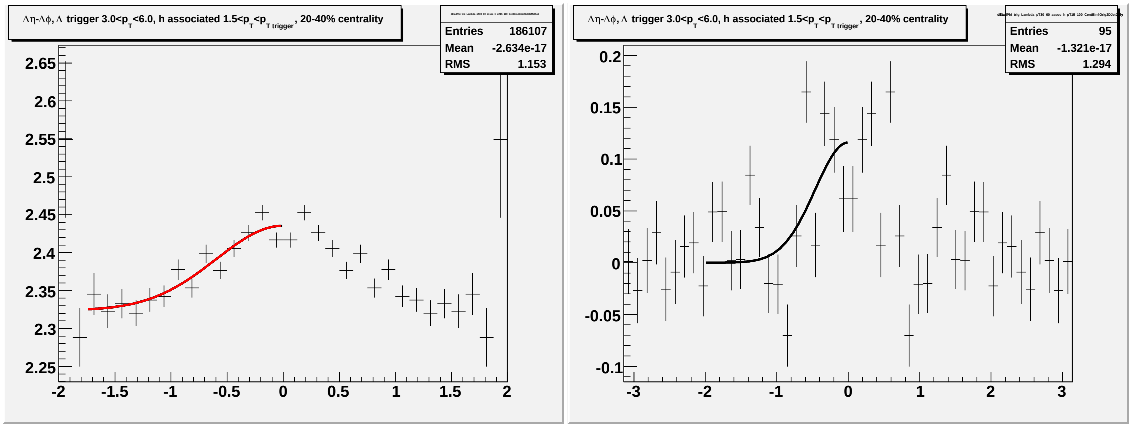

Figure F.10: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

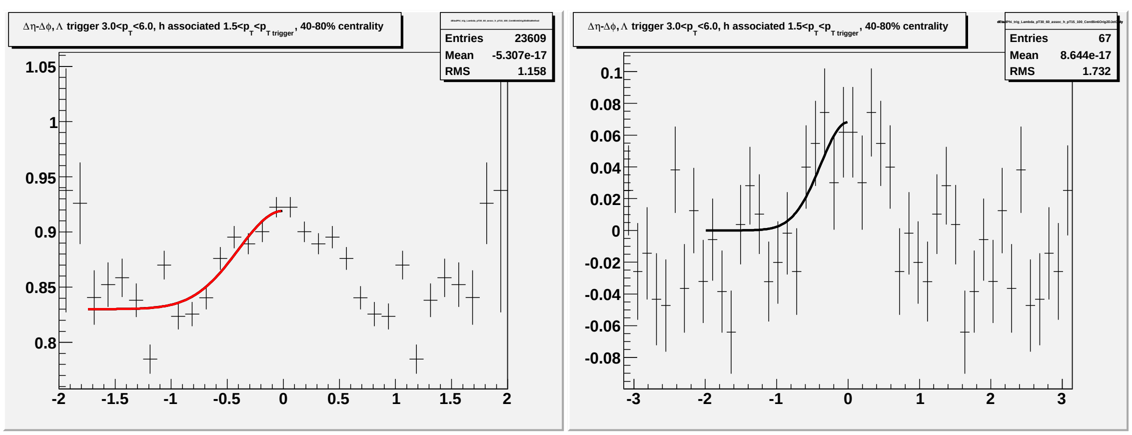

Figure F.11: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

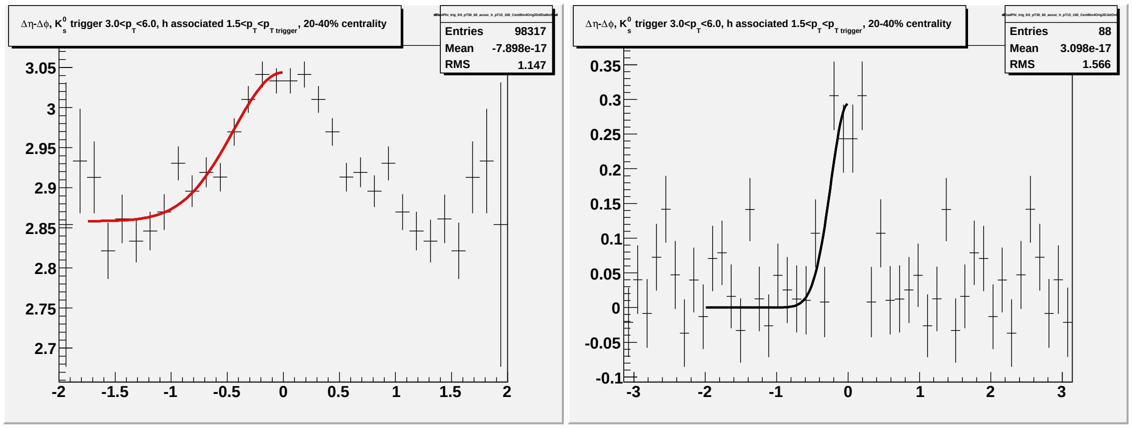

Figure F.12: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta \,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta \,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta \,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

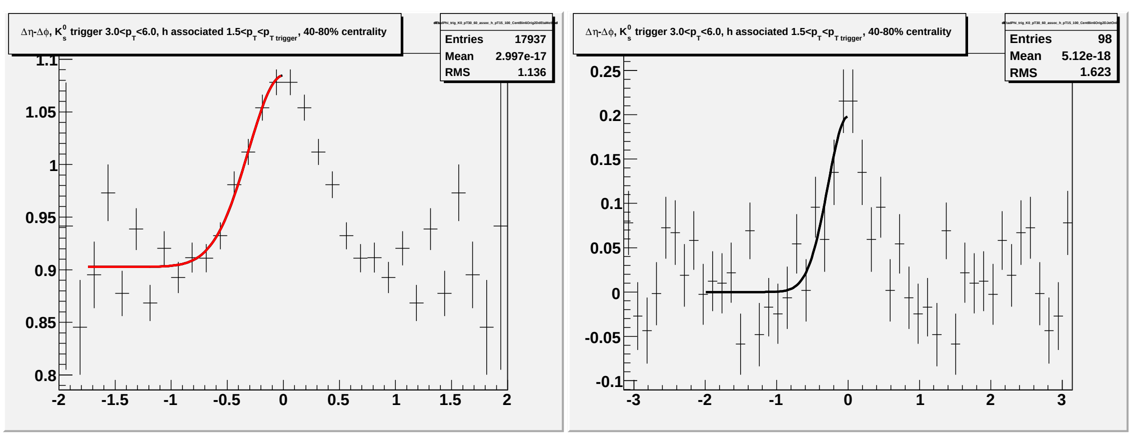

Figure F.13: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta \,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta \,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta \,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for K_S^0 trigger particles.

Appendix G

$p_T^{associated}$ dependence for identified trigger particles

$Cu+Cu$**- use fit on highest** p_T K_S^0 **and** Λ **points.** $Au+Au$ **40-80% mid-**p_T **and highest** p_T **lambda use fit and change fit range, highest** p_T K_S^0**.** $d+Au$ **mid-**p_T Λ**.**

The default method is **fitting due to the large residual track merging effect in most bins.**

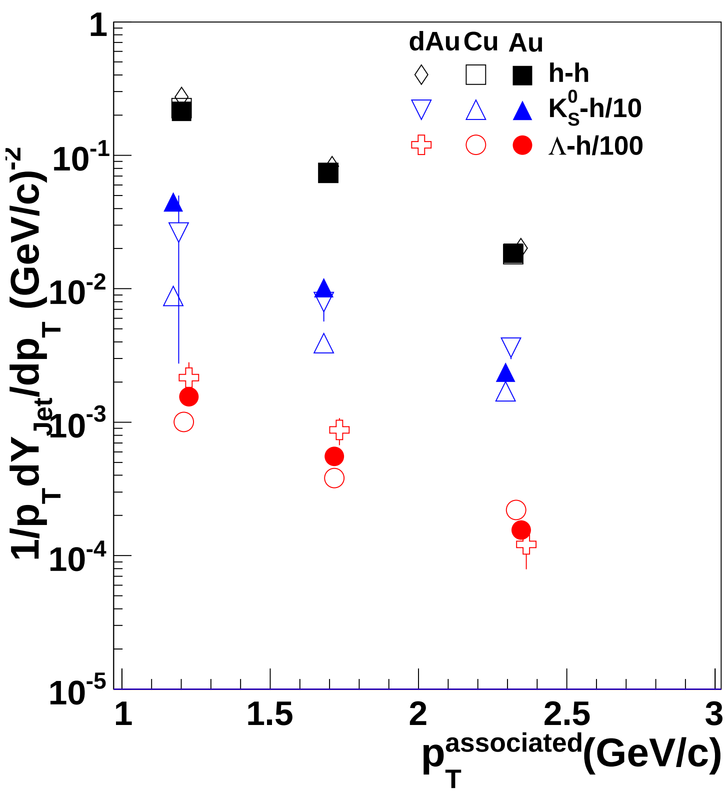

Figure G.1: $p_T^{associated}$ dependence of Y_{Jet} for $3.0 < p_T^{trigger} < 6.0$ GeV/c in 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV for unidentified hadrons, Λ, K_S^0, and Ξ trigger particles. The line is from a fit of the h-h data to A$e^{-p_T/k}$. The identified trigger particle data have been horizontally offset for visibility.

G.A Fits for $p_T^{associated}$ dependence with identified trigger particles in $d + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV

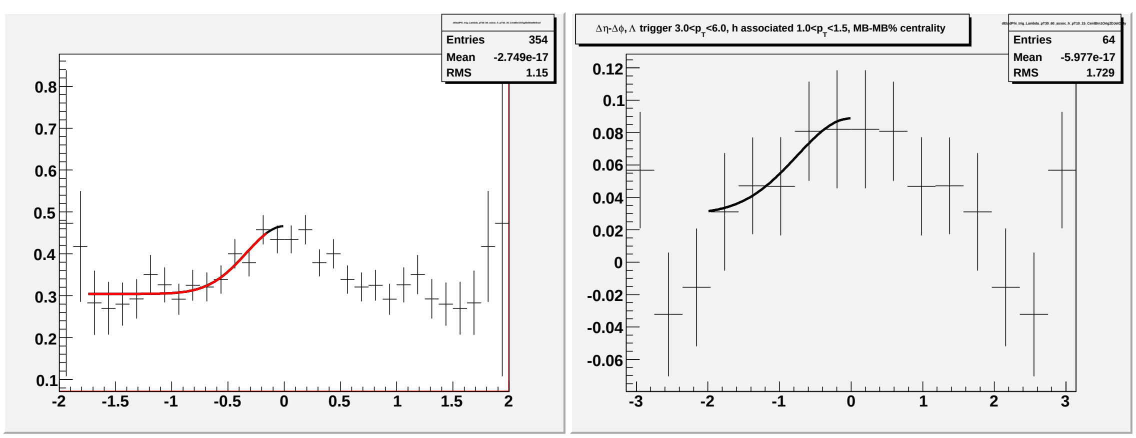

Figure G.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} <$ 1.5 GeV/c for identified Λ triggers.

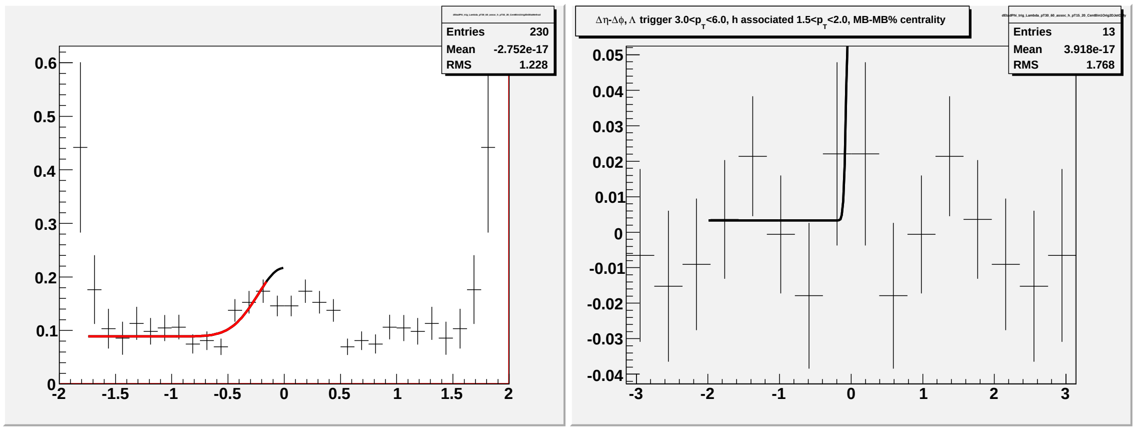

Figure G.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c for identified Λ triggers.

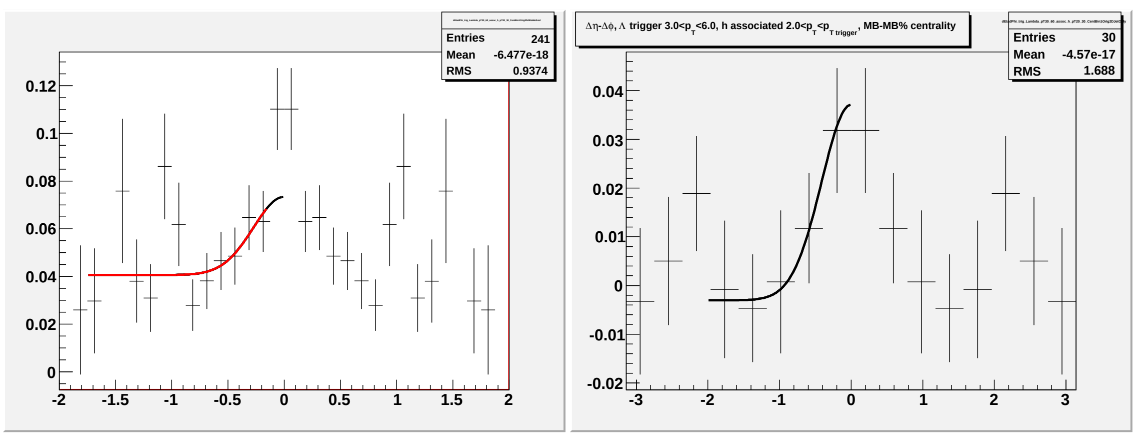

Figure G.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c for identified Λ triggers.

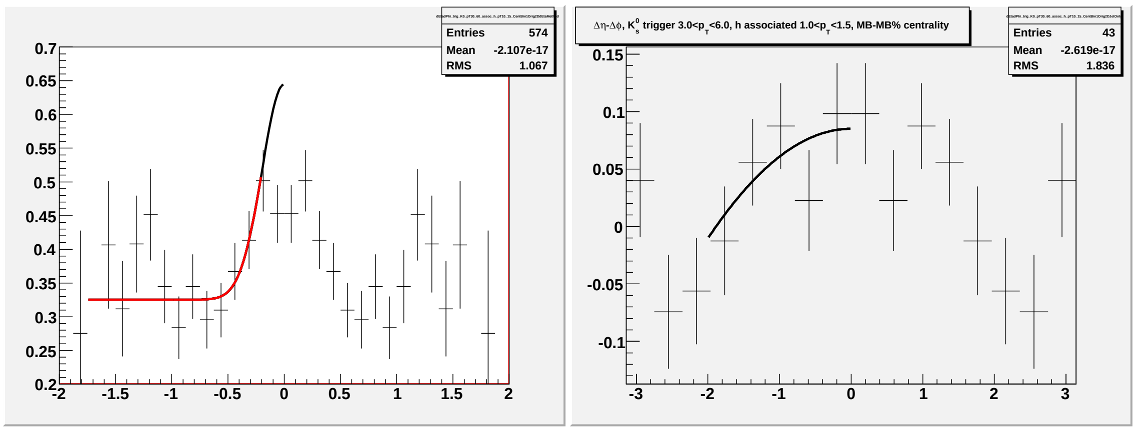

Figure G.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c for identified K_S^0 triggers.

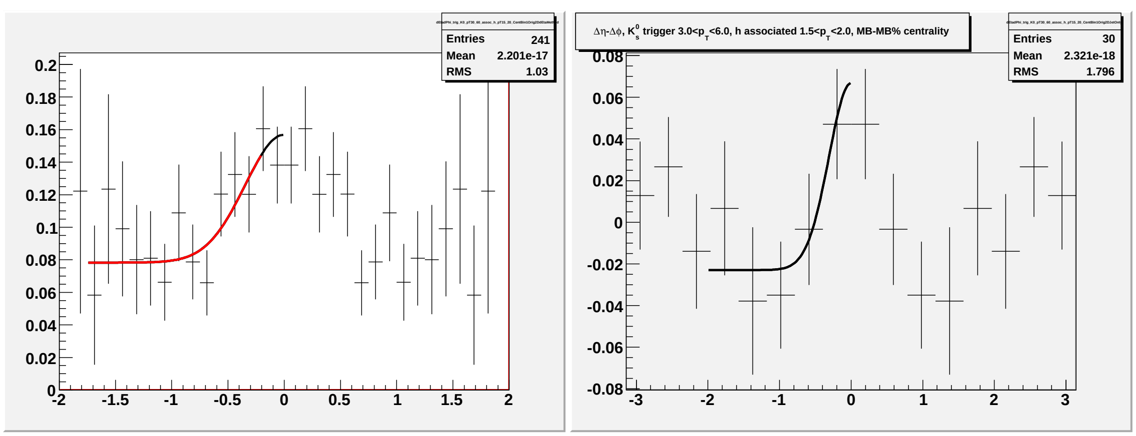

Figure G.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c for identified K_S^0 triggers.

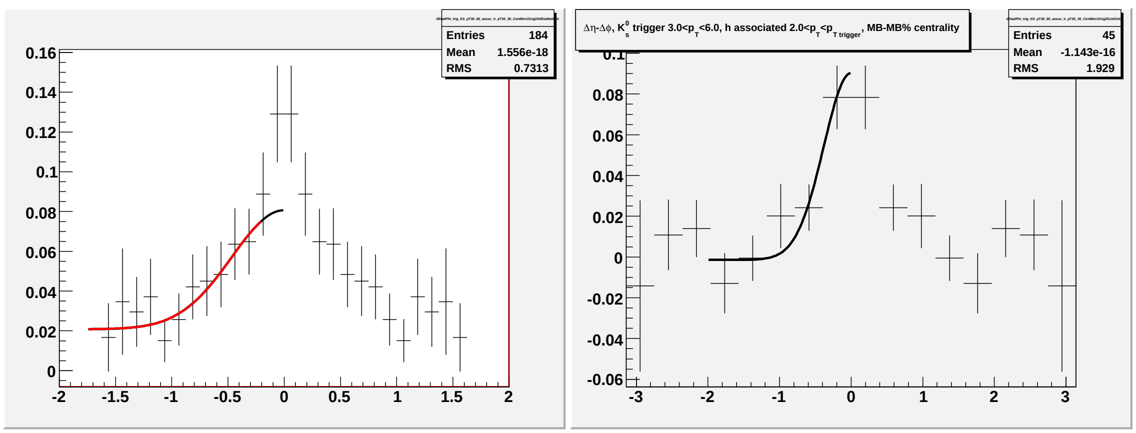

Figure G.7: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} <$ 3.0 GeV/c for identified K_S^0 triggers.

G.B Fits for $p_T^{associated}$ dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV for identified trigger particles

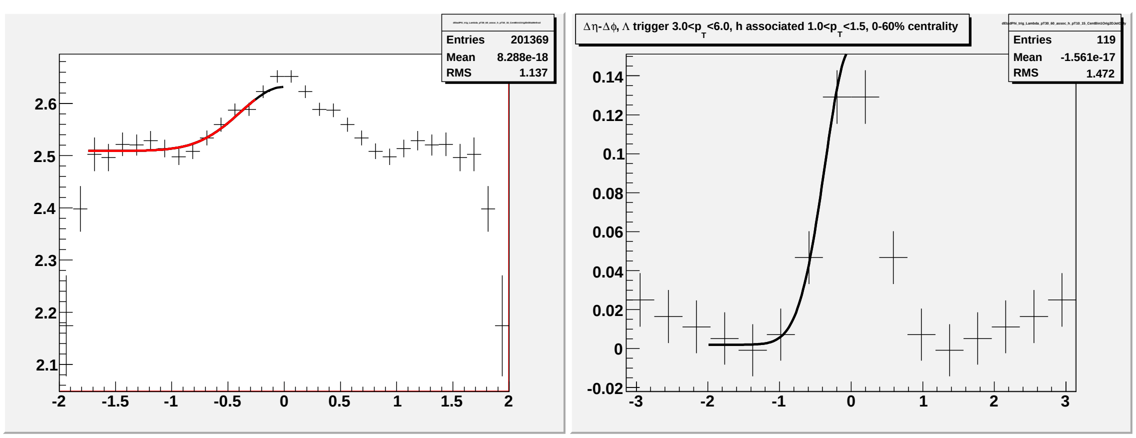

Figure G.8: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

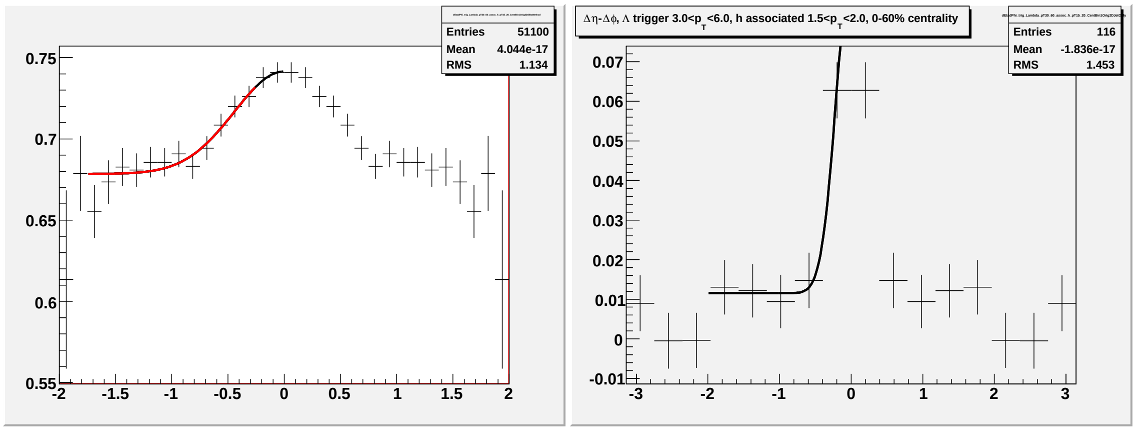

Figure G.9: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

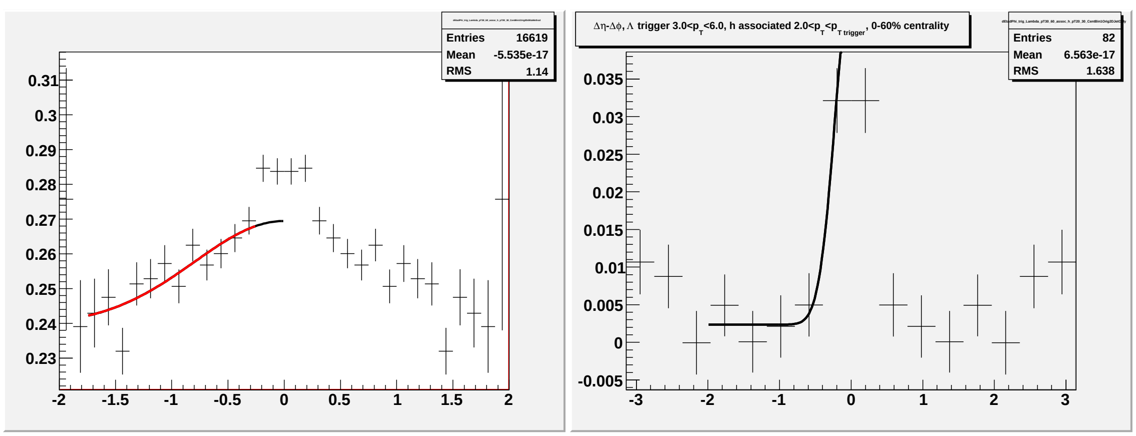

Figure G.10: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

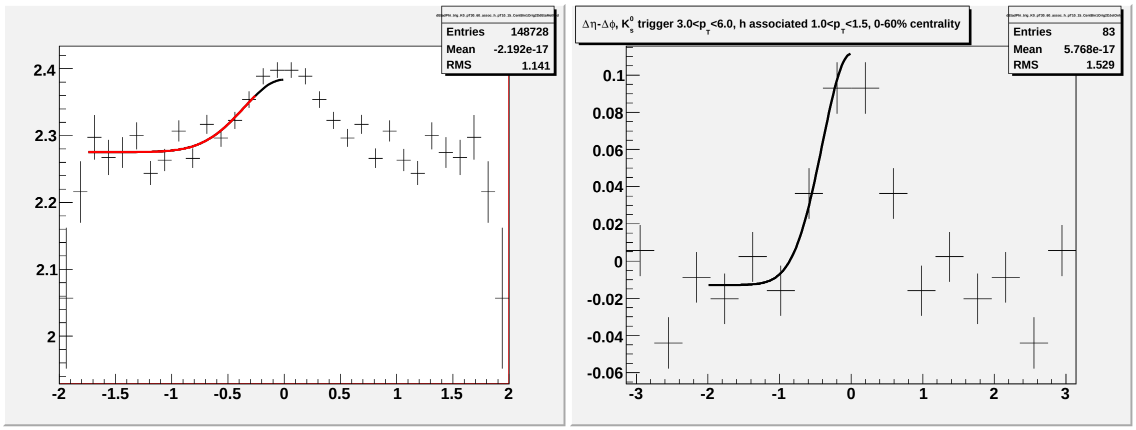

Figure G.11: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c for K_S^0 trigger particles.

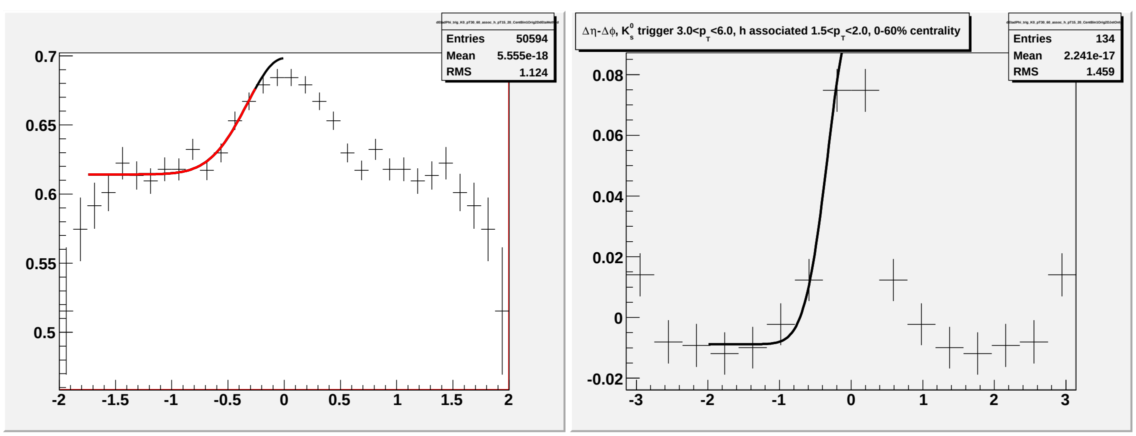

Figure G.12: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c for K_S^0 trigger particles.

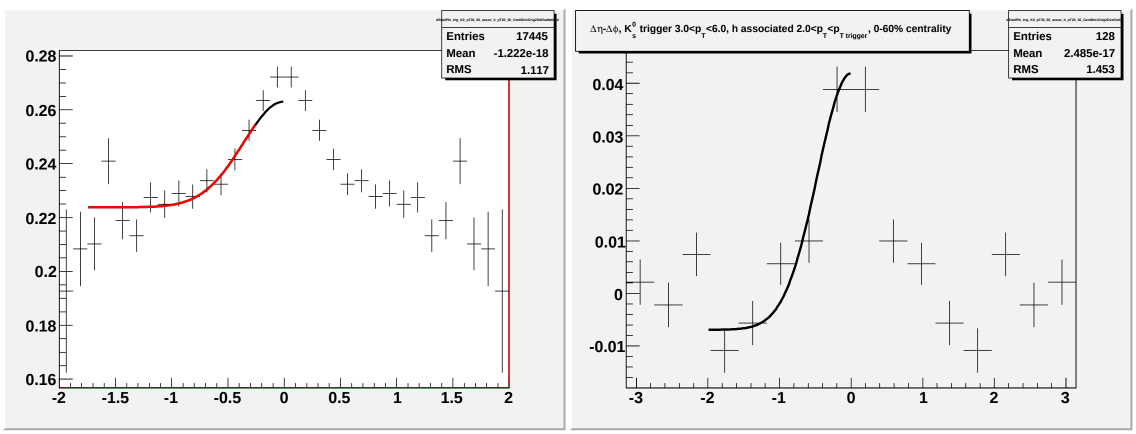

Figure G.13: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta \,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta \,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta \,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c for K_S^0 trigger particles.

G.C Fits for $p_T^{associated}$ dependence in 40-80% $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV for identified trigger particles

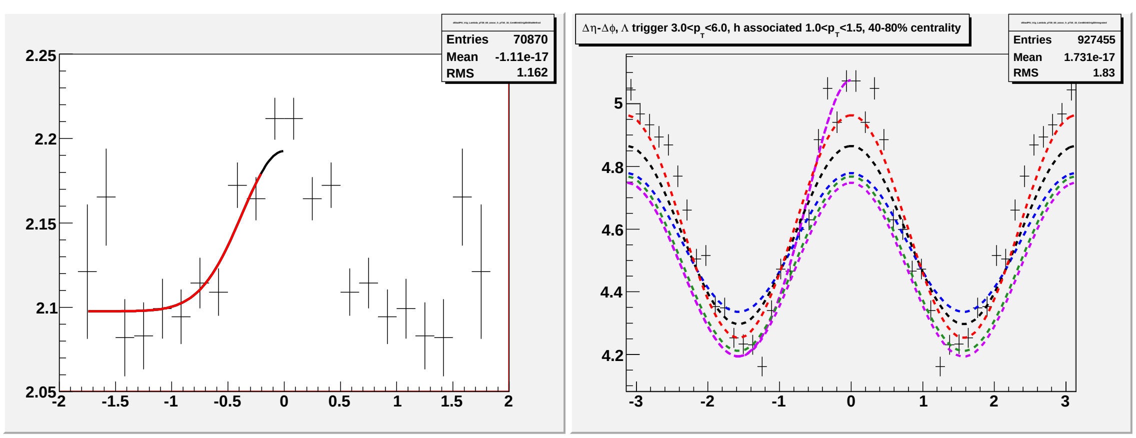

Figure G.14: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

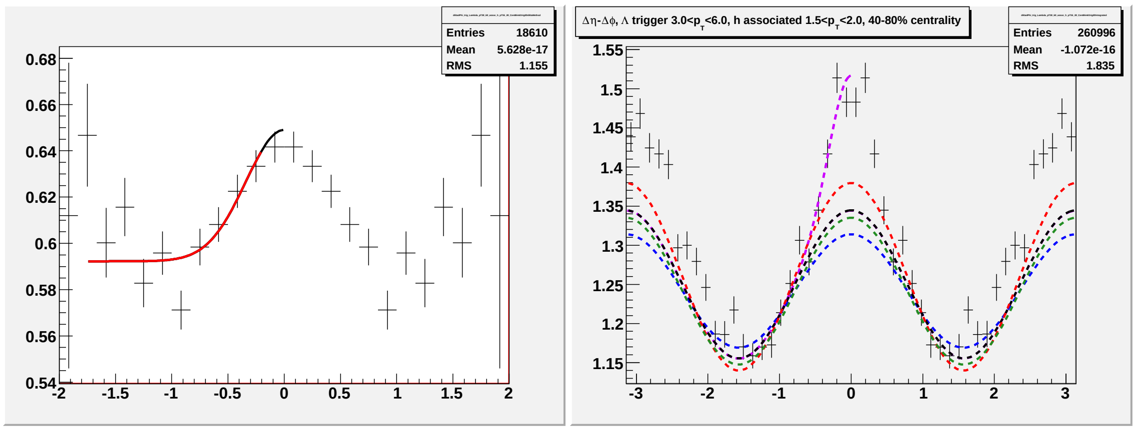

Figure G.15: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta$ $| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta$ $| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 $< p_T^{associated} < 2.0$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

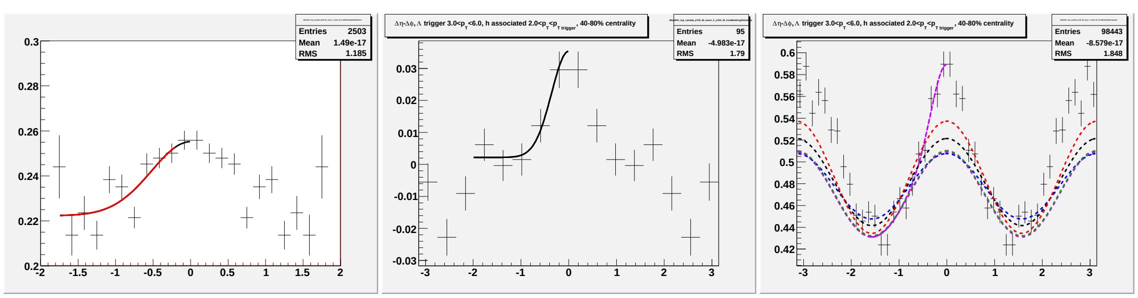

Figure G.16: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta$ $| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta$ $| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 2.0 $< p_T^{associated} < 3.0$ GeV/c for Λ and $\bar{\Lambda}$ trigger particles.

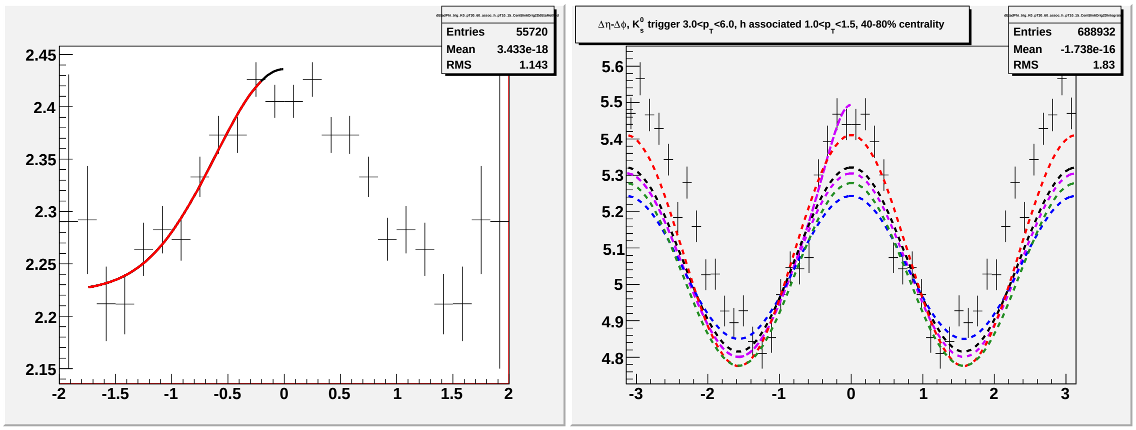

Figure G.17: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c for K_S^0 trigger particles.

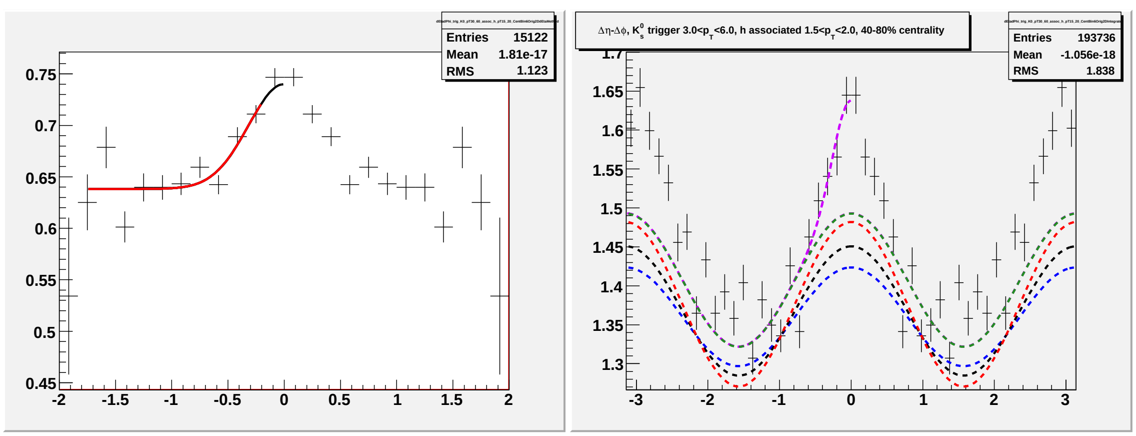

Figure G.18: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c for K_S^0 trigger particles.

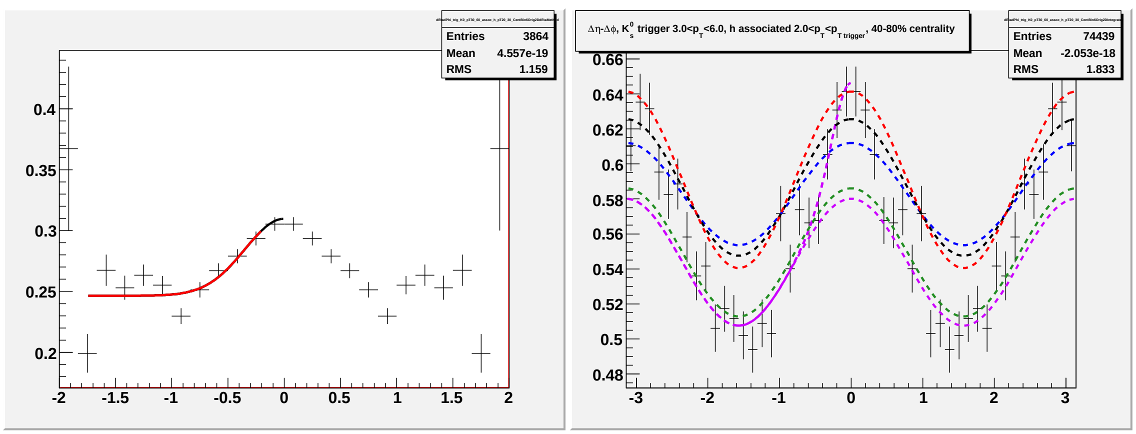

Figure G.19: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c for K_S^0 trigger particles.

Appendix H

$p_T^{associated}$ dependence for identified associated particles

peripheral $Au+Au$ **lowest** p_T K_S^0 **need to use fit**

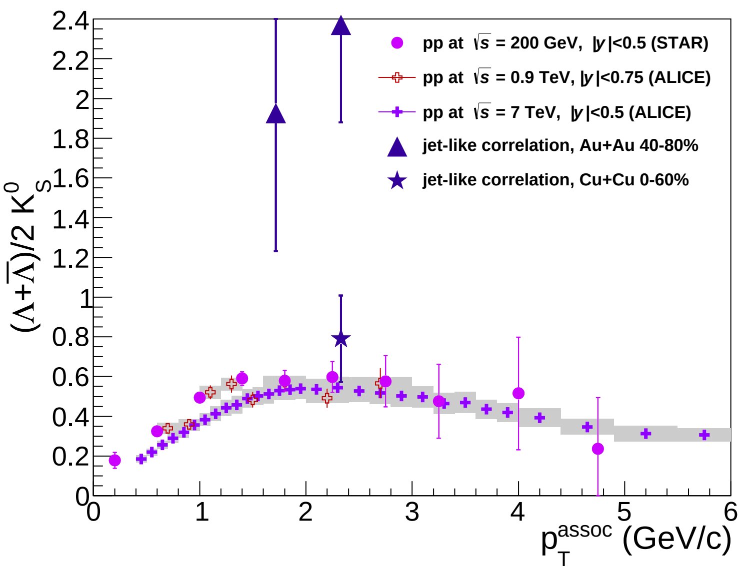

Figure H.1: Composition of the jet-like correlation compared to the inclusive spectra

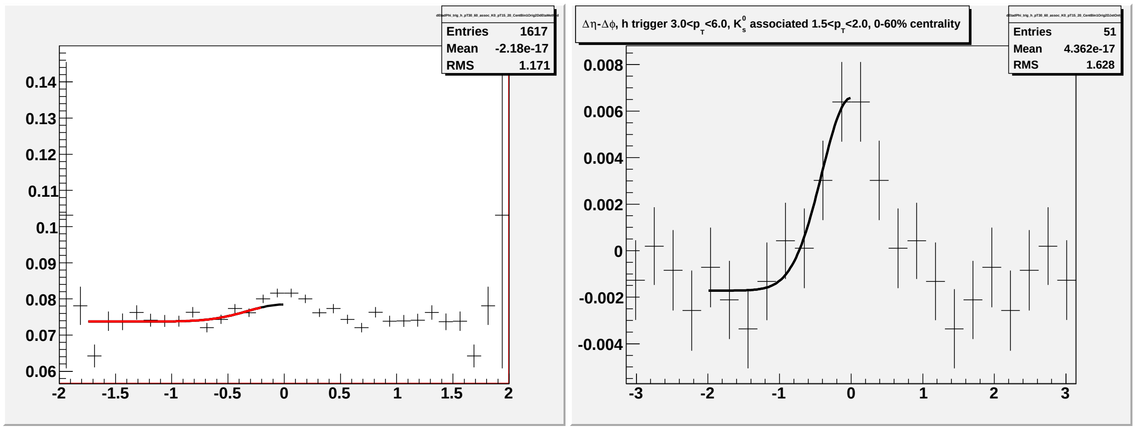

Figure H.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

H.A Fits for $p_T^{associated}$ dependence for identified associated particles in Cu+Cu collisions at $\sqrt{s_{NN}}$ = 200 GeV

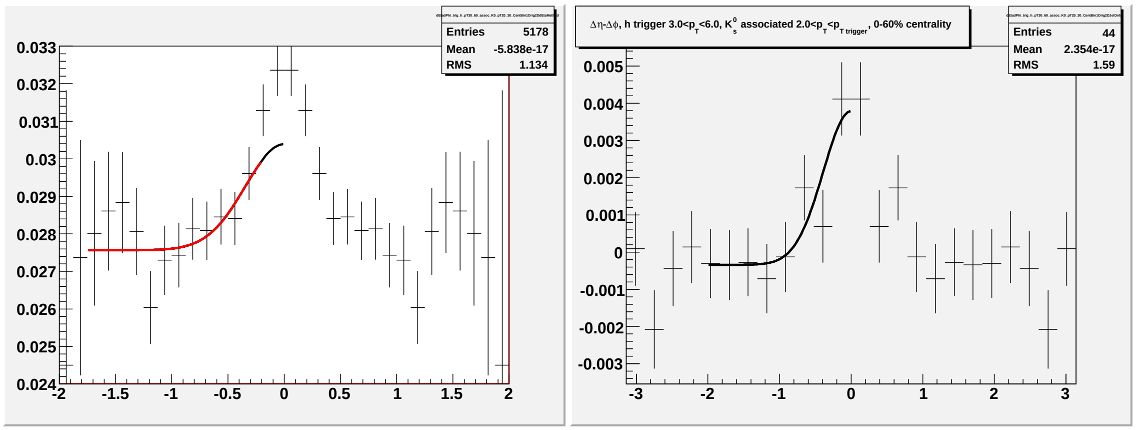

Figure H.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

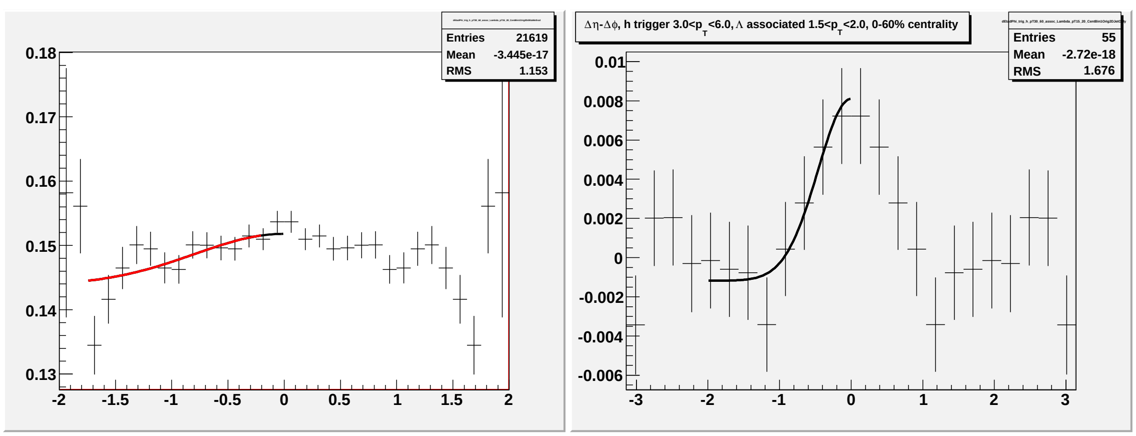

Figure H.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

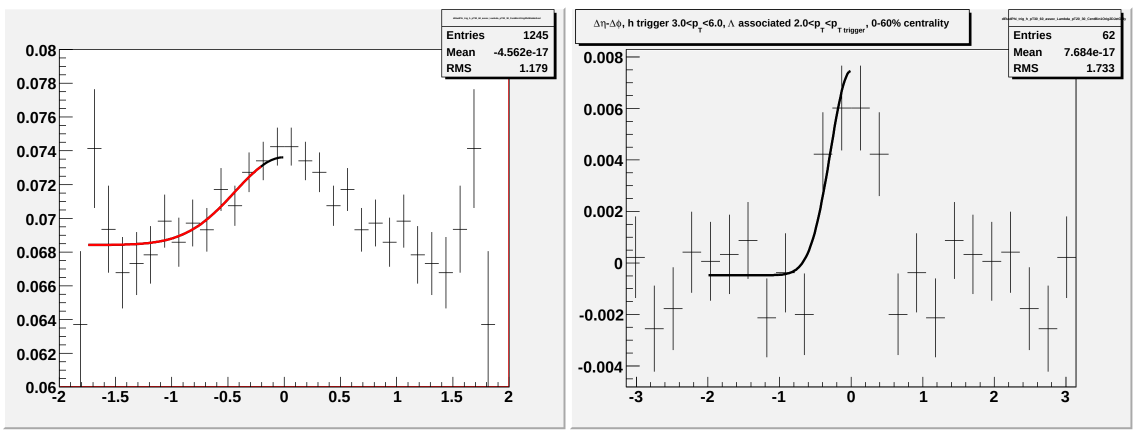

Figure H.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} <$ 3.0 GeV/c.

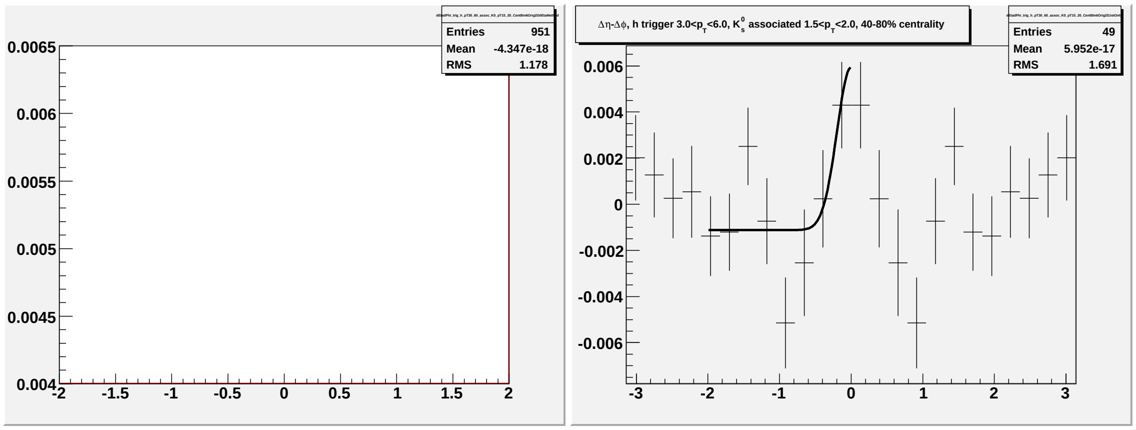

Figure H.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} <$ 2.0 GeV/c.

H.B Fits for $p_T^{associated}$ dependence for identified associated particles in 40-80% $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV

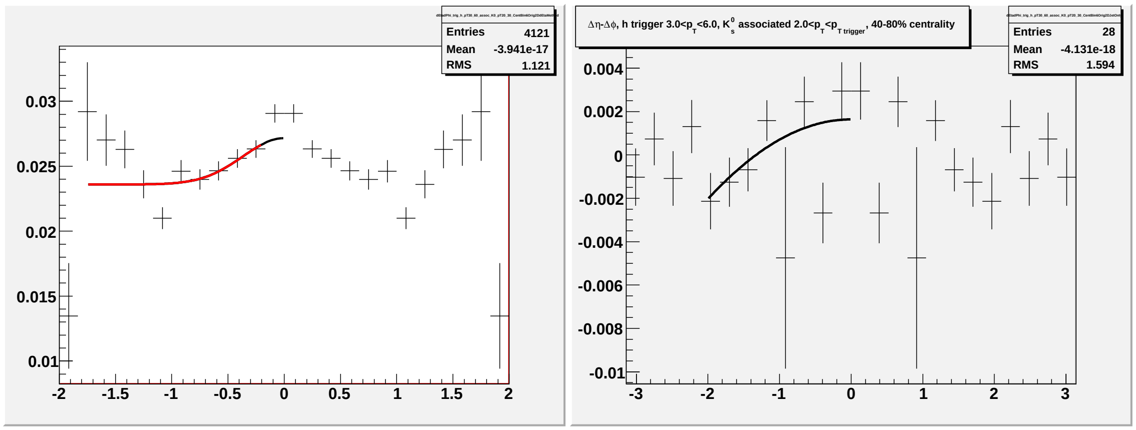

Figure H.7: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

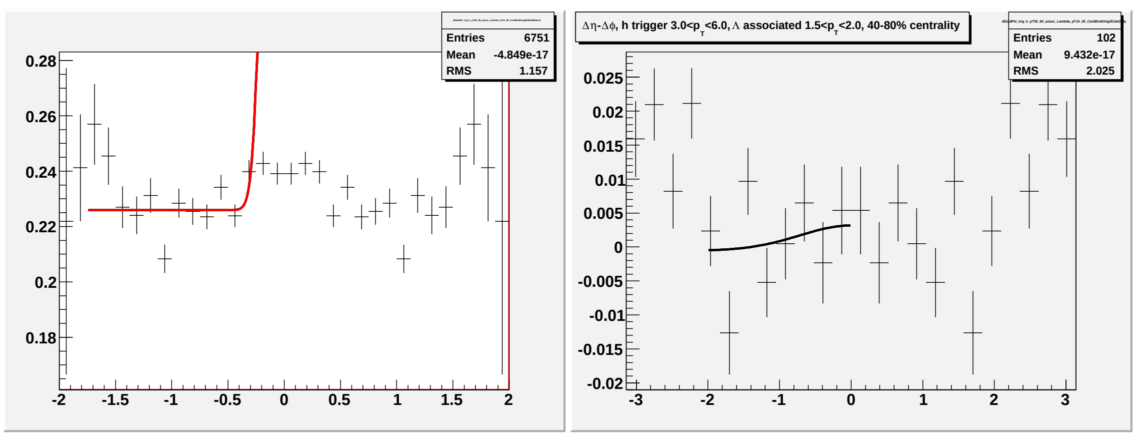

Figure H.8: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

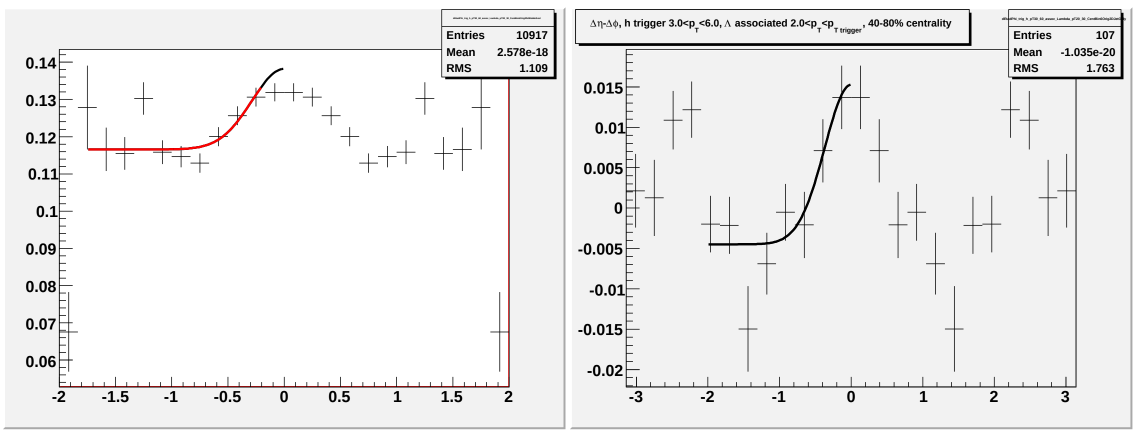

Figure H.9: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} <$ 3.0 GeV/c.

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